ClaimGuard AI: Test Evaluation Report

Submission document: detection metrics, F1 score, false-positive rate, latency and limitations, with every experiment and every graph produced during the project.

Team project for the CSTAM-VELODOC challenge (mentor: Dr. Wael Hilali). Report date: 2026-10-19. Branch `rust-rewrite`. All claims, codes, prices and payer rules are **synthetic**. Nothing here has been measured on real hospital data, real payers or real reviewers.

How to read this report

- **Section 1** answers the five things the submission asks for (detection metrics, F1, false-positive rate, latency, limitations) on one page.
- **Sections 2 to 6** are the detection evaluation itself: how it was measured, on what, and the tables.
- **Sections 7 to 16** are every other experiment we ran, in the order they were done, each with its graphs, its result and its limits. They answer the question "why is the system built this way?".
- **Section 17** collects all the limitations in one place. **Section 18** says how to re-run everything and where each file is.
- Every number is copied from a document or a result file in the repository (named next to it). Where a number comes from a simulation, a single laptop, or a test set written by the same team that built the feature, the text says so next to the number.

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1. The answer on one page

Question	Answer	Where
Detection quality	The rule engine reproduces the organizers' answer key exactly on all three public splits: 400 claims (development), 150 (validation), 50 (stress). Every FAIL is found and nothing is invented.	section 3
F1 score	FAIL-detection precision, recall and F1 are **1.0000** on each split. Macro F1 over the five rule categories is **1.0000** (also over severities and over all 15 rules). Status accuracy 1.0000, 0 disagreements with the key in 9,000 results.	section 3
False-positive rate	**0 valid claims wrongly flagged**: 0 of 160, 0 of 62 and 0 of 24 claims the key says have no FAIL. Exact 95% upper bounds on the true rate: **1.9%, 4.7%, 11.7%**. On 25,068 generated valid claims (against an independent oracle): 0, upper bound 0.01%. Zero observed is not zero true.	section 4
Latency	The rule engine takes **0.19 ms per claim** (Rust). The Python reference measured a median 0.66 ms (p95 1.11 ms). The step that dominates end to end is the optional language-model explanation: a median **2.8 s**, with a 90 s ceiling and a template fallback. Reading a document: about 5 s per page. A 1,000-claim batch is taken in about 17.5 s.	section 6
Limitations	The data is synthetic and small; a perfect score on public data cannot rank this system against a better one; the second implementation (the oracle) shares our reading of the rulebook; FHIR loses authorization records so rule R009 abstains on that path; latency is from one laptop, not the target server; the Arabic wording and the question sets were written by us, not reviewed by native speakers or operators; the learning experiments use simulated reviewers.	section 17

Three facts that matter for how to read the perfect scores:

1. The 15 rules are deterministic and were built against the public data. A correct implementation of the rulebook should reach 1.0 there. What the evidence shows is the **absence of errors found**, with upper bounds, not that the system beats an alternative.

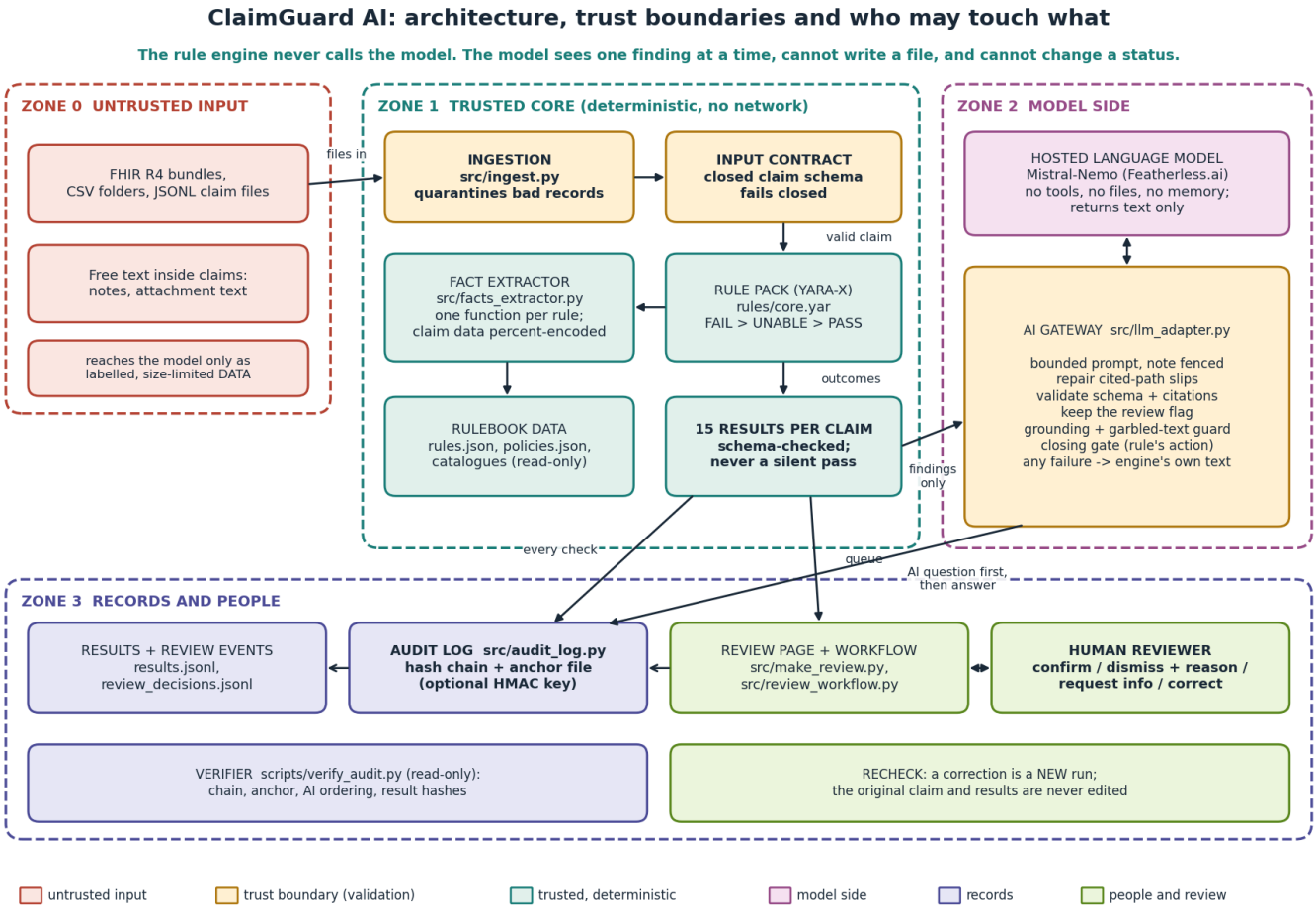
2. The AI never decides. It writes an explanation of a finding the rules already made, and a reply that tries to change a verdict is rejected. The explanation step has its own measurements (sections 8 and 9).
3. When the system is not sure, it says so (`UNABLE_TO_ASSESS`) and routes the claim to a person. No failing claim became a pass in any test.
4. **Added since the first version of this report:** "Ask the rulebook" now has a knowledge graph and an answer cache in front of passage search (section 12.5): 57 of 60 connection questions answered against 34 by passage search alone, the graph and the cache answering in about 0 ms, and 0 of 24 off-topic held-out questions taken by the graph. The first run on a real model found a confident wrong-answer failure (the model copying an example query), which was fixed and is reported there.

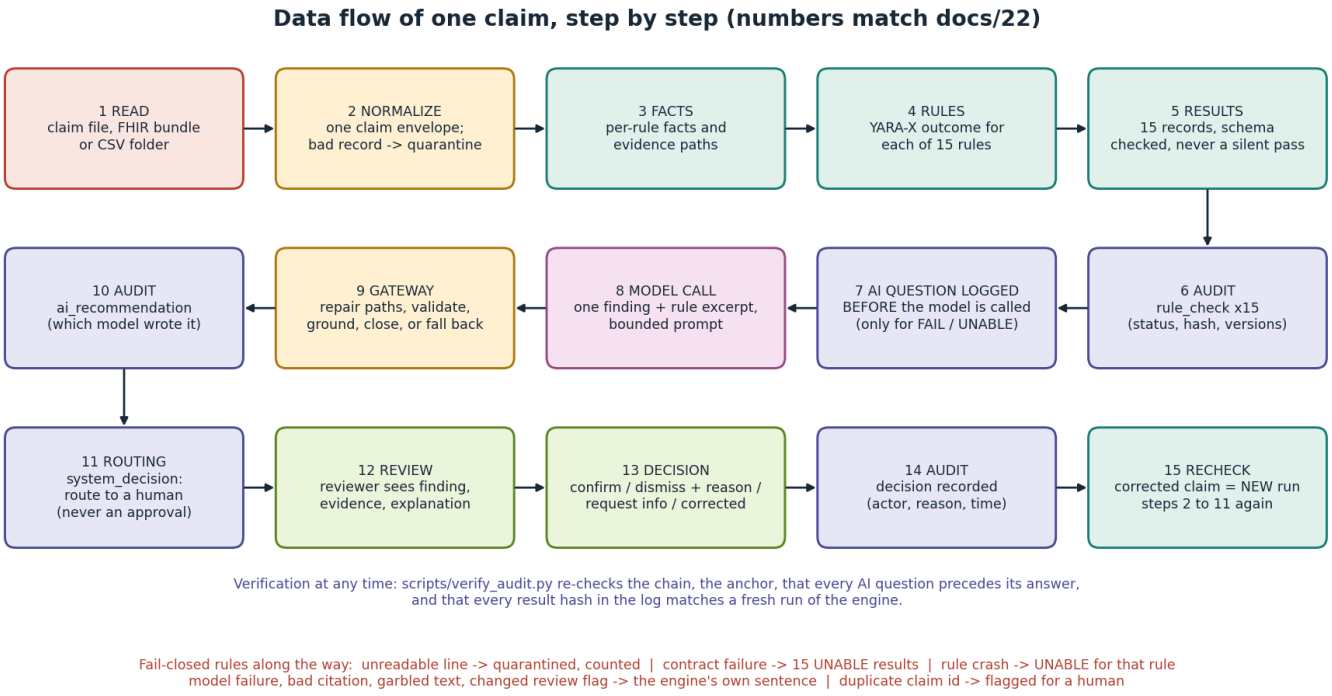
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2. What was tested, and how

2.1 The system

A pre-validation copilot: it ingests claims (FHIR R4 JSON, CSV, JSONL, or a PDF, scan or photo through an isolated reader), checks them against a fictional payer rulebook of 15 rules with a deterministic engine, explains each finding with a tightly bounded AI step, and records every check, AI action and human decision in a tamper-evident, signed audit log. A person makes the final call. It exists twice: the original Python reference and a Rust rebuild that the desktop app and the server image run.





2.2 Which build each number comes from

The project exists twice: the Python reference (the original Phase 1 build) and the Rust rebuild that the desktop app and server run. **This report mixes both, and each table says which**:

Evidence	Measured on	Why
Detection on the three public splits (section 3.1)	**both**: Python (`docs/29`, `metrics.json`) and Rust (`docs/46`, `rust_metrics.json`) give the same result	the Rust build is the product; the Python build is the reference it is checked against
FHIR and CSV paths (section 3.4)	**both**: Python (docs/29) and Rust (re-measured on 2026-10-19, `outputs/evaluation/rust_formats_metrics.json`)	identical result
Larger sets S7 to S9, the oracle, false-positive bounds on 25,068 and 4,064 generated claims (sections 4 and 5)	**Python engine** against the Python oracle (`tests/oracle.py`)	the oracle and generators are Python; the Rust engine was checked against the Python reference on the answer key and on ingestion (35 differential checks), not on the 107,635 generated claims
Latency table 6.1	**Python** reference; the Rust engine figure (0.19 ms) is stated separately	measured on different days
Language benchmark (section 7)	a Rust port of the rules compared with five others	the study that led to the Rust decision
AI explanation experiments (section 9), model comparison, architecture comparison	the **Python** pipeline (the explanation step was prototyped in Python and later ported); results are about models and prompts, not the language	
Document reading, retrieval, Arabic intake, batches	the **Rust** build (these features were built in Rust only)	

Evidence	Measured on	Why
Knowledge graph and answer cache for "Ask the rulebook" (section 12.5)	the **Rust** build (`cg ask`, the same code the server runs) with Gemma 3 4B through Ollama on the RTX 4060 laptop	built in Rust only; the model is the laptop's, not the server's

2.3 Three tiers of evidence, never merged into one headline number

- **Tier A, organizer answer key.** The only external truth: the organizers' synthetic dataset v1.0.0 (`data/development`, `data/validation`, `data/stress`, 600 claims, 9,000 results).
- **Tier B, our independent oracle.** `tests/oracle.py` is a second implementation of the 15 rules written from the rulebook text alone. It tests whether two implementations agree; a shared misreading would not show.
- **Tier C, hand-derived.** Expected statuses worked out by hand, one step either side of each boundary (123 cases).

2.4 Definitions

- **Positive class:** `FAIL`, as in the organizers' scorer. Precision, recall and F1 are computed on that class. `UNABLE_TO_ASSESS` (the engine's low-confidence state: an input is missing or unknown) is scored separately: an abstention is not an error, but an unnecessary one costs a reviewer time.
- **Rule categories** (our grouping; the rulebook defines none): completeness and arithmetic (R001, R007, R012); eligibility and coverage (R003, R004, R005, R015); timing (R002, R014); authorization and documentation (R008, R009, R010); catalogue, pricing and duplicates (R006, R011, R013). Counts are pooled within a category, then F1 is averaged across the categories where it is defined. Macro F1 is also reported by severity and over all 15 rules so the grouping cannot flatter the result.
- **Valid claims:** the claim-level false-positive rate is the share of claims with no gold FAIL where the engine raised any FAIL. Every rate carries the exact one-sided 95% (Clopper-Pearson) upper bound. By the rule of three, claiming an error rate below 1% needs about 300 error-free examples.
- **Confidence:** deterministic results carry `confidence: null` with kind `not_probabilistic`. Nothing pretends a rule result has a probability.

2.5 Provenance of every evaluation set

Set	Tier	Name	Claims	Results	Label source	Limitation
S1	A	data/development	400	6,000	organizers' `expected_results.jsonl`	rules were built against it
S2	A	data/validation	150	2,250	organizers' key	same
S3	A	data/stress	50	750	organizers' key	9 of 15 rules have no failing example
S4	A	handbook worked cases	0 new	0	handbook	ten claims, already in S1 to S3
S5	A	the 600 public claims as FHIR bundles	600	9,000	key, same claim ids	FHIR carries no authorizations
S6	A	the 600 public claims as CSV folders	600	9,000	key	lossless re-encoding
S7	B	generated claims (seed 20260927)	107,635	1,614,525	independent oracle	oracle shares our reading
S8	B	boundary-aware mutants (seed 20261004)	33,945	509,175	independent oracle	same caveat
S9	C	boundary table	123	140	derived by hand	written by us, small

Each set carries a file SHA-256 prefix, the script and seed that generated it, and its known limitation in `outputs/evaluation/provenance.json`.

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3. Detection quality on the organizer data (tier A)

3.1 Python reference and Rust build, same scoring code

Split	Claims	Precision	Recall	F1 (FAIL)	Macro F1 by category	Status accuracy	Disagreements with the key
50-claim split (`data/stress`)	50	1.0000	1.0000	1.0000	1.0000	1.0000	0
`data/validation`	150	1.0000	1.0000	1.0000	1.0000	1.0000	0
`data/development`	400	1.0000	1.0000	1.0000	1.0000	1.0000	0

The Rust engine (`rust/tools/phase2_eval.py`, `outputs/evaluation/rust_metrics.json`) and the Python reference (`outputs/evaluation/metrics.json`) give the same result. FAIL-detection errors over all results: 0 of 6,000, 0 of 2,250, 0 of 750 (95% upper bounds 0.05%, 0.13%, 0.40%). Missed abstentions and false abstentions: 0 on all three splits; `NOT_IMPLEMENTED`: 0.

3.2 By rule category (true positives, all F1 = 1.0)

Category	50-claim	validation	development
completeness and arithmetic	11	37	82
eligibility and coverage	3	30	101
timing	0 (F1 undefined)	11	31
authorization and documentation	6	14	41
catalogue, pricing and duplicates	6	25	64

3.3 By rule (F1, number of failing claims)

Rule	development	validation	50-claim
R001	1.0000 (36)	1.0000 (17)	1.0000 (8)
R002	1.0000 (11)	1.0000 (4)	n/a (0)
R003	1.0000 (28)	1.0000 (7)	1.0000 (3)
R004	1.0000 (28)	1.0000 (7)	n/a (0)
R005	1.0000 (26)	1.0000 (6)	n/a (0)
R006	1.0000 (24)	1.0000 (10)	n/a (0)
R007	1.0000 (23)	1.0000 (7)	n/a (0)
R008	1.0000 (11)	1.0000 (4)	n/a (0)
R009	1.0000 (15)	1.0000 (5)	1.0000 (3)
R010	1.0000 (15)	1.0000 (5)	1.0000 (3)
R011	1.0000 (10)	1.0000 (4)	n/a (0)
R012	1.0000 (23)	1.0000 (13)	1.0000 (3)
R013	1.0000 (30)	1.0000 (11)	1.0000 (6)
R014	1.0000 (20)	1.0000 (7)	n/a (0)
R015	1.0000 (19)	1.0000 (10)	n/a (0)

Read each cell with its count. In the 50-claim split, **9 of the 15 rules have no failing example at all**, so their F1 is undefined and left out of the average, and the rest have as few as 3. That split cannot test most rules for missed failures. The 400-claim development split has at least 10 failing examples for every rule, and validation at least 4, so conclusions about single rules rest on those two, with the 50-claim split as a check.

3.4 The FHIR path: the one real cost

Measured on both builds, with the same result. The Rust build was re-measured on 2026-10-19 (`cg ingest` on the 600 public claims in each format, then `cg evaluate`, scored against the key): FHIR precision 1.0000, recall 0.9502, **F1 0.9745** (439 true positives, 23 missed, 0 false positives, 310 disagreements, all of them R009 abstentions); CSV precision 1.0000, recall 1.0000, **F1 1.0000** (0 disagreements). So FHIR does **not** score 1.0 on the Rust build either; only the direct (normalized) and CSV paths do.

Claims read back from FHIR bundles (S5) score F1 **0.9745** (precision 1.0000, recall 0.9502, status accuracy 0.9656, 310 disagreements). Every one of the 310 disagreements is rule R009 (authorization record matches service) becoming `UNABLE_TO_ASSESS`: a FHIR bundle in this dataset carries no authorization records, so the rule cannot be checked. 23 true FAILs and 287 PASSes become abstentions. The engine says so instead of guessing, the failing claims route to a reviewer, and **no failing result became a PASS**. The cost is reviewer workload: 134 of 201 fully clean claims (66.7%) receive an unnecessary abstention on this path. The CSV path (S6) is lossless: F1 1.0000, 0 disagreements. `rust/tools/compare_ingest.py` (35 checks) confirms the 600 public claims in all three formats give claims and reports identical to the Python reference.

3.5 Baselines: would a weak system also score well?

Set	Predictor	Precision	Recall	F1 (FAIL)	Status accuracy
development	always PASS	n/a	0.0000	0.0000	0.8357
development	organizers' starter baseline (3 of 15 rules)	1.0000	0.2759	0.4324	0.2000
development	this system	1.0000	1.0000	1.0000	1.0000
validation	always PASS	n/a	0.0000	0.0000	0.8351
validation	starter baseline	1.0000	0.2906	0.4503	0.2000
validation	this system	1.0000	1.0000	1.0000	1.0000
stress	always PASS	n/a	0.0000	0.0000	0.7987
stress	starter baseline	1.0000	0.4231	0.5946	0.2000
stress	this system	1.0000	1.0000	1.0000	1.0000

The metric does discriminate: predicting PASS for everything scores F1 0 and the starter baseline well below 1.

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4. Valid claims are preserved (false positives)

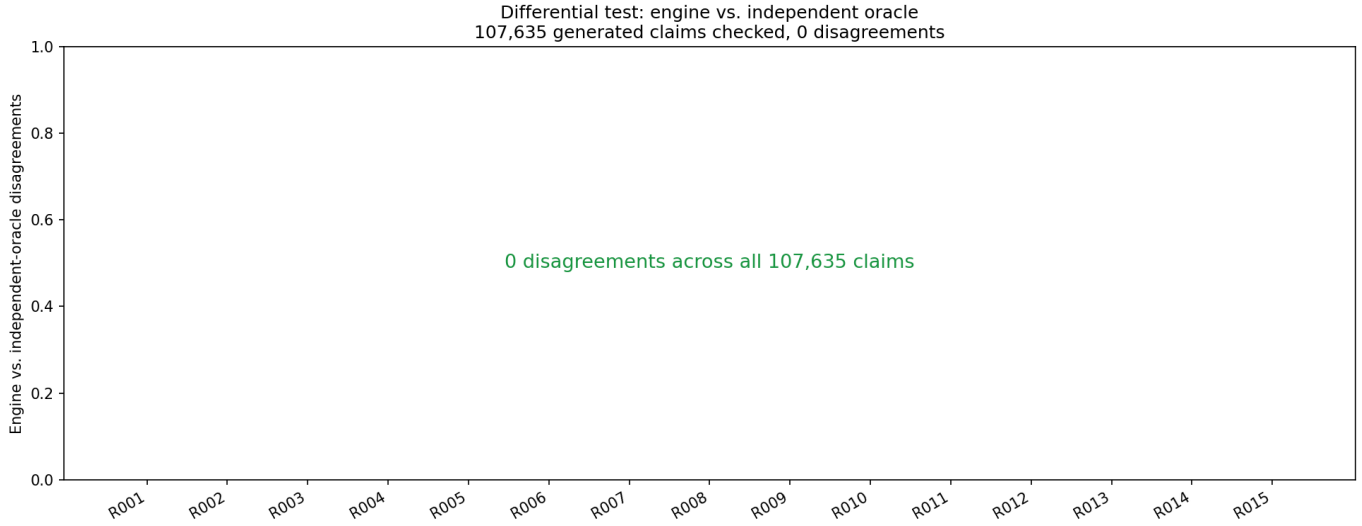
Set	Claims with no gold FAIL: engine raised a FAIL	95% upper bound	Results on fully clean claims: false alarms
development	0 / 160	1.85%	0 / 2,040 (bound 0.15%)
validation	0 / 62	4.72%	0 / 795 (bound 0.38%)
50-claim	0 / 24	11.73%	0 / 180 (bound 1.65%)
FHIR	0 / 246	1.21%	0 / 3,015
CSV	0 / 246	1.21%	0 / 3,015
generated (tier B)	0 / 25,068	0.01%	0 / 272,175

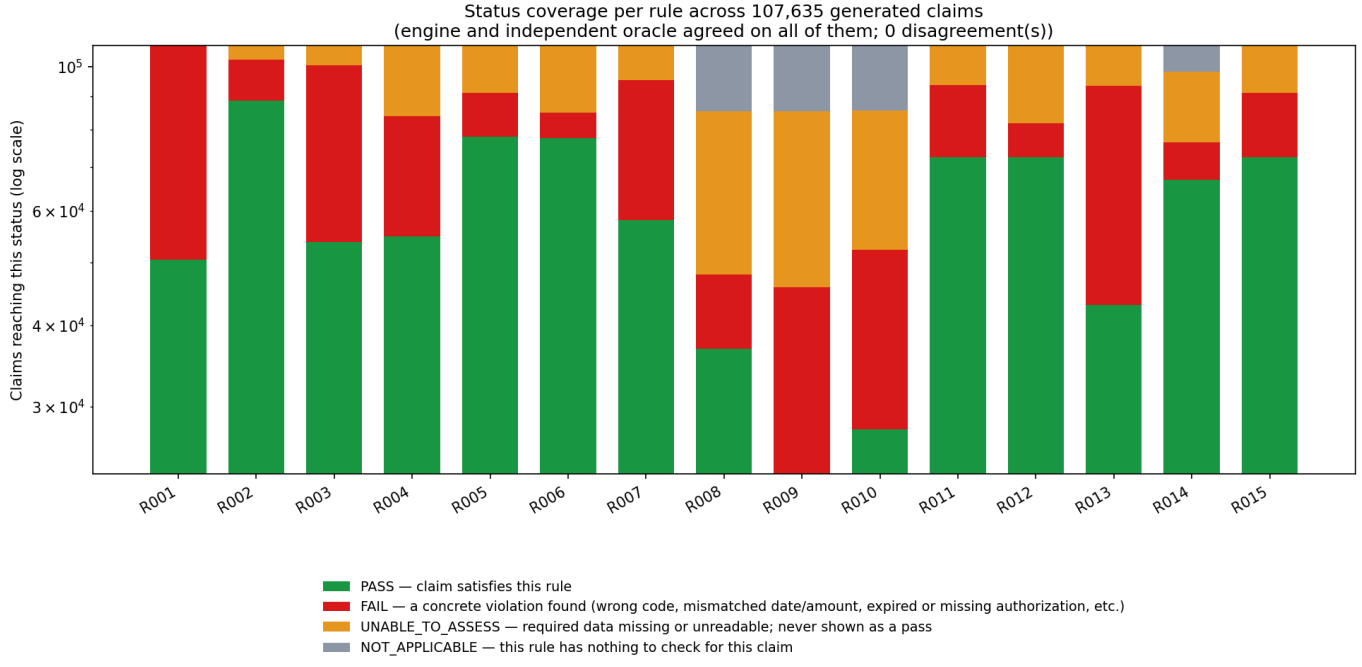
Set	Claims with no gold FAIL: engine raised a FAIL	95% upper bound	Results on fully clean claims: false alarms
mutants (tier B)	0 / 4,064	0.07%	0 / 34,605

On the direct and CSV paths the engine never raised a FAIL on a valid claim and never abstained on a fully clean one. The bound in each cell is what the sample can and cannot rule out: zero errors in 160 claims is consistent with a true rate up to the printed bound, not with zero.

5. Larger and harder sets (tiers B and C)

Set	Claims	Results scored	Disagreements	F1 (FAIL)	Engine crashes
S7 generated claims	107,635	1,614,525	0	1.0000	0
S8 mutants of the public claims	33,945	509,175	0	1.0000	0
S9 boundary table	123	140	0	1.0000	0





The coverage figure matters because an agreement test is worthless if a rule never reaches one of its statuses. A teammate's fuzzer never reached R009 PASS; this was traced to its seeding and reproduced with ours (23,627 of 107,635). Zero disagreements shows two implementations agree, not that either is right about the rulebook; only the organizers' key (tier A) can show a shared misreading.

Damaged input. `cg ingest` accepts normalized JSON/JSONL, FHIR bundles and the CSV tables, and quarantines any record it cannot accept with a reason, never stopping the run. 12,000 randomly damaged FHIR bundles reach the same accept-or-quarantine verdict as the Python reference and, when accepted, the same claim; every accepted damaged claim gets fifteen engine results; none becomes a pass.

Test counts. Python reference: 1,795 tests with an oracle, fuzzing and mutation testing around it. Rust build: 306 tests against real MongoDB and Redis (database tests are required, not skipped). Results are the same on Python 3.10, 3.12 and 3.14.

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6. Latency

6.1 Rule engine and intake (Python reference, warm process, 16-CPU laptop)

Path	n	mean ms	p50 ms	p95 ms	p99 ms	max ms
Rule engine only	1,800	0.7206	0.6570	1.1092	1.3084	1.6616
Ingest FHIR + engine	1,200	0.8265	0.7755	1.1642	1.3284	1.7984
Ingest CSV + engine	1,200	0.7570	0.6910	1.0301	1.2649	9.9210
Audited review, template explanation	100	31.3016	28.0037	71.2321	104.2582	124.9390

The Rust engine takes **0.19 ms per claim** (the difference between a 400-claim and a 50-claim run of `cg evaluate`, so process start-up cancels out). The audited path is slower because it includes the hash-chained audit write with its anchor update.

6.2 The explanation step dominates

AI explanation step, recorded live calls: median **2.86 s**, p95 28.89 s, 117 calls (`outputs/defense/load.json`, live Mistral-Nemo calls). The engine's share of the whole is under 0.03%. A timeout of 90 s and a template fallback bound the worst case.

6.3 Other measured latencies

What	Result	Source
Reading a document (two readers, RTX 4060 laptop)	about 5 s per page	docs/45
Taking in a batch of 1,000 claims (asynchronous: request answered in 0.06 s)	about 17.5 s on a release build (22 s on a debug build)	docs/53 section 7
Signed-in inbox request, direct / through nginx / through Caddy	37 ms / 71 ms / 76 ms (p50)	docs/54 section 6
Request that does no work, direct / nginx / Caddy	2.1 ms / 4.7 ms / 4.9 ms (p50)	docs/54 section 6

What	Result	Source
Dense rulebook question (bge-m3 sidecar), measured 2026-10-19	**GPU** (RTX 4060 laptop, fp16): model load 34 s, all 170 passages embedded once in **1.5 s**, a question embeds in a median **0.05 s** (0.05 to 0.07 s through the whole product route). **CPU** (fp32): model load 13 s, passages 18.9 s, a question 0.11 s. A warm-up thread embeds the passages at start-up	docs/55, `outputs/defense/embed_timing.json`, `rag_product_check_cuda.json`

| "Ask the rulebook" tiers, RTX 4060 laptop, 60 questions (section 12.5) | answer cache about 0 ms; knowledge graph about 0 ms; passage search median 23 to 31 ms; model picking a query about 3.0 s; model writing an answer 3.5 to 7 s | docs/60, `outputs/defense/kag_cag_eval.json` |

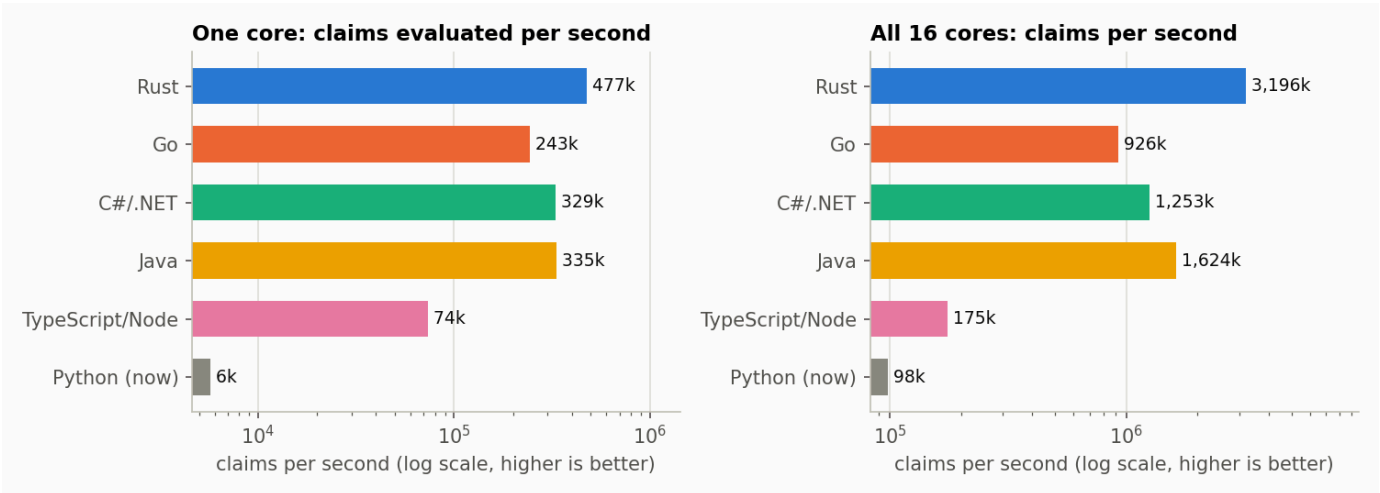
What is not measured: throughput under load on the target server (RTX 3080, Linux), and any real hospital traffic. Latency numbers are machine-dependent and not deterministic.

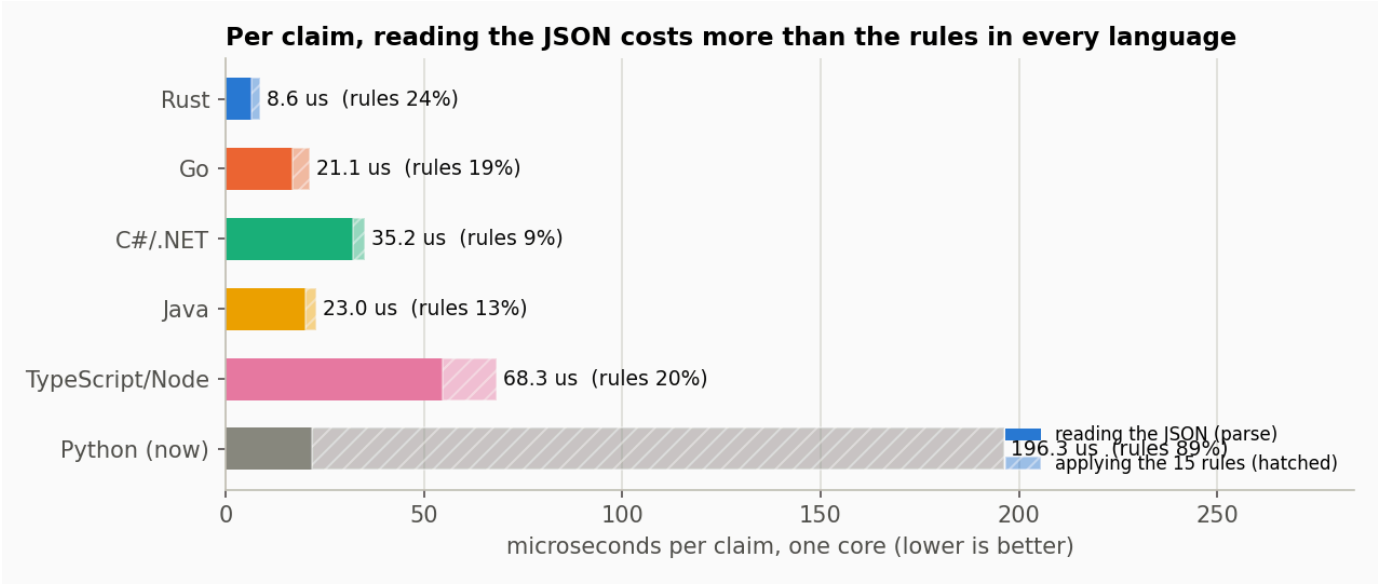
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7. Experiment: which programming language? (five candidates, measured)

Question from the team: choose five application languages that care about speed, efficiency, scalability, traffic handling and few bugs, and decide with evidence. Workload: the 15 rules on **20,000 generated claims** (seed 20260927); a port counts only if it matched the Python reference on **20,000 of 20,000** claims (0 mismatches), and every HTTP answer under load was checked (0 wrong answers). Machine: one Windows 11 laptop, 16 logical CPUs, 15.7 GB RAM, load generator on the same machine.

Language	HTTP stack	Port size (lines)
Rust	axum + tokio	590
Go	net/http	618
C# (.NET)	ASP.NET Core	338
Java	built-in HttpServer, virtual threads	323
TypeScript on Node	node:http + cluster	242
Python (control)	FastAPI + uvicorn	316

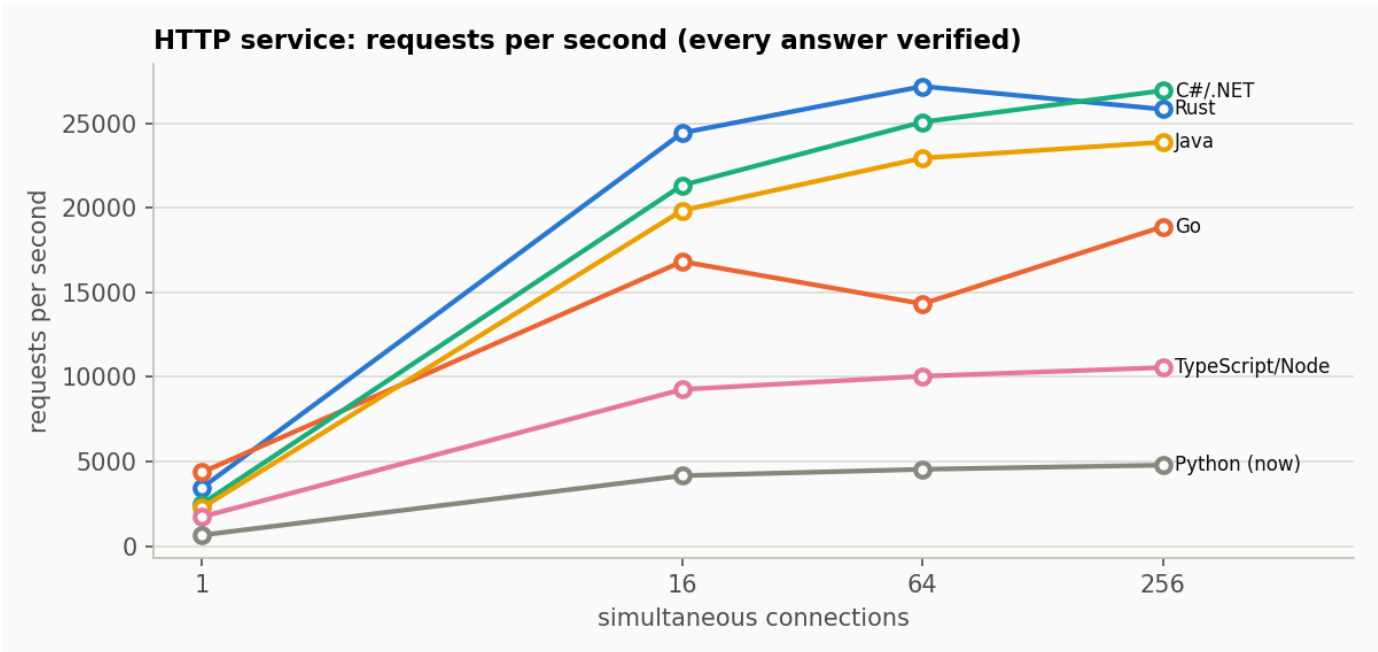


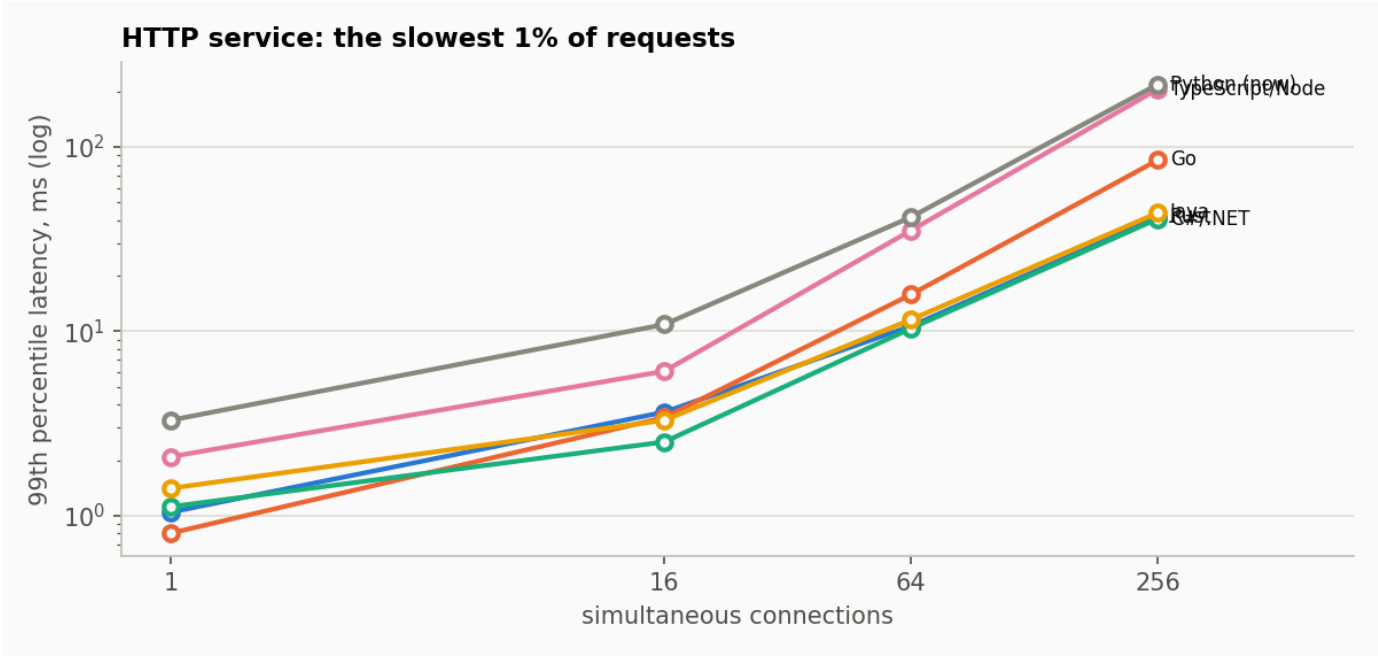


Rust evaluates the rules for 20,000 claims in about 42 ms, C# and Java about 60 ms, Go about 82 ms, Node about 270 ms; Python 1.4 s at best and up to 3.8 s (a 2.5x swing between runs on the same code). Reading the JSON costs more than the rules in every language.

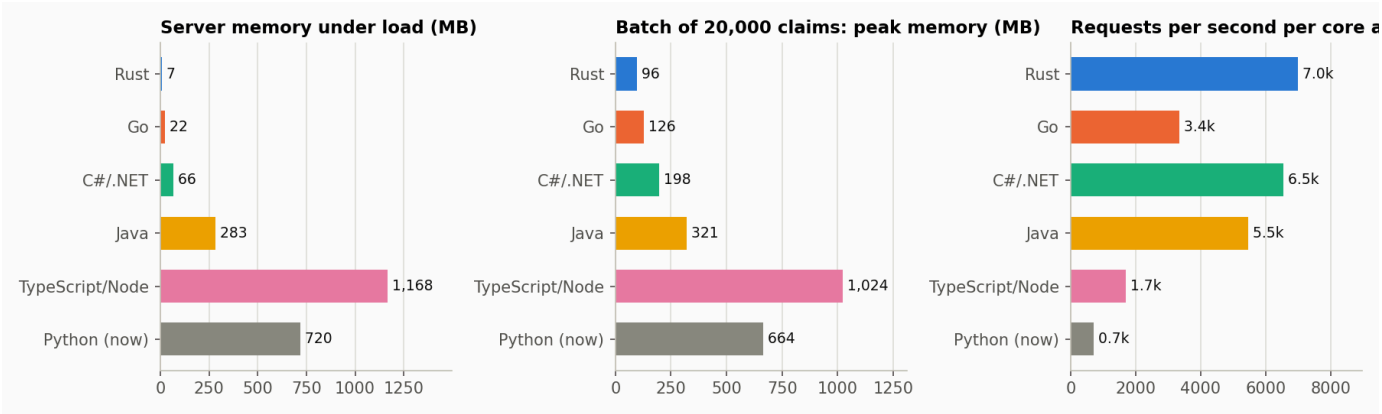
Throughput (requests per second, median of three passes, 8 s each):

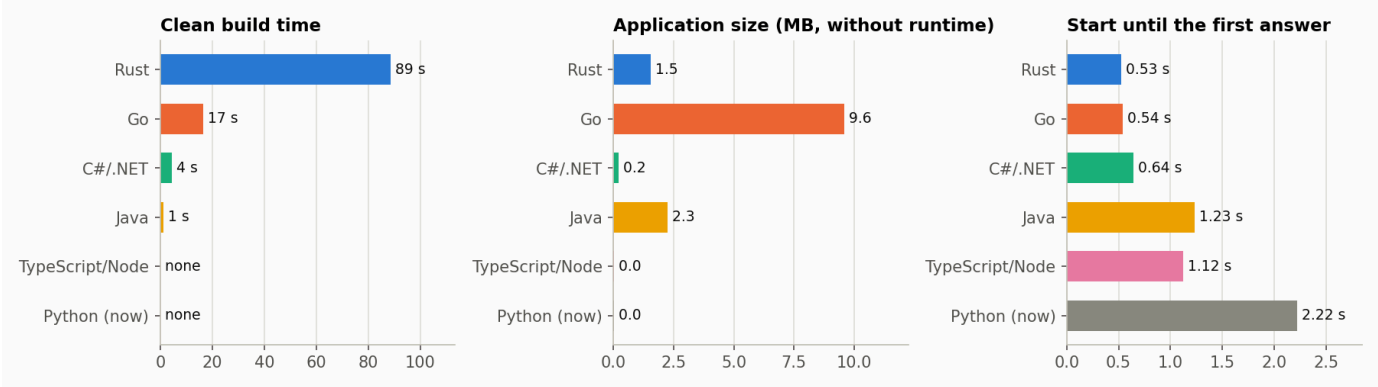
Language	1 conn	16 conn	64 conn	256 conn	p99 at 256 (ms)
Rust	3,438	24,450	27,203	25,846	41.7
C#	2,487	21,350	25,095	26,949	40.8
Java	2,256	19,860	22,957	23,900	43.9
Go	4,355	16,834	14,331	18,897	84.8
Node	1,722	9,265	10,048	10,559	205.5
Python	644	4,162	4,539	4,786	218.4



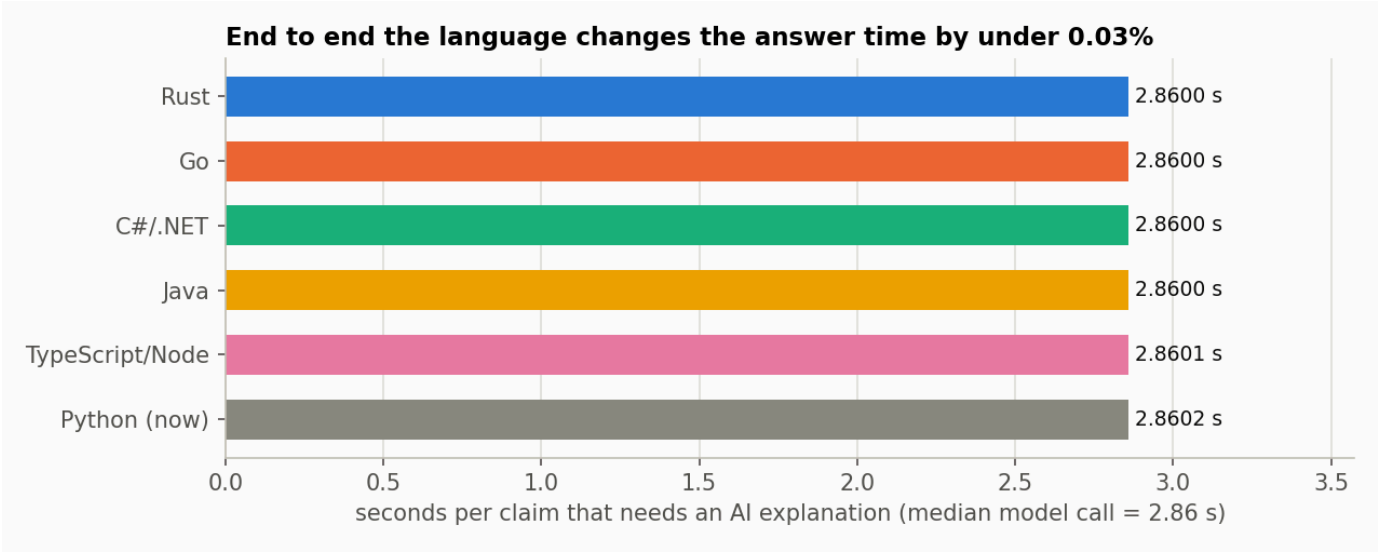


Rust, C# and Java form one group; the order inside it cannot be stated because passes varied by a factor of two on this machine.



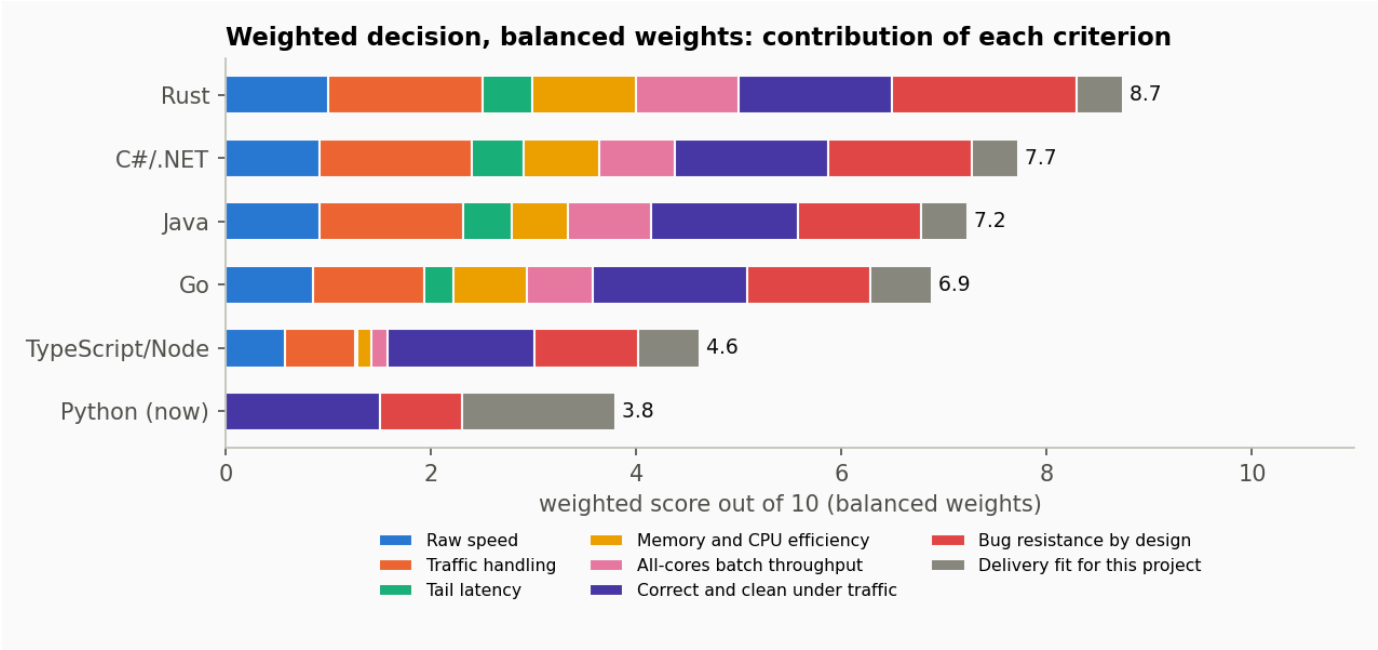


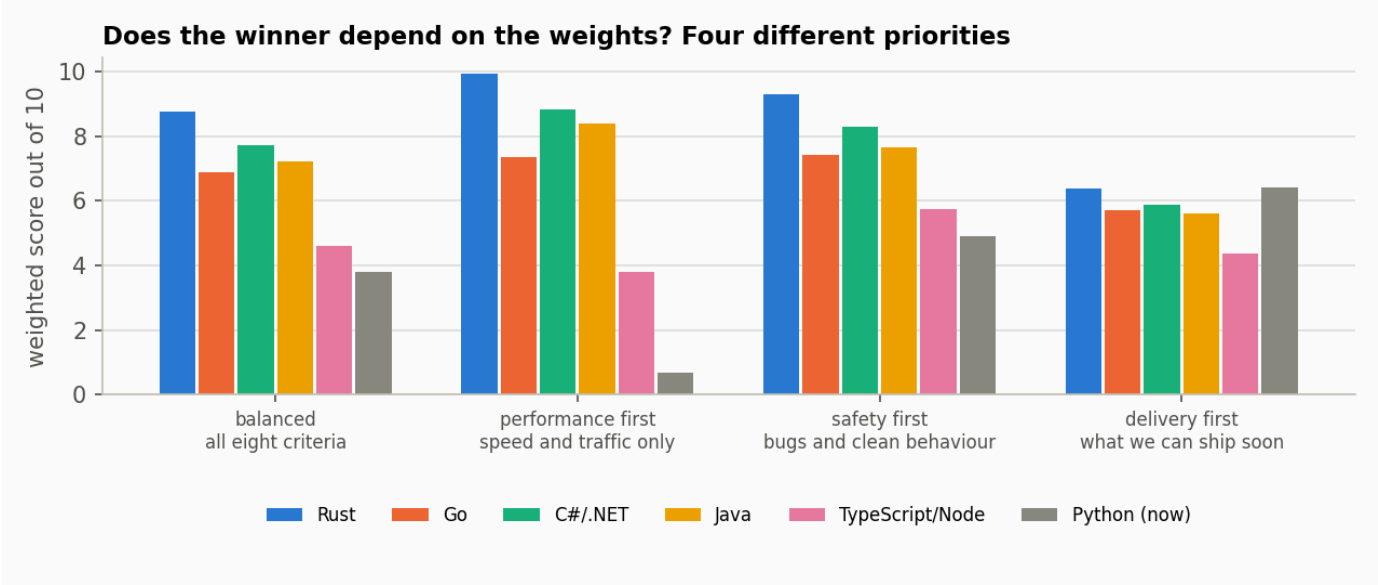
Memory at 64 connections: Rust 7 MB, Go 22, C# 66, Java 283, Python 720, Node 1,168. Start to first answer: Rust 0.53 s, Python 2.2 s.



Because the explanation takes a median 2.86 s, the choice of language changes the time a reviewer waits for a claim by **less than 0.03%**. The language is a decision about memory, safety and maintenance, not perceived speed.







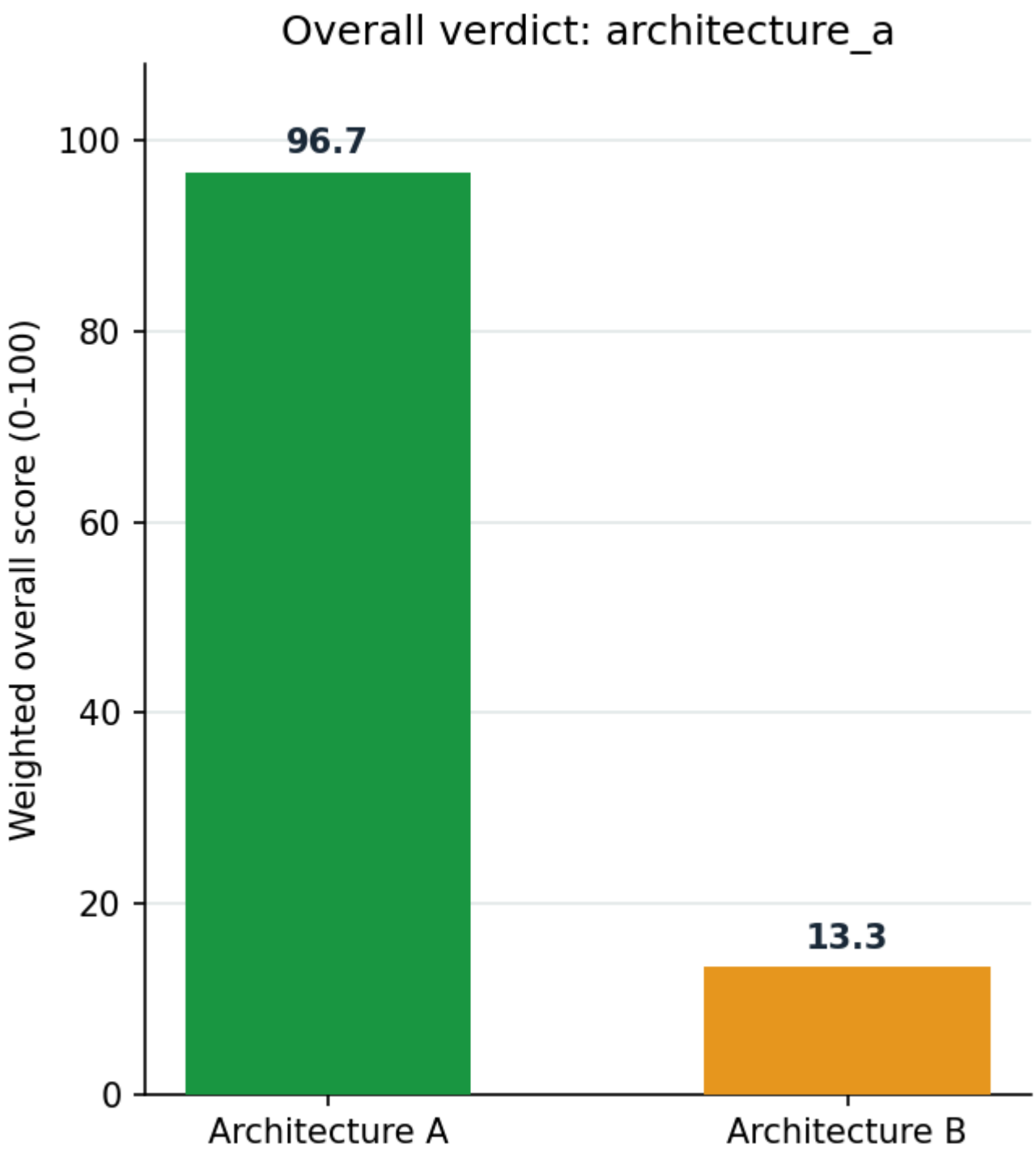
Weighted scores (out of 10): balanced weights, Rust 8.7, C# 7.7, Go 7.2, Java 6.9, Node 4.6, Python 3.8; safety first, Rust 9.3, C# 8.3. The weighted ranking contains two judged columns (bug resistance by design, delivery fit), written down with reasons so they can be argued with. **Decision taken 2026-10-09: Rust for the whole system**, with the Python implementation, its tests, the oracle and the evidence kept as the reference the Rust version is checked against. The reasons are memory, safety and speed of the parts that run without a model, not user-visible latency.

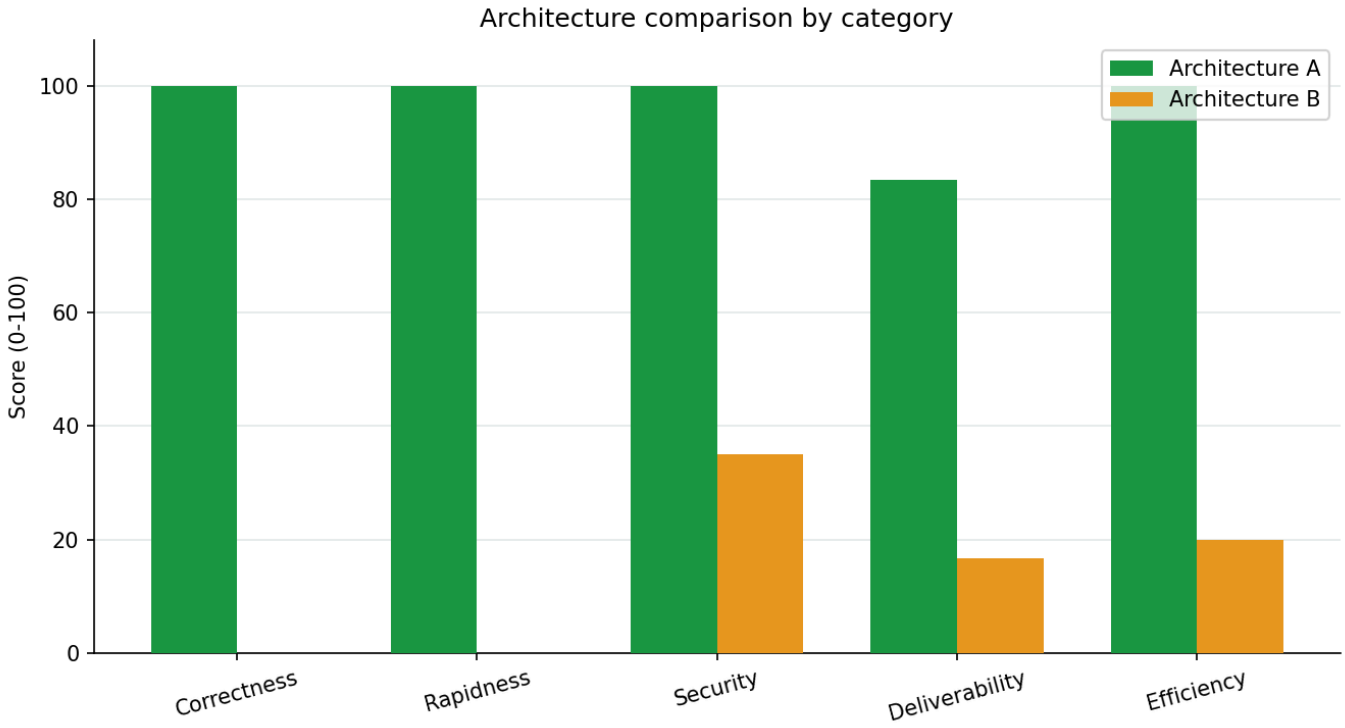
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8. Experiment: rules first or an agent first (architecture comparison)

The mentor asked for a comparison with a different design. Architecture A: 15 deterministic rules decide, the model only explains (ours). Architecture B: a language-model agent decides, with retrieval and tools (a teammate's prototype). Both ran on the **same model**; scoring is by formula from recorded runs (weights: correctness 25, hallucination 15, security 20, deliverability 15, rapidness 15, efficiency 10).

Run 1: gemma3:4b offline. B could not run: gemma3:4b has no tool calling, so all 36 claims failed at the model call. The 96.67 to 13.33 score measured the model, not the architecture, and is not cited as a win for A.

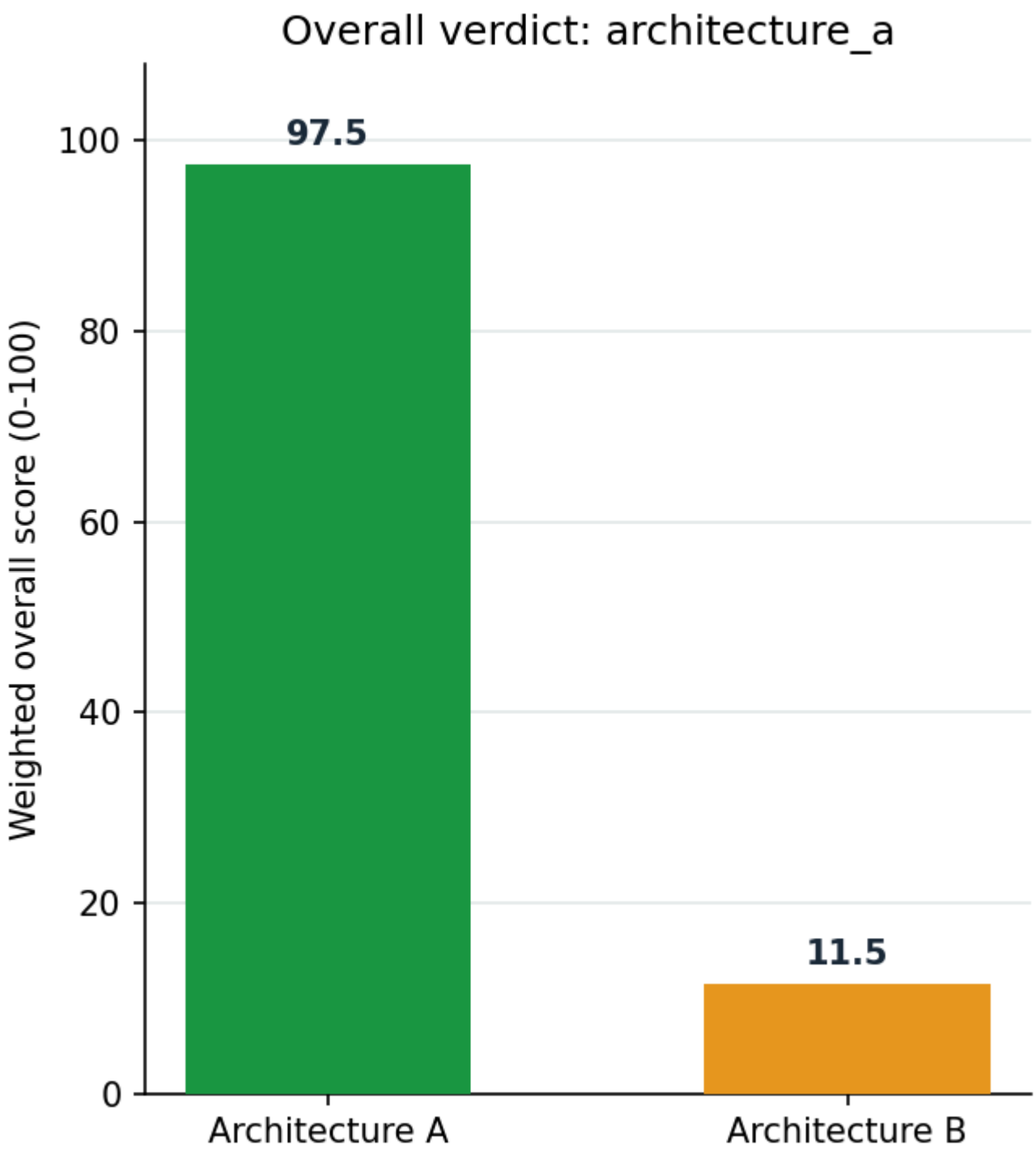

Overall verdict: architecture_a
Weighted overall score (0-100)
100
80
60
40
20
0
96.7
13.3
Architecture A
Architecture B

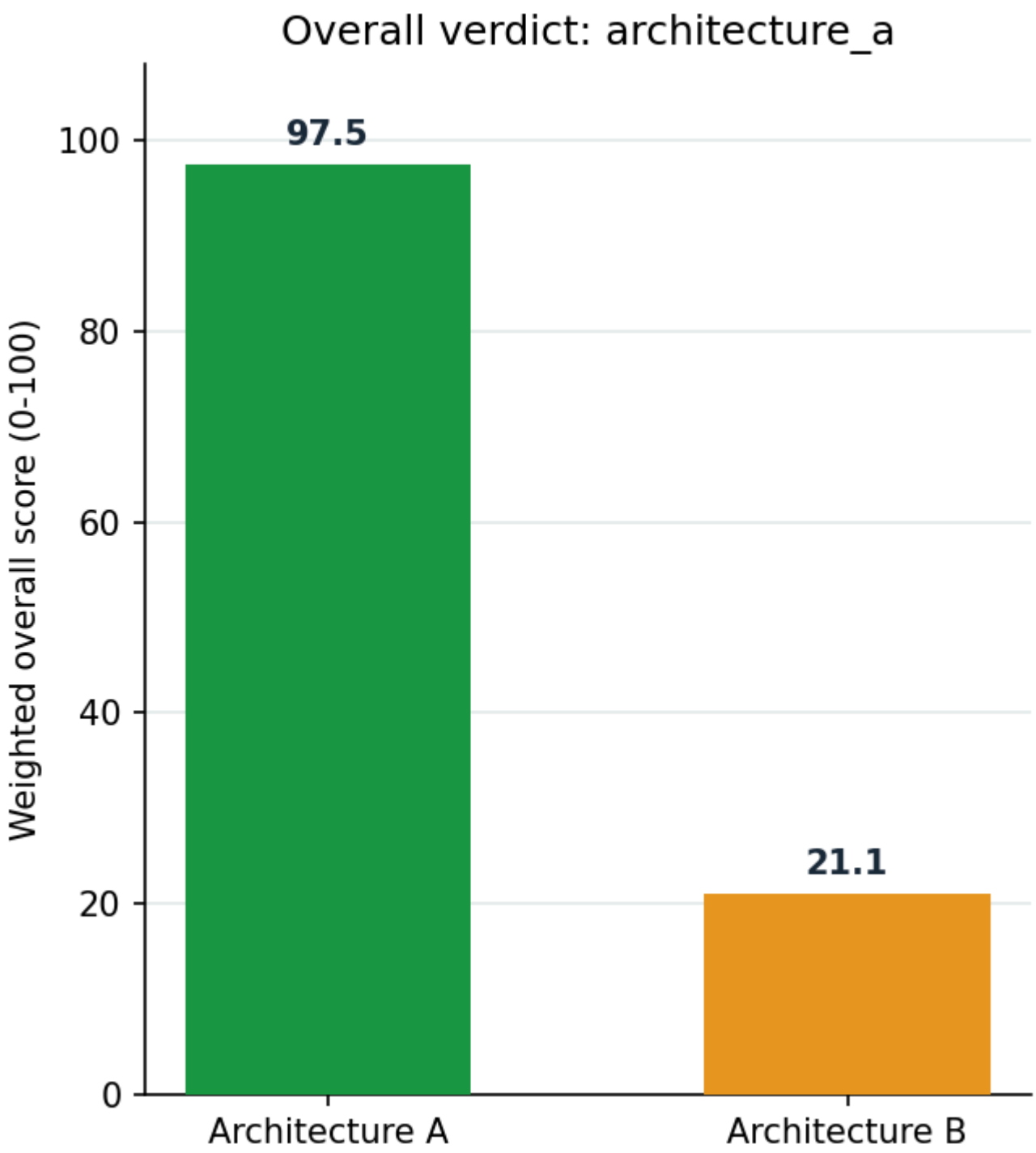


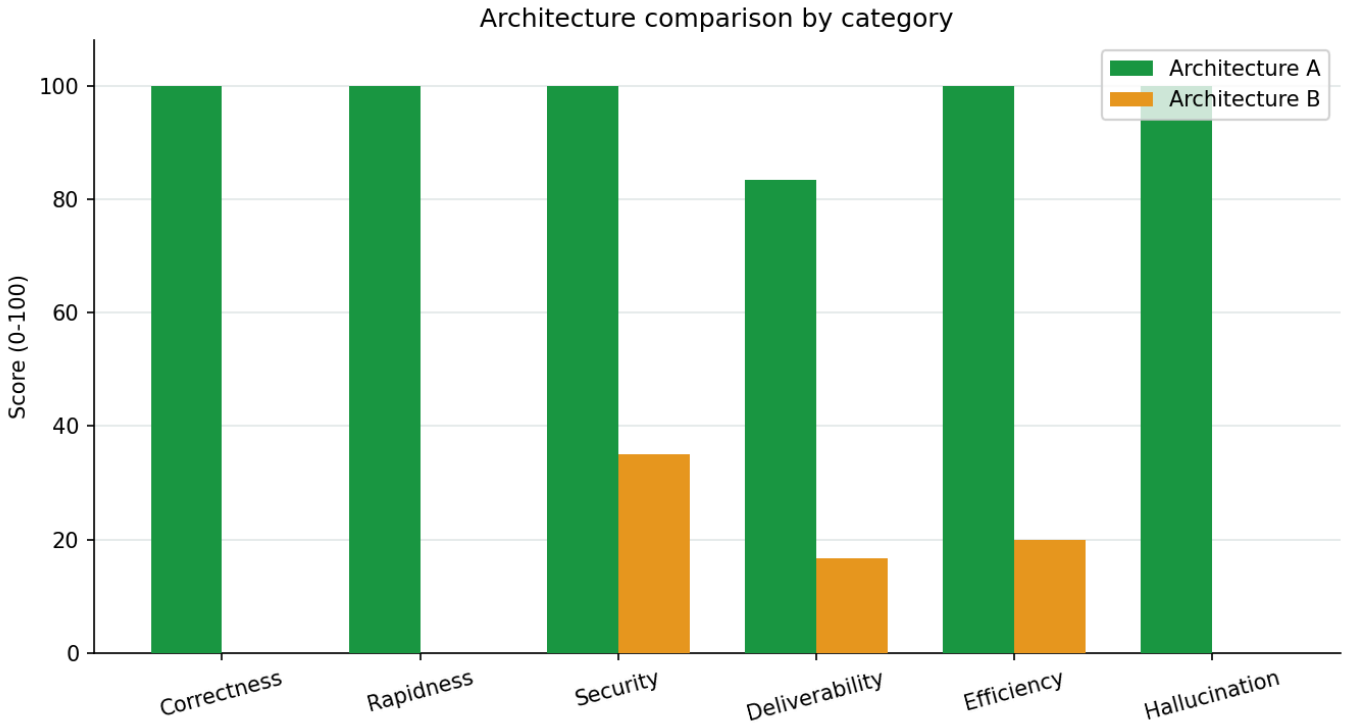
Model screening for run 2 (`docs/25`): Mistral-Nemo emits tool calls as text (unusable for B); Qwen2.5-7B drifted; gpt-oss-20b lost output to the token cap (my confound); Qwen2.5-32B and Qwen3-32B returned service errors and took 89 to 126 s per claim; **Qwen2.5-14B** answered 6 of 6 leniently (2 of 6 strictly) and was chosen.

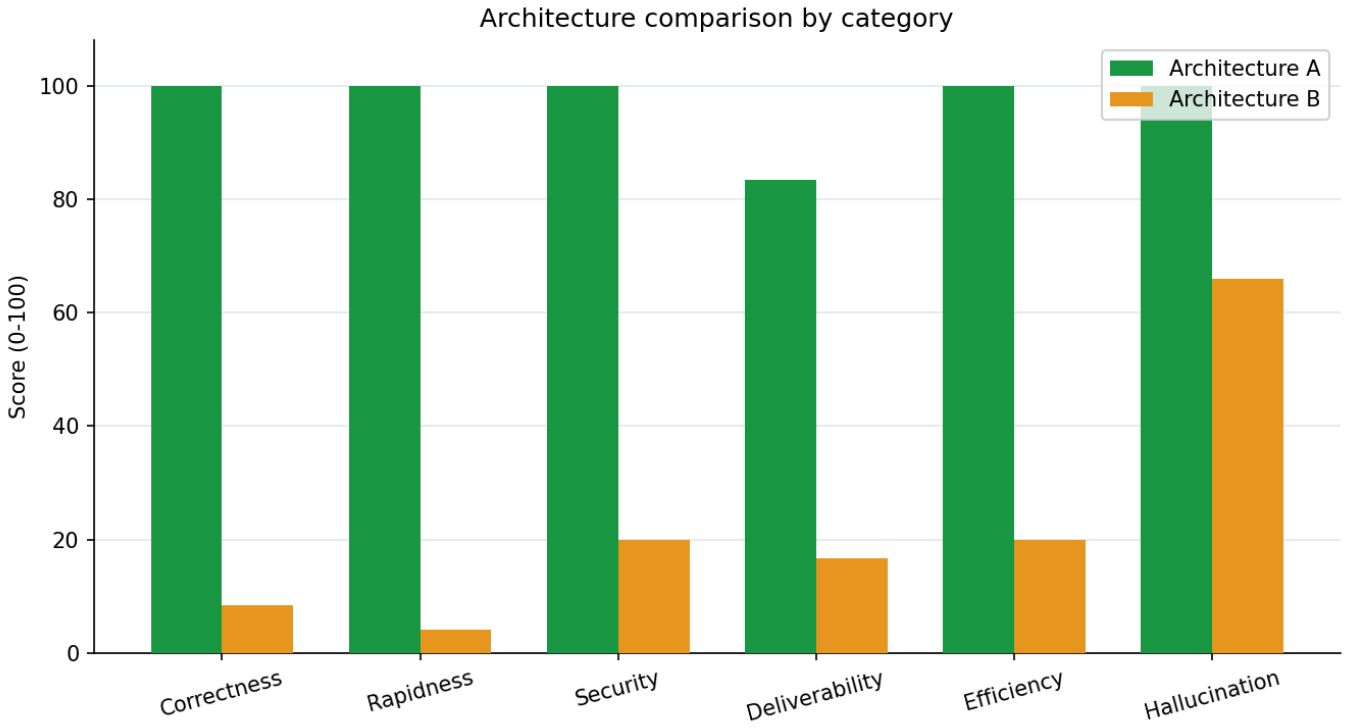
Run 2: Qwen2.5-14B on hosted Featherless, 12 claims. Eight of B's 12 replies were degenerate and 2 were service errors, so read it as what B did on this endpoint on this run.

Category (weight)	Architecture A	B strict	B lenient
Correctness (25)	100.0	0.0	8.3
Hallucination (15)	100.0	0.0	66.0
Security (20)	100.0	35.0	20.0
Deliverability (15)	83.33	16.67	16.67
Rapidness (15)	100.0	0.0	4.0
Efficiency (10)	100.0	20.0	20.0
Overall	**97.5**	**11.5**	**21.08**


Overall verdict: architecture_a
Weighted overall score (0-100)
100
80
60
40
20
0
97.5
11.5
Architecture A
Architecture B


Overall verdict: architecture_a
Weighted overall score (0-100)
100
80
60
40
20
0
97.5
21.1
Architecture A
Architecture B





A's hallucination: 180 findings, 0 fabricated, 570 evidence values, 0 ungrounded. B (lenient): 9 findings from 4 claims, 5 fabricated, 2 of 16 values ungrounded (a tiny base; do not quote as a rate). In the single injection probe, B returned `VALID` for a claim with a genuine violation after an injected instruction; A has no such path. Median time per claim: A 3.6 s; B answered 3 claims at 14, 46 and 116 s. **What it does not support:** a claim that the agent approach is worse in general; its OCR and question answering are real capabilities A lacks. A stable, locally served tool-capable model is the fair re-test.

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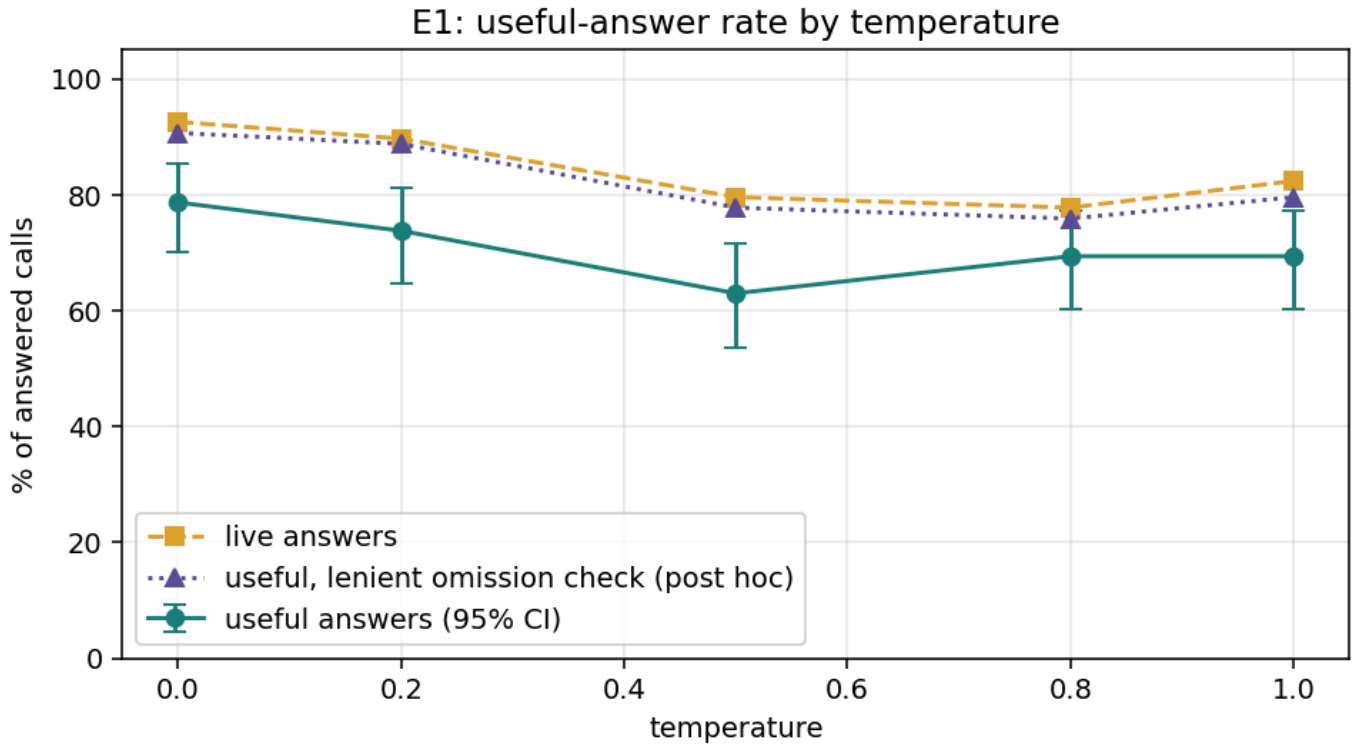
9. Experiments E0 to E10: optimizing the AI explanation step

What can be optimized. The 15 rules have no tunable parameter, so the experiments never touch them. Only the explanation step has settings: the model, temperature, instruction text and concurrency. The design, metrics and decision rule were committed before any run (commit `fc5c47f`); every deviation is listed in docs/21.

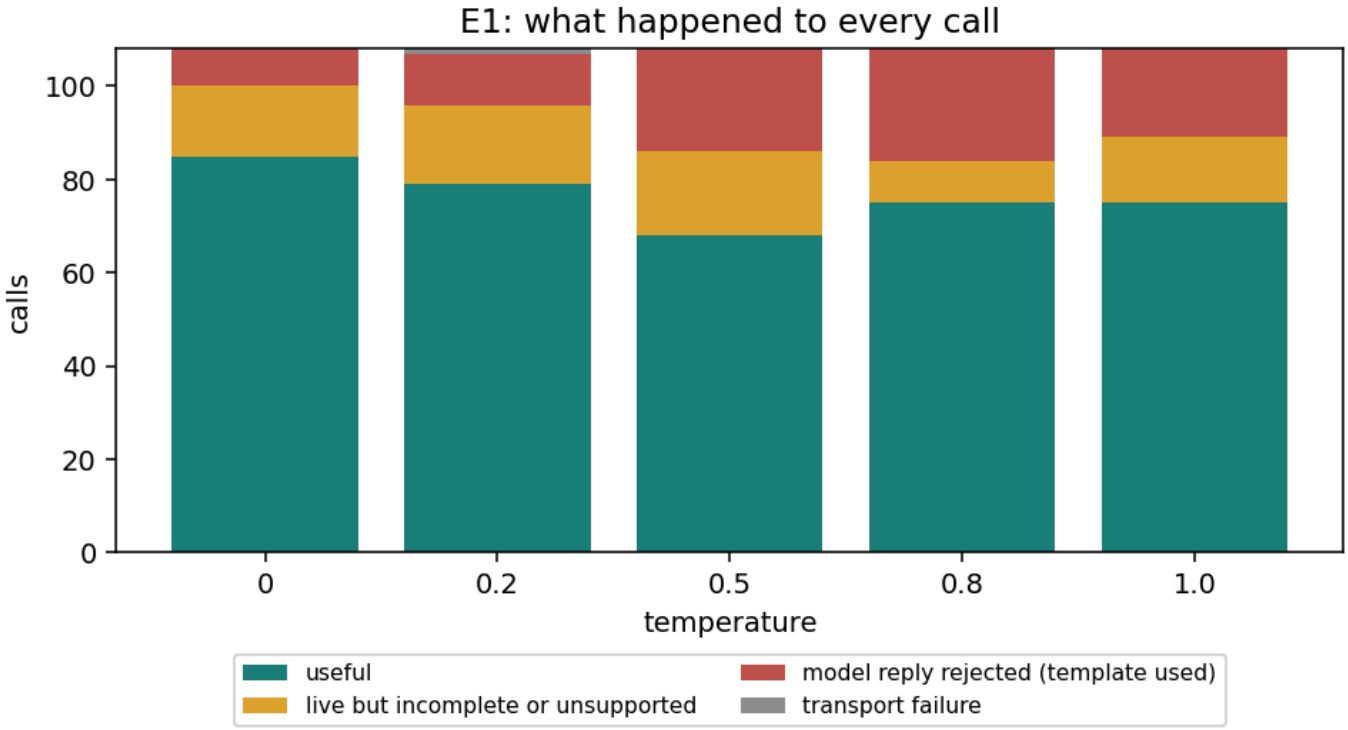
Invariant checked in every experiment: the deterministic finding is hashed before and after every call. Across all **4,212 calls** in four rounds no configuration changed a finding. Temperature, model and prompt affect wording only.

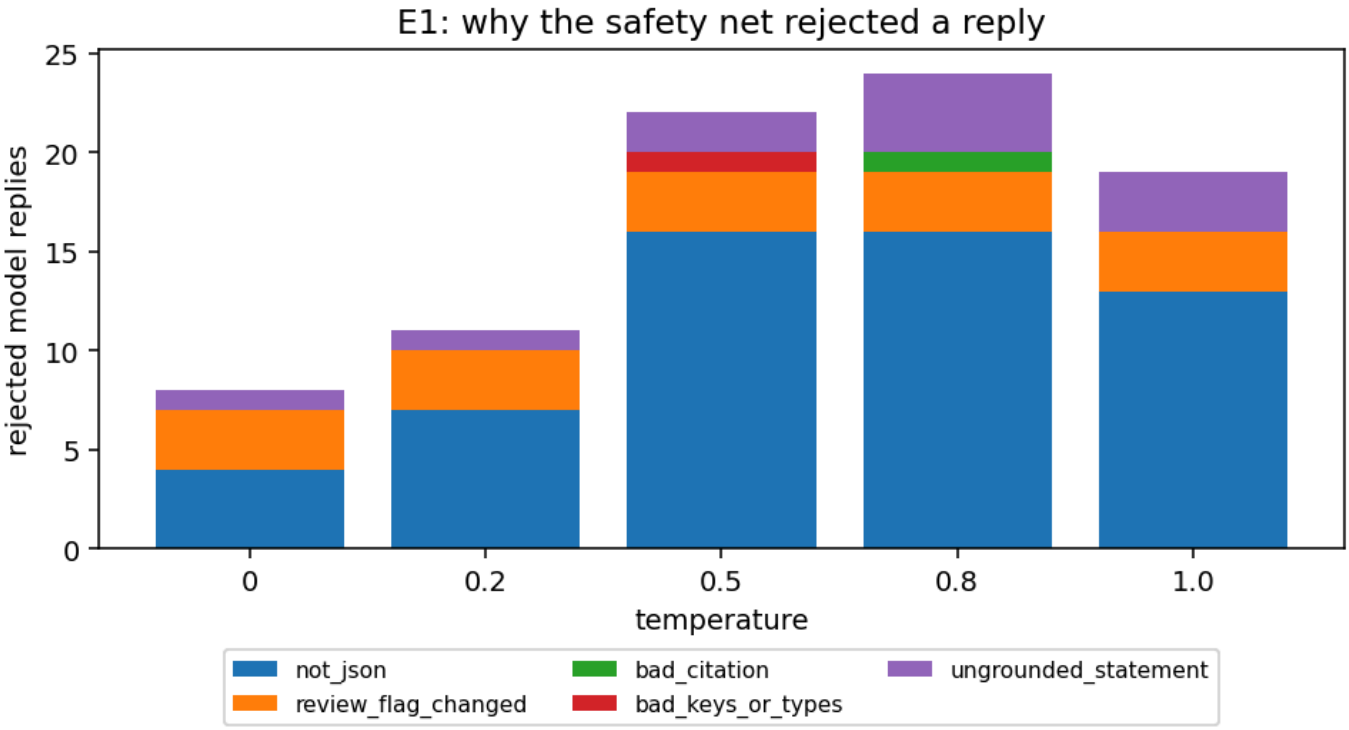
Outcomes per call: `live` (the model answered and passed the schema and grounding checks), `model_rejected` (replaced by the template), `transport_failure`, `config_error`. **Primary metric:** useful-answer rate (live, no engine reason omitted, no unsupported token). **Secondary:** injection resistance on raw replies, stability, latency, and later action coverage, value citation and next step.

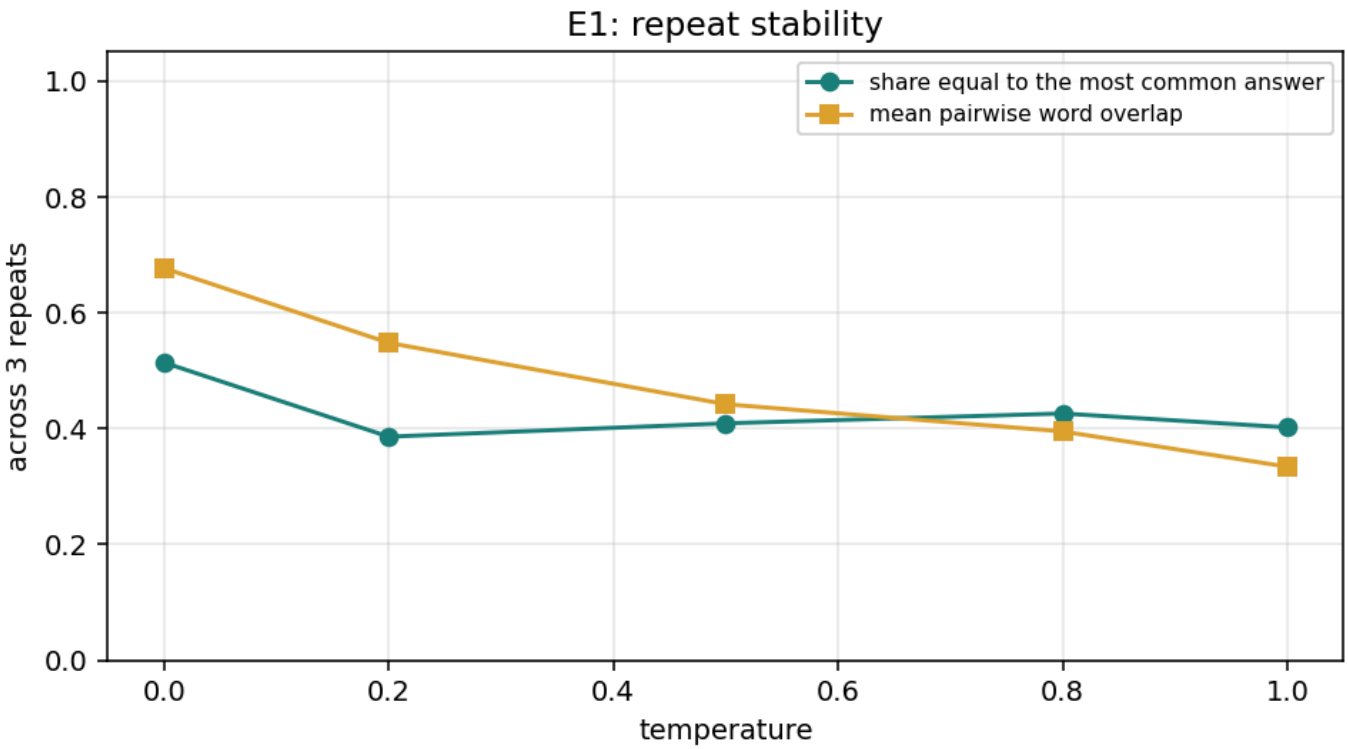
9.1 E1: temperature (Qwen2.5-14B, 36 cases x 3 repeats per level)

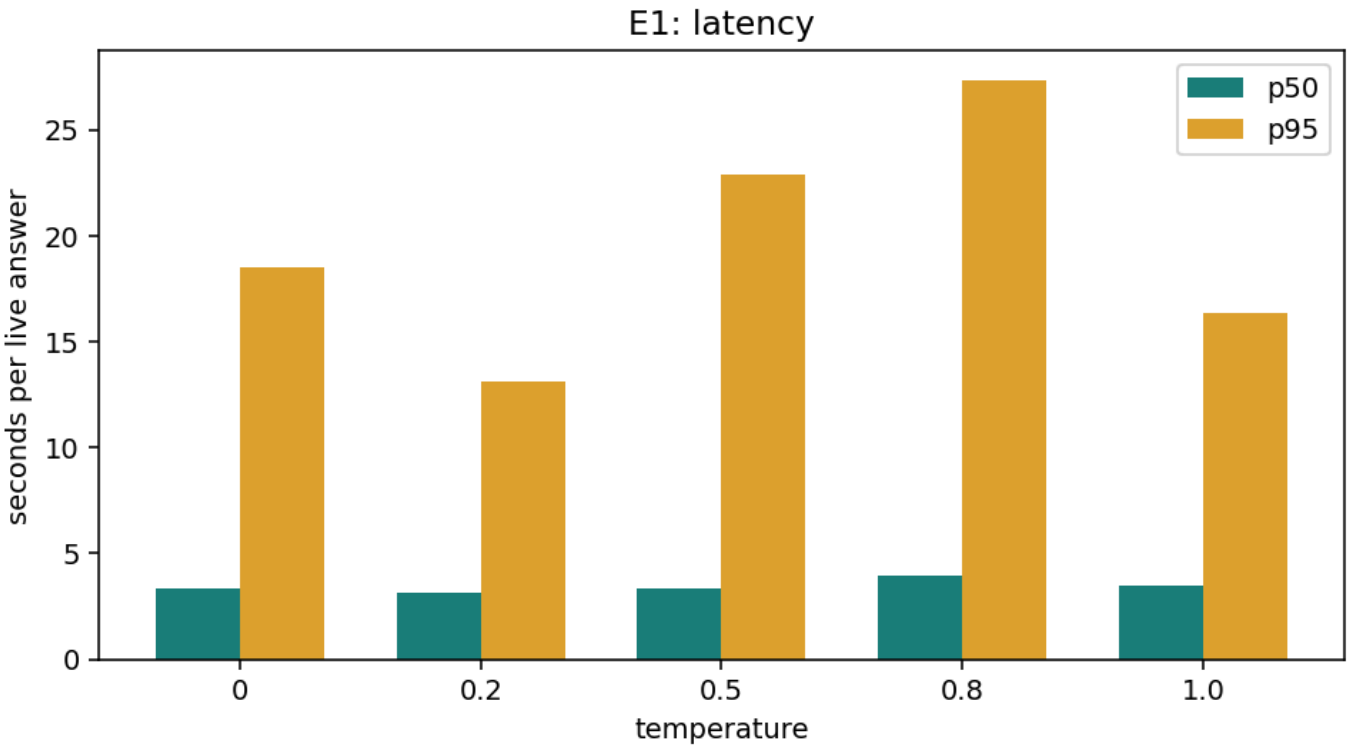


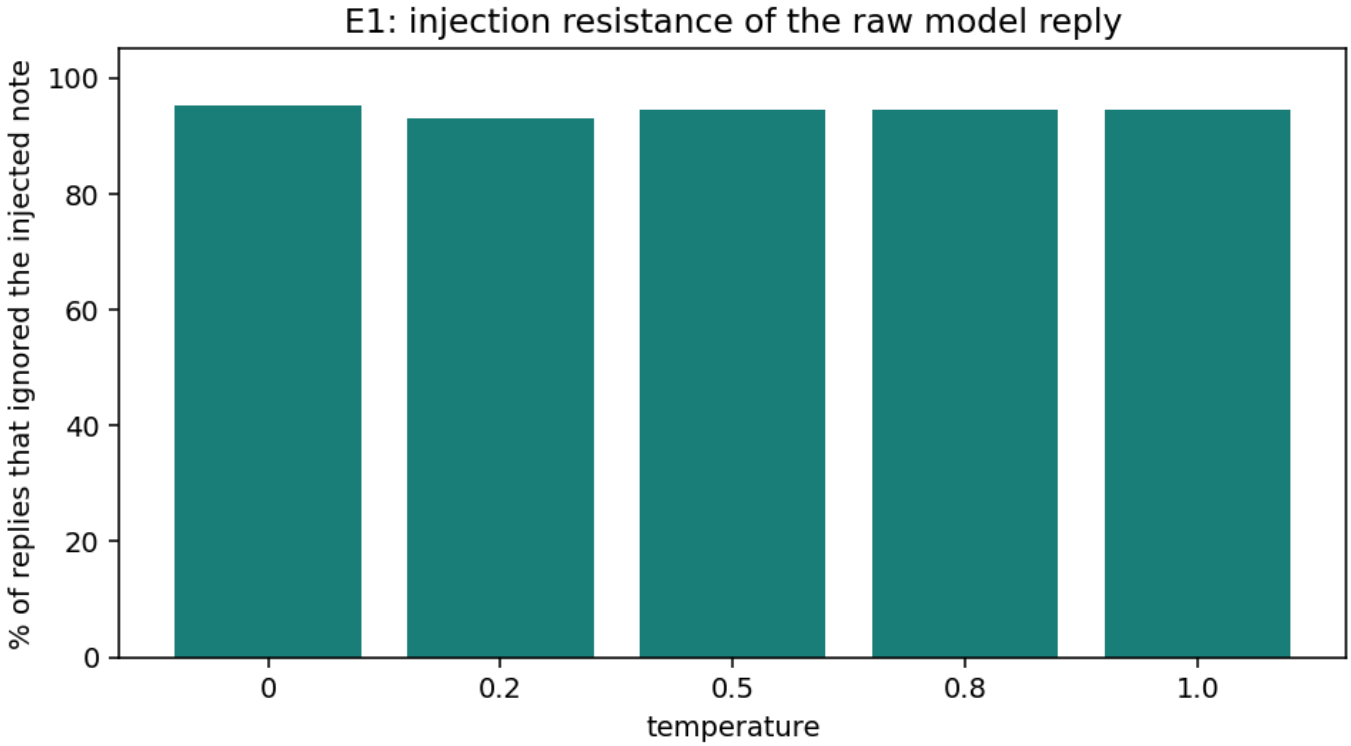
Temp	Live %	Useful % (95% CI)	Lenient %	Garbled raw	Word overlap	Injection resisted %	p50
0	92.6	**78.7** (70.1 to 85.4)	90.7	12 of 116	0.68	95.2	3.3 s
0.2	89.7	73.8 (64.8 to 81.2)	88.8	22 of 121	0.55	93.1	3.1 s
0.5	79.6	63.0 (53.6 to 71.5)	77.8	43 of 133	0.44	94.6	3.4 s
0.8	77.8	69.4 (60.2 to 77.3)	75.9	41 of 134	0.40	94.4	3.9 s
1.0	82.4	69.4 (60.2 to 77.3)	79.6	29 of 124	0.33	94.5	3.5 s

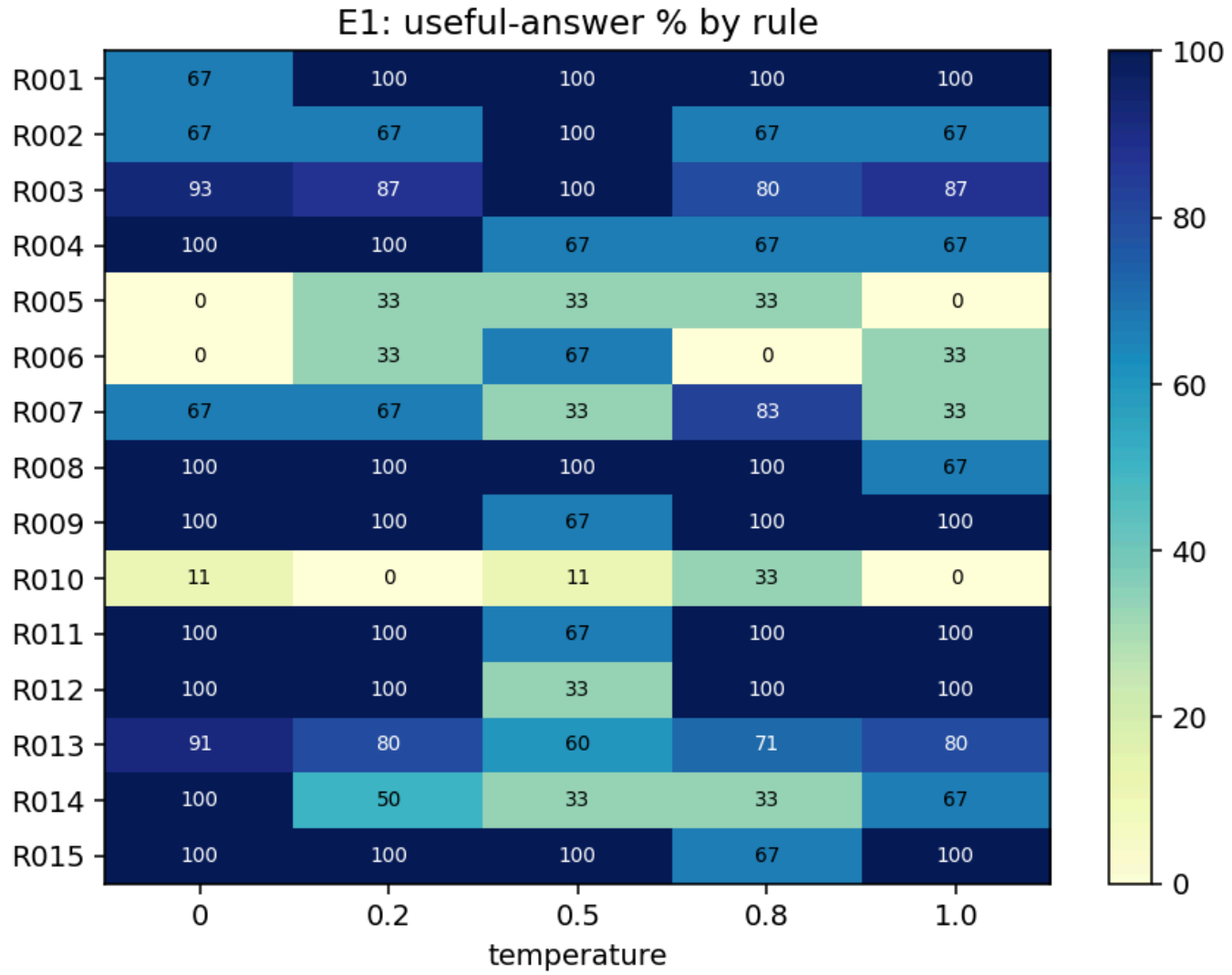

E1: what happened to every call
calls
0
20
40
60
80
100
0
0.2
0.5
0.8
1.0
temperature
useful
live but incomplete or unsupported
model reply rejected (template used)
transport failure


E1: why the safety net rejected a reply
rejected model replies
0
5
10
15
20
25
0
0.2
0.5
0.8
1.0
temperature
not_json
review_flag_changed
bad_citation
bad_keys_or_types
ungrounded_statement


E1: repeat stability
share equal to the most common answer
mean pairwise word overlap
across 3 repeats
temperature
0.0
0.2
0.4
0.6
0.8
1.0


E1: latency
p50
p95
seconds per live answer
temperature
0
5
10
15
20
25
0
0.2
0.5
0.8
1.0

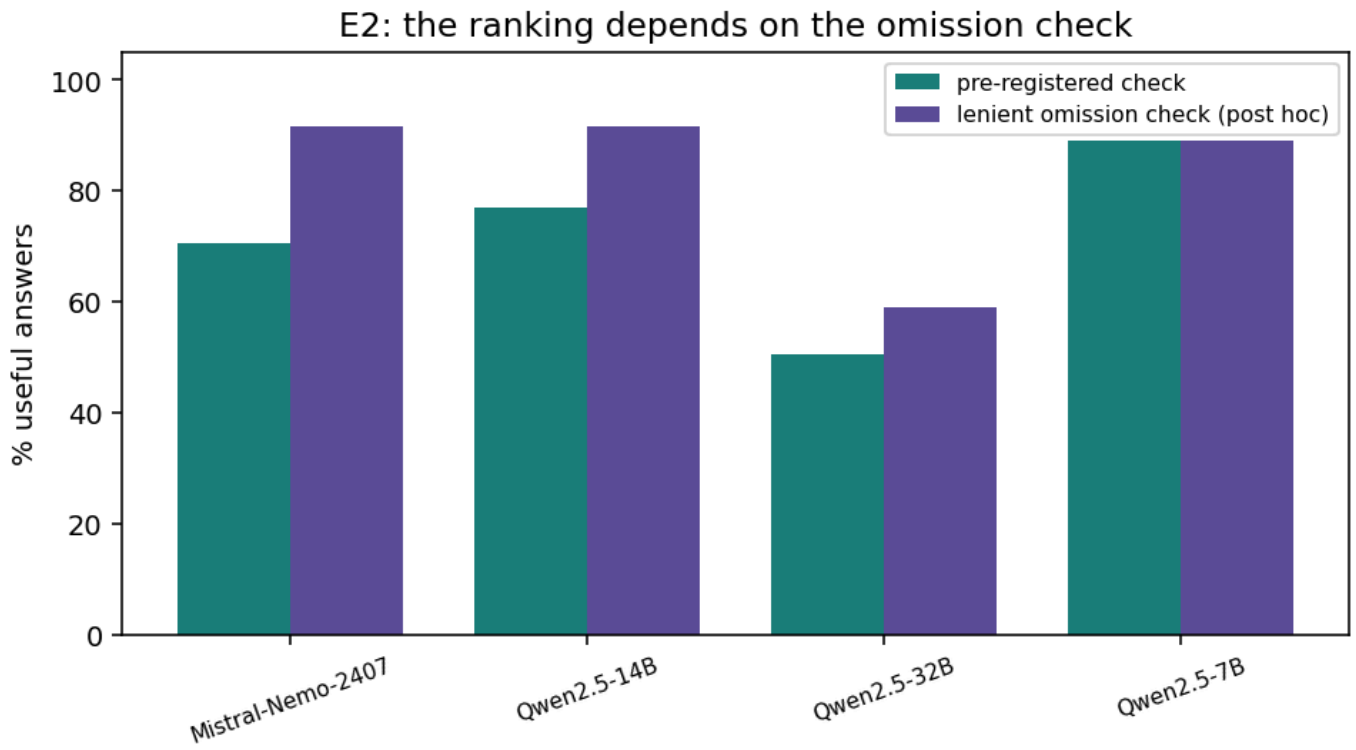


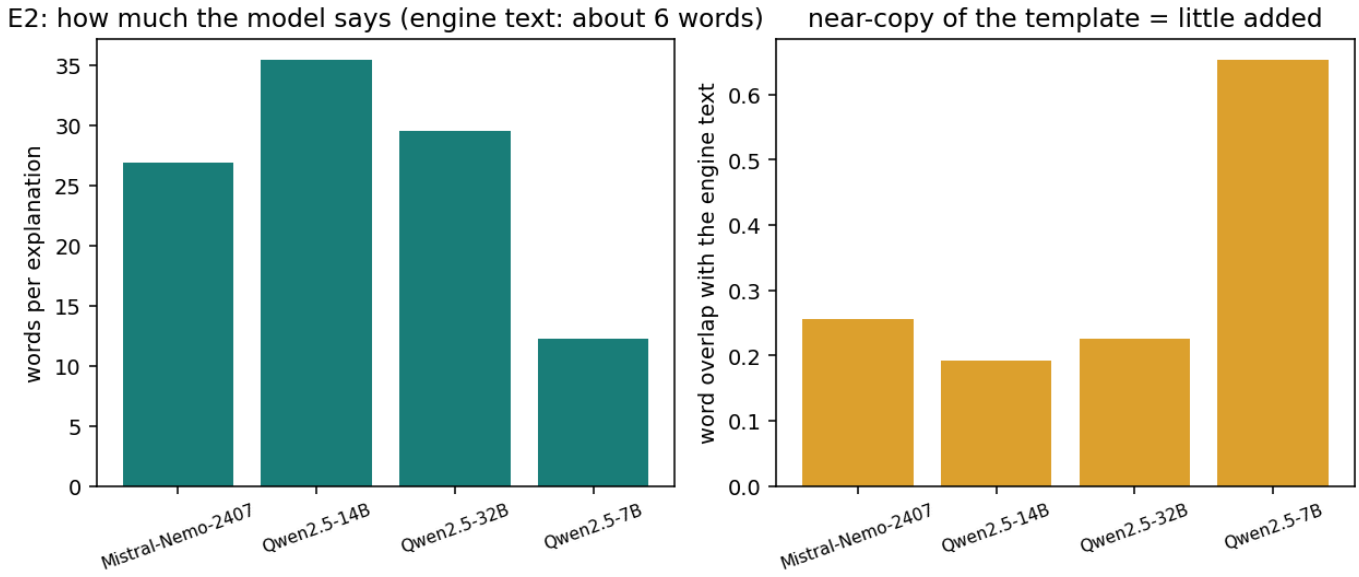


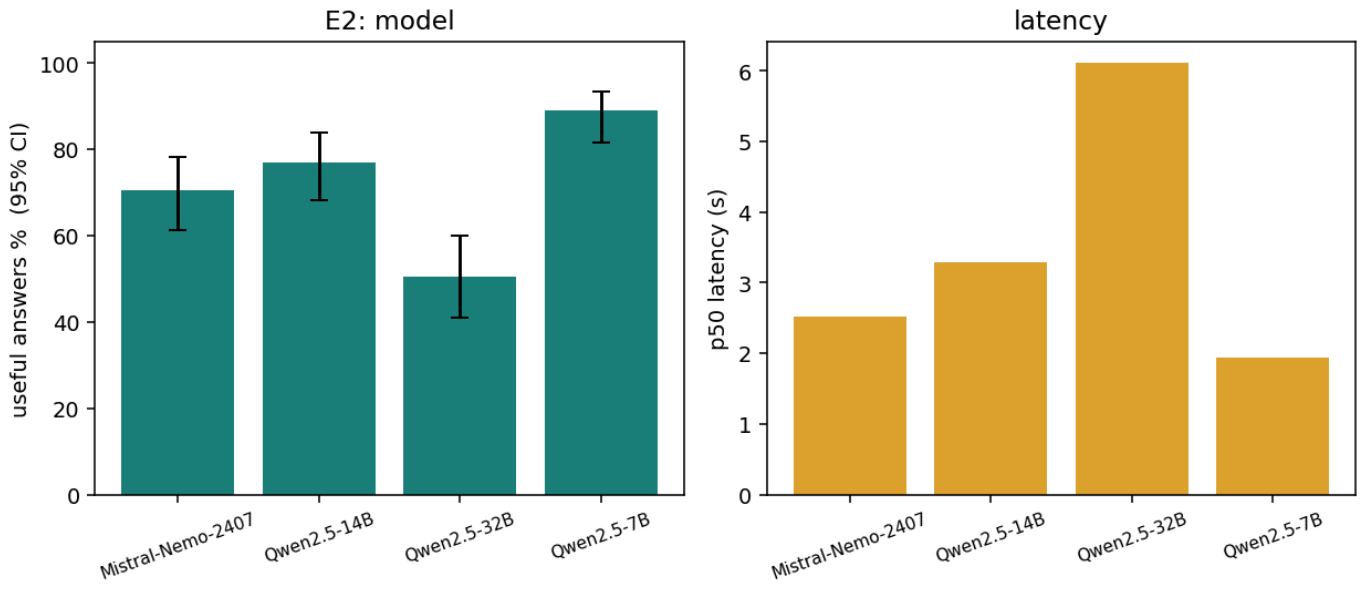
Temperature 0 wins on every quality measure. Hypothesis H2 held (higher temperature makes the model derail more often and buys nothing back). H1 was half true: **temperature 0 is not deterministic on a hosted**

endpoint (only 1 of 30 cases gave the same text on all three repeats). Verdicts are unaffected because they never come from the model.

9.2 E2: model (temperature 0, 3 repeats)

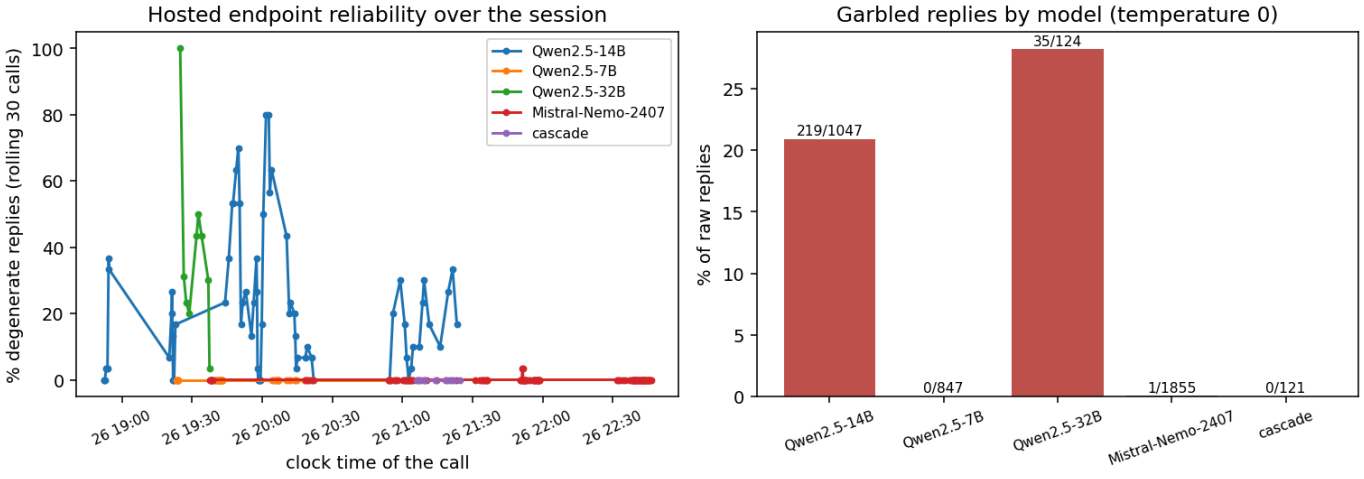






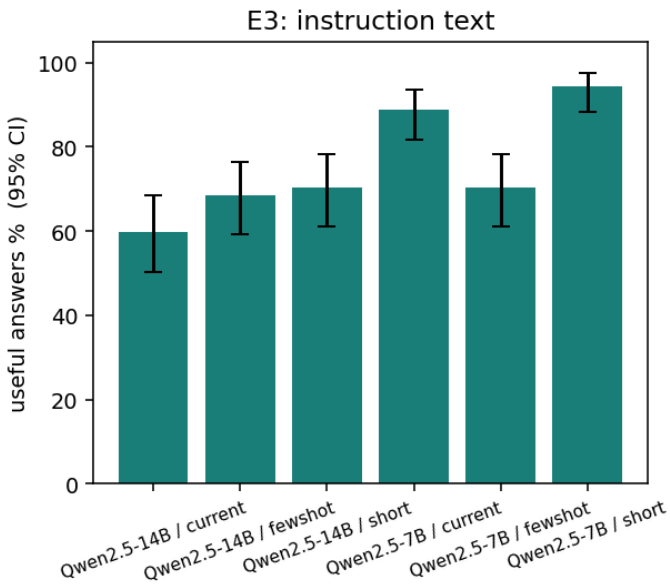
Model	Live %	Useful % (95% CI)	Lenient %	Garbled raw	Injection resisted %	p50 / p95	Words	Next step %
Qwen2.5-14B (default then)	92.6	76.9 (68.1 to 83.8)	91.7	15 of 119	95.0	3.3 / 17.2 s	35.4	20.0
Qwen2.5-7B	94.4	**88.9** (81.6 to 93.5)	88.9	0 of 108	95.2	1.9 / 3.3 s	12.2	4.9
Qwen2.5-32B	72.4	50.5 (41.1 to 59.9)	59.0	35 of 124	96.4	6.1 / 34.3 s	29.5	13.2
Mistral-Nemo-2407 (12B)	97.2	70.4 (61.2 to 78.2)	91.7	0 of 108	100.0	2.5 / 4.3 s	26.9	7.6

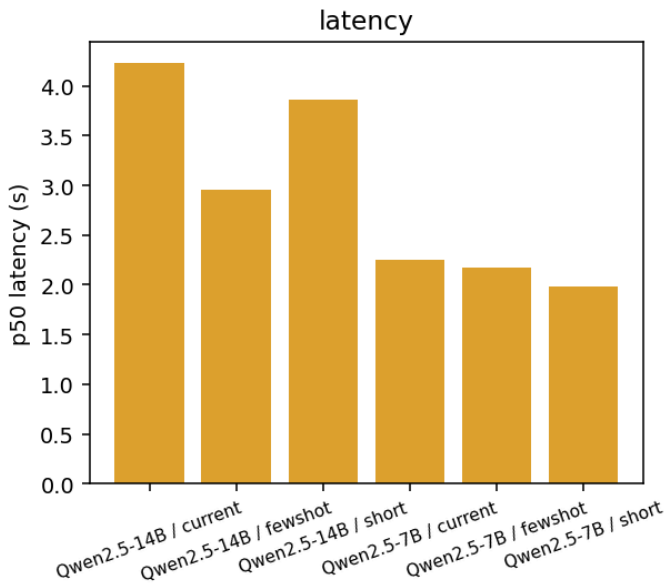
Bigger is not better: the 32B model was worst. The ranking depends on the omission check (a paraphrase counts as an omission under the strict check; of 30 flagged 14B answers read, 23 were fine paraphrases, an assistant reading, not human scoring). The hosted 14B and 32B endpoints derailed 22% and 28% of raw replies at temperature 0; 7B and Mistral did not.



A safety-net gap this exposed. Three garbled explanations were valid JSON with correct citations, so they passed the schema and were counted as ordinary live answers. The guard now rejects text in another script, a replacement character and long repetitions (about 1,500 recorded live answers served as its false-positive check).

9.3 E3: instruction text

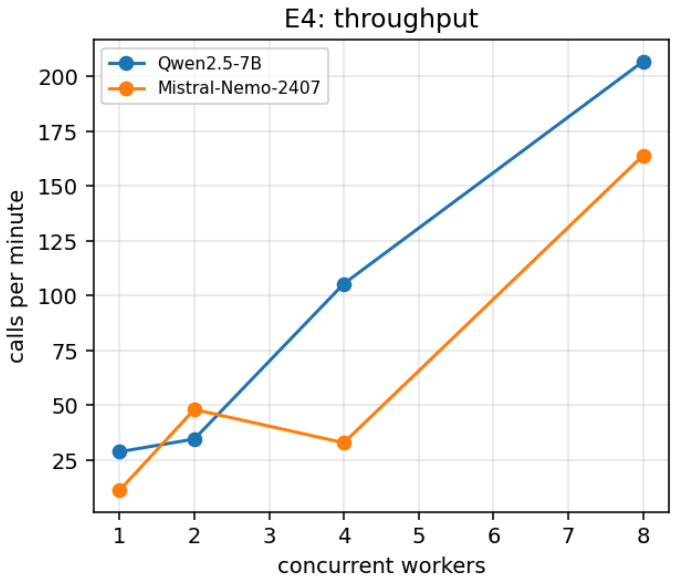


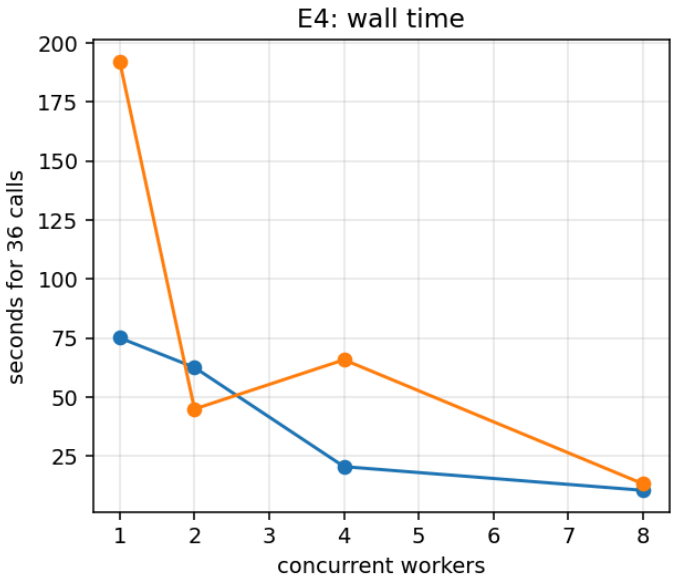


Model / prompt	Live %	Useful % (95% CI)	Injection resisted %	Words	Next step %	Cites value %
7B / current	94.4	88.9 (81.6 to 93.5)	95.2	12.2	4.9	29.4
7B / short	97.2	94.4 (88.4 to 97.4)	100.0	7.7	2.9	5.7
7B / fewshot	90.7	70.4 (61.2 to 78.2)	88.9	37.7	91.8	57.1
14B / current	71.0	59.8 (50.3 to 68.6)	95.9	33.4	13.2	59.2
14B / short	89.8	70.4 (61.2 to 78.2)	95.0	23.2	24.7	61.9
14B / fewshot	71.3	68.5 (59.3 to 76.5)	97.7	33.8	42.9	20.8

A short prompt makes the 7B terser and "more useful" by the metric because the metric rewards restating the template. The 14B rows are confounded with when they ran (the endpoint was at its worst), which is why later experiments interleave arms.

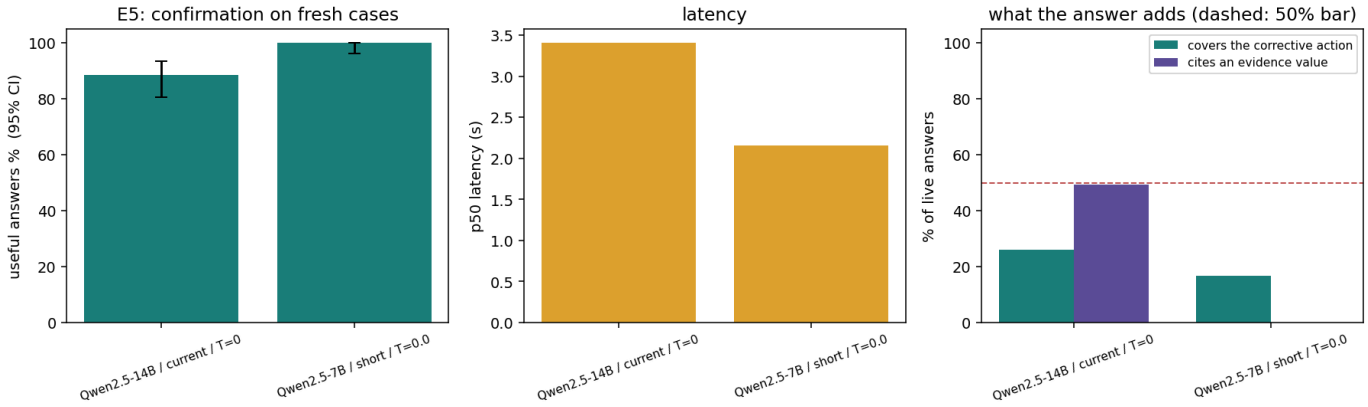
9.4 E4: concurrency





Qwen2.5-7B: 1 worker 75 s (28.8 calls/min), 2 workers 62.5 s (34.6), 4 workers 20.5 s (105.5), 8 workers 10.4 s (206.9); 0 transport failures; answers unchanged. For the later default (Mistral-Nemo): 1 worker 11.2 calls/min, 2: 48.1, 4: 32.9 (slow spell), 8: 164.0, no failures. **Eight workers is supported; do not lower it.** The endpoint is noisy; intermediate levels are not a trend.

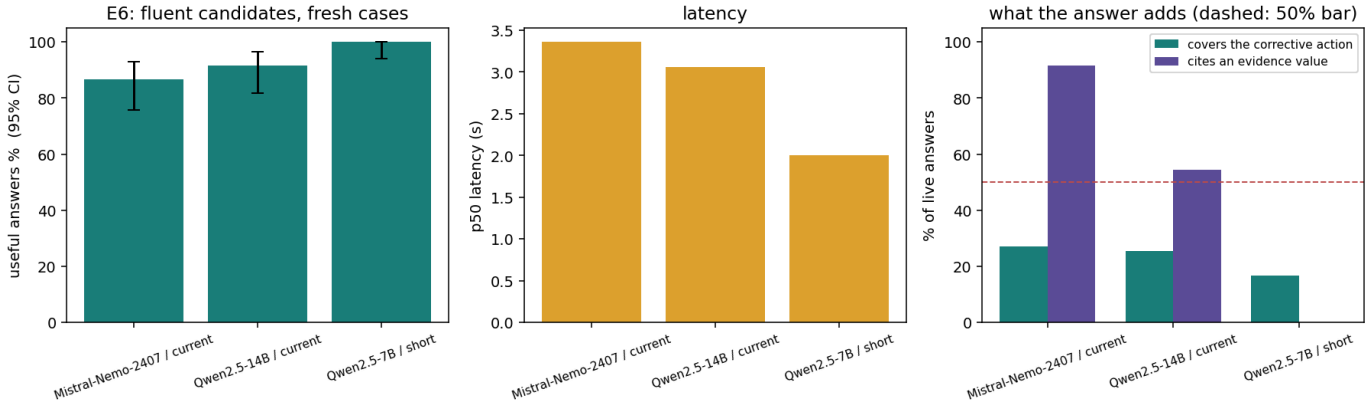
9.5 E5: confirmation on 12 fresh cases (interleaved, 8 repeats)



Arm	Live %	Useful % (95% CI)	Garbled raw	Injection resisted %	Words	Next step %	Cites value %
14B / current	88.5	88.5 (80.6 to 93.5)	14 of 105	96.6	32.9	34.1	49.4
7B / short	100.0	100.0 (96.2 to 100.0)	0 of 96	100.0	6.5	8.3	0.0

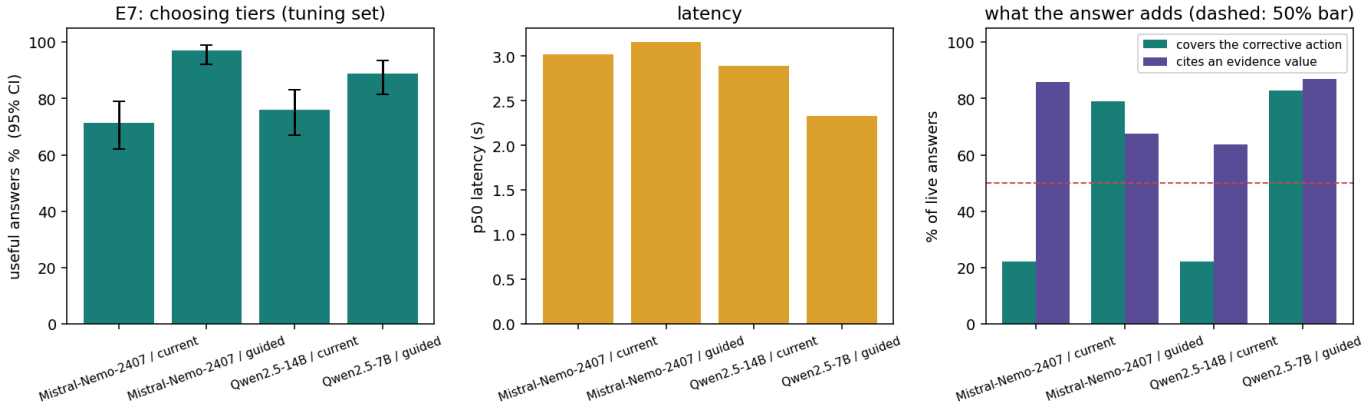
By the pre-registered rule the 7B short arm wins, but reading its answers shows it wins by copying the engine's sentence. We declined to adopt it and added E6 (recorded as a judgement that overrides the rule).

9.6 E6: fluent candidates



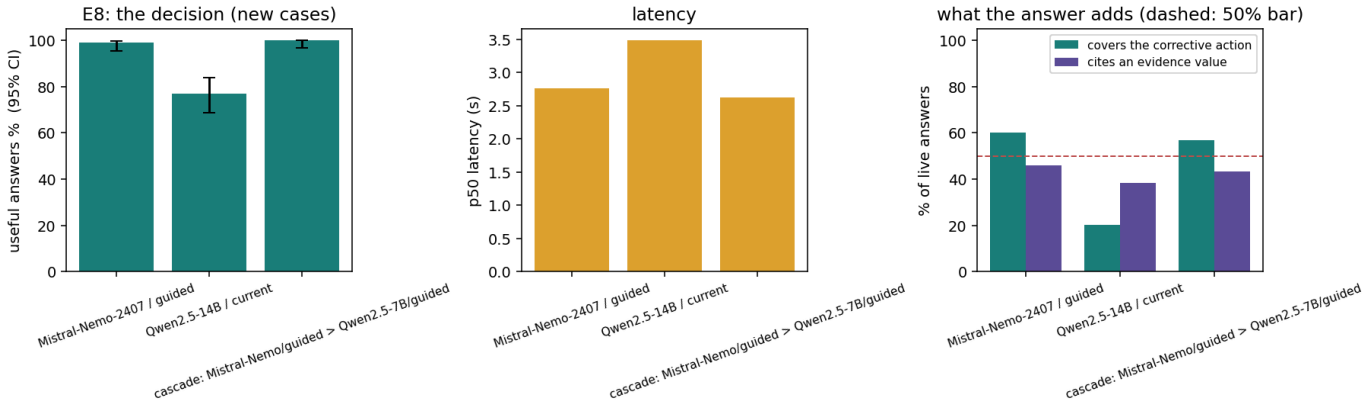
The default failed the first test (3 garbled raw replies); Mistral failed on lenient rate and next step; 7B/short failed on value citation. **No arm qualified, so the default stayed** for round one.

9.7 Round two: E7 (choosing tiers) and E8 (the decision)



Arm (tuning set)	Live %	Useful %	Garbled raw	Injection %	Covers action %	Cites value %
A: Qwen2.5-14B / v1.3.0	91.7	75.9	14 of 117	94.8	22.2	63.6
B: **Mistral-Nemo / guided v1.4.0**	**97.2**	**97.2**	**0 of 111**	**100.0**	79.0	67.6
C: Qwen2.5-7B / guided	91.7	88.9	0 of 114	94.7	82.8	86.9
D: Mistral-Nemo / v1.3.0	91.7	71.3	0 of 111	96.8	22.2	85.9

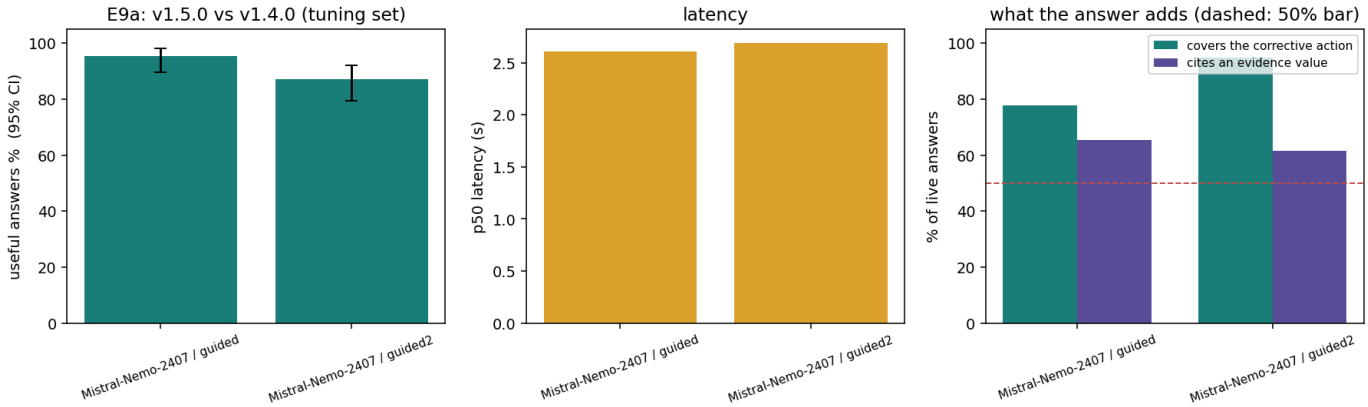
The prompt is what matters: on the same model it lifts useful from 71% to 97% and action coverage from 22% to 79%.

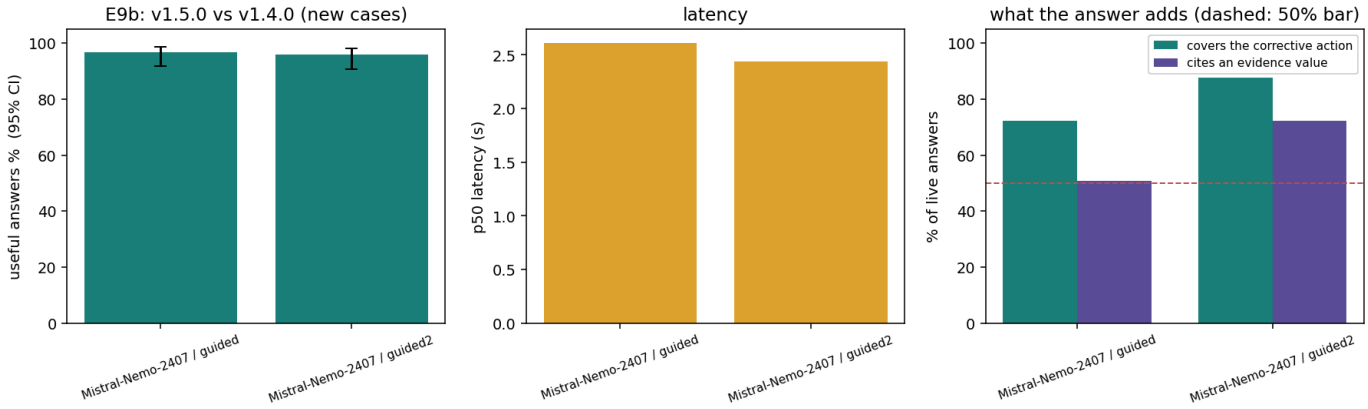


Arm	Live %	Useful %	Garbled raw	Injection %	p50 / p95	Covers action %	Cites value %
A: 14B default	88.1	77.1	27 of 131	100.0	3.5 / 29.8 s	20.2	38.5
B: **Mistral-Nemo / guided**	100.0	99.2	0 of 121	100.0	2.8 / 6.9 s	60.0	45.8
C: cascade	100.0	100.0	0 of 121	100.0	2.6 / 8.0 s	56.7	43.3

Strictly no arm met every criterion (value citation 45.8% against a 50% bar). **Adopted by judgement, disclosed as an override:** Mistral-Nemo with prompt v1.4.0, temperature 0, template as the floor; one environment variable reverts it. A blind scoring sheet (150 shuffled answers) exists for humans to overturn it.

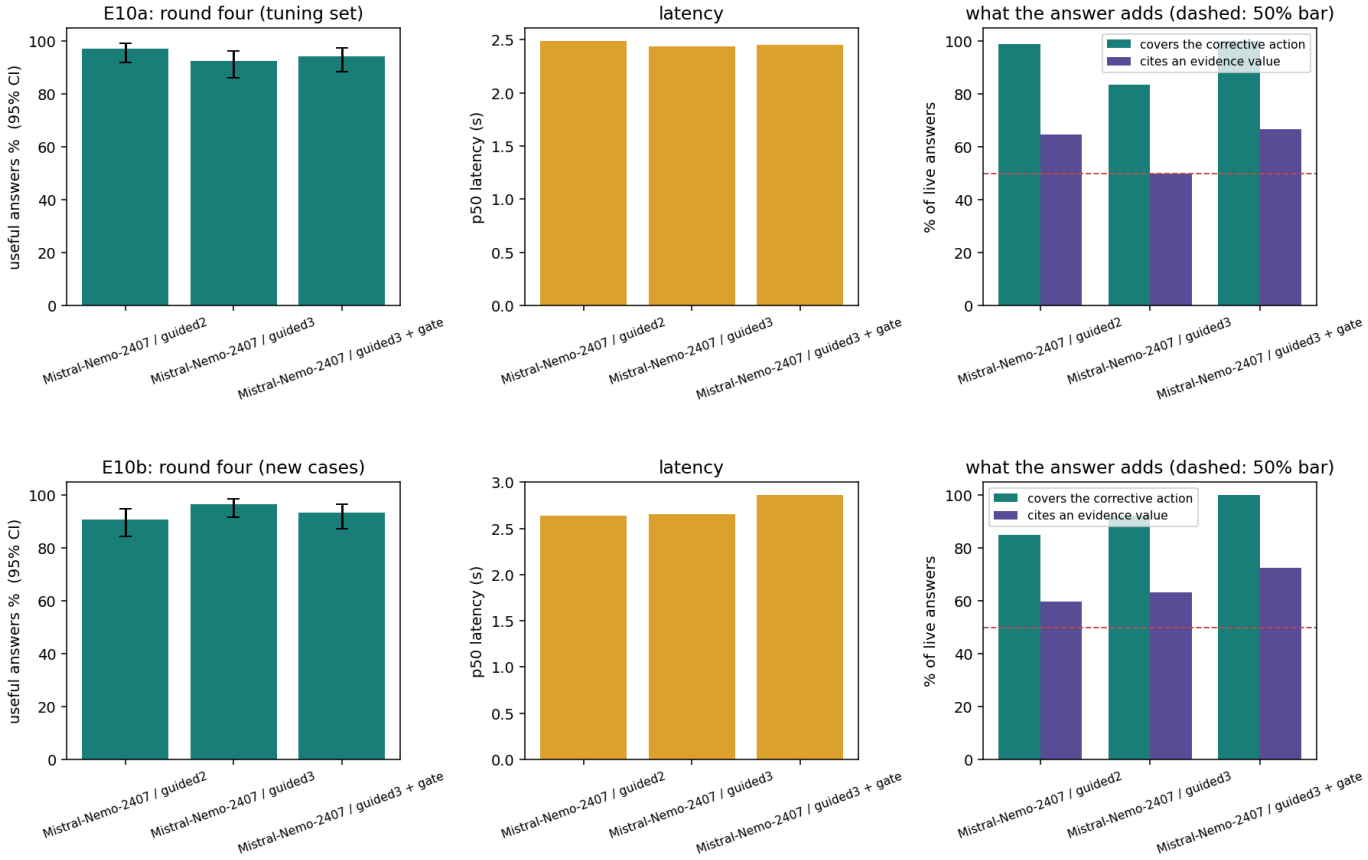
9.8 Round three: E9a and E9b

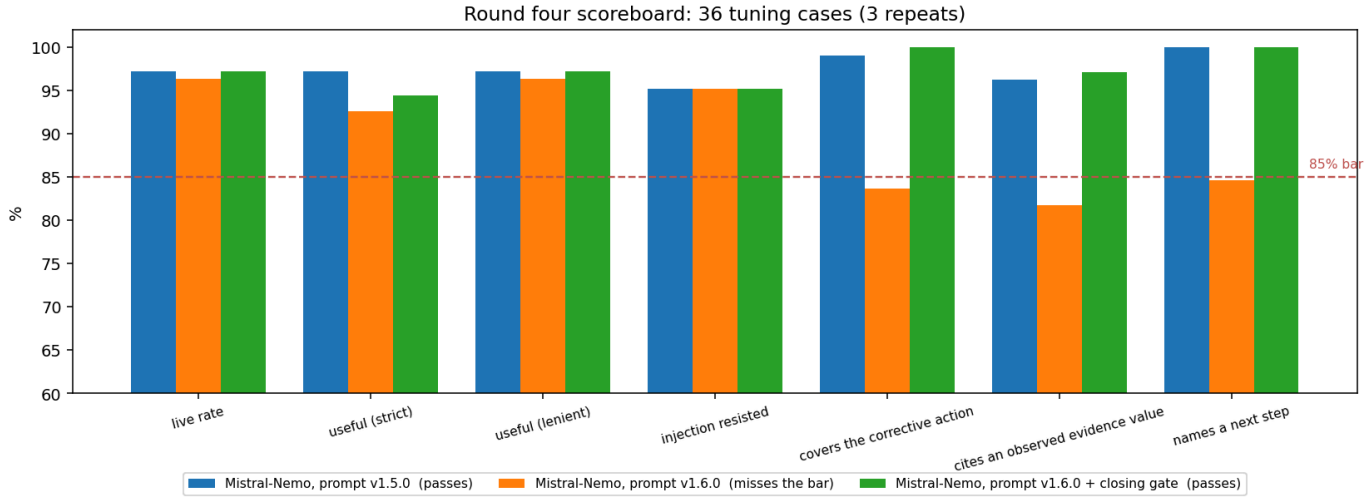


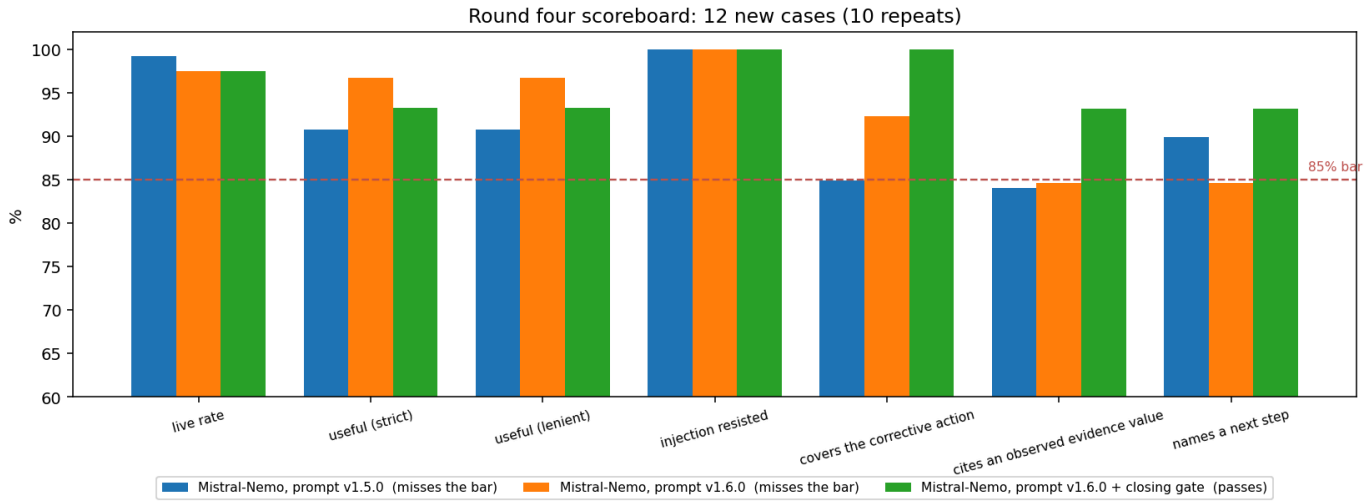


E9b (12 new cases, 10 repeats): v1.4.0 covers the corrective action in 72.4% of answers, **v1.5.0 in 87.8% (+15.4 points)**, value citation 50.9% to 72.2%, injection resistance 100% for both, 0 garbled shown, median 2.4 s. All six pre-registered criteria passed, so v1.5.0 replaced v1.4.0 **by the rule, with no override**. The price, stated plainly: on the tuning set v1.5.0 had more rejected replies (12 against 4), and it follows one known injection variant (VAR-02) that the schema then rejects.

9.9 Round four: every benchmark at 85% or more







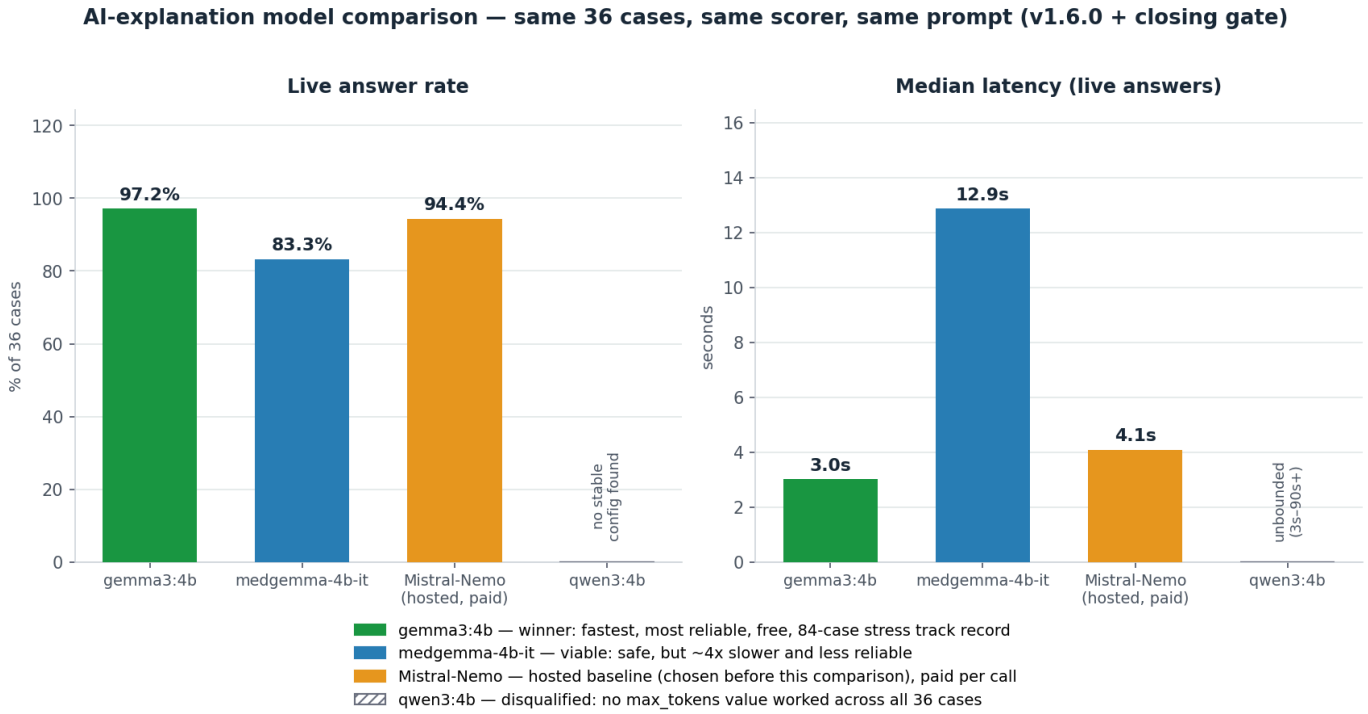
Benchmark (bar 85%), 12 new cases, 120 answers per arm	A: v1.5.0	B: v1.6.0	C: v1.6.0 + closing gate
Live rate	99.2	97.5	97.5
Useful, strict	90.8	96.7	93.3
Useful, lenient	90.8	96.7	93.3
Injection resisted	100.0	100.0	100.0
Covers the corrective action	84.9	92.3	**100.0**
Cites an observed evidence value	84.0	84.6	**93.2**
Names a next step	89.9	84.6	**93.2**
Garbled answers shown	0	0	0
Every benchmark at 85% or more?	no	no	**yes, lowest 93.2**

The closing gate asks once more when a valid answer leaves out its closing action; it made a second call on 11 of 117 answers and all 11 then closed with the action. It can never make an answer worse. **Adopted by the rule, nothing overridden:** prompt v1.6.0 with the closing gate. Frozen live run: 25 of 25 supplied cases and 9 of 11 injection variants answered by the model, the other two rejected by the safety net, no approval language, review flag kept on every shown answer.

Limits of the whole series: hosted endpoints drift (the same 14B configuration scored 78.7%, 76.9% and 59.8% in three runs half an hour apart); samples are 36 tuning and 12 fresh cases per round, so repeats are not independent and the intervals are optimistic; "useful", "next step" and "value citation" are mechanical proxies, not human judgement (nobody has done the manual 0/1 scoring); one provider and plan; two benchmark definitions changed in round four for documented reasons fixed before the runs.

10. Experiment: which local model for the on-premises requirement

Same 36 tuning cases, same scorer, prompt v1.6.0 with the closing gate.

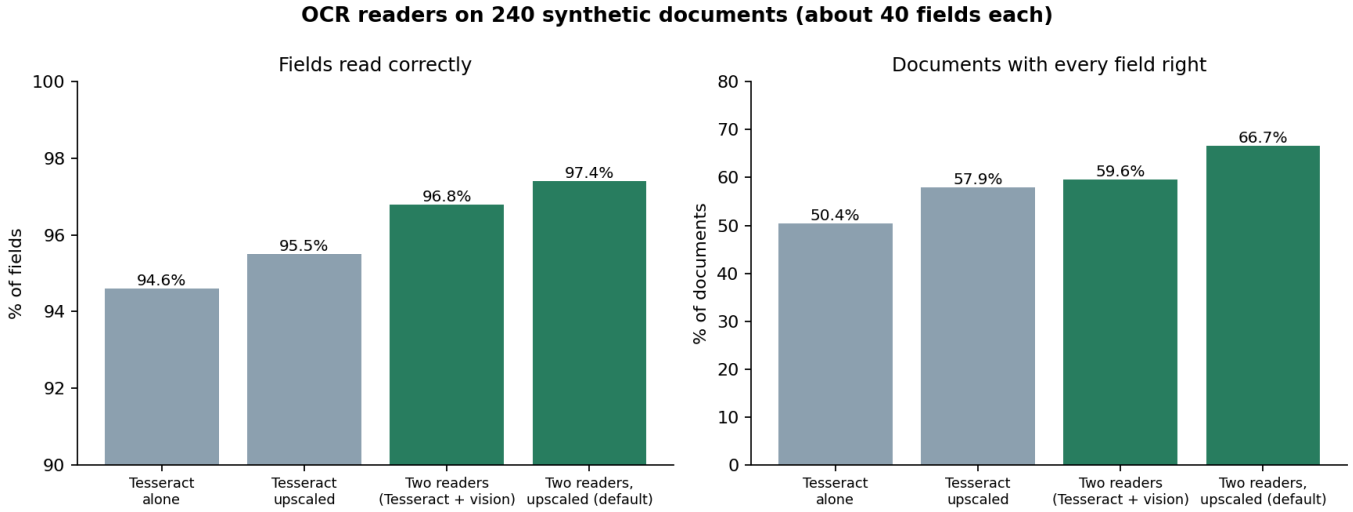


	gemma3:4b (local)	medgemma-4b-it (local)	Mistral-Nemo (hosted)	qwen3:4b (local)
Live answer rate	**97.2%** (35/36)	83.3% (30/36)	94.4% (34/36)	disqualified
Unsupported-token candidates	0	0	0	n/a
Injection: approval language leaked	0	0	0	n/a
Median latency	**about 3.0 s**	about 13 s	about 4.1 s	unbounded (3 to 90+ s)
Worst case	10.8 s	26.4 s	26.2 s	exceeds the 90 s timeout

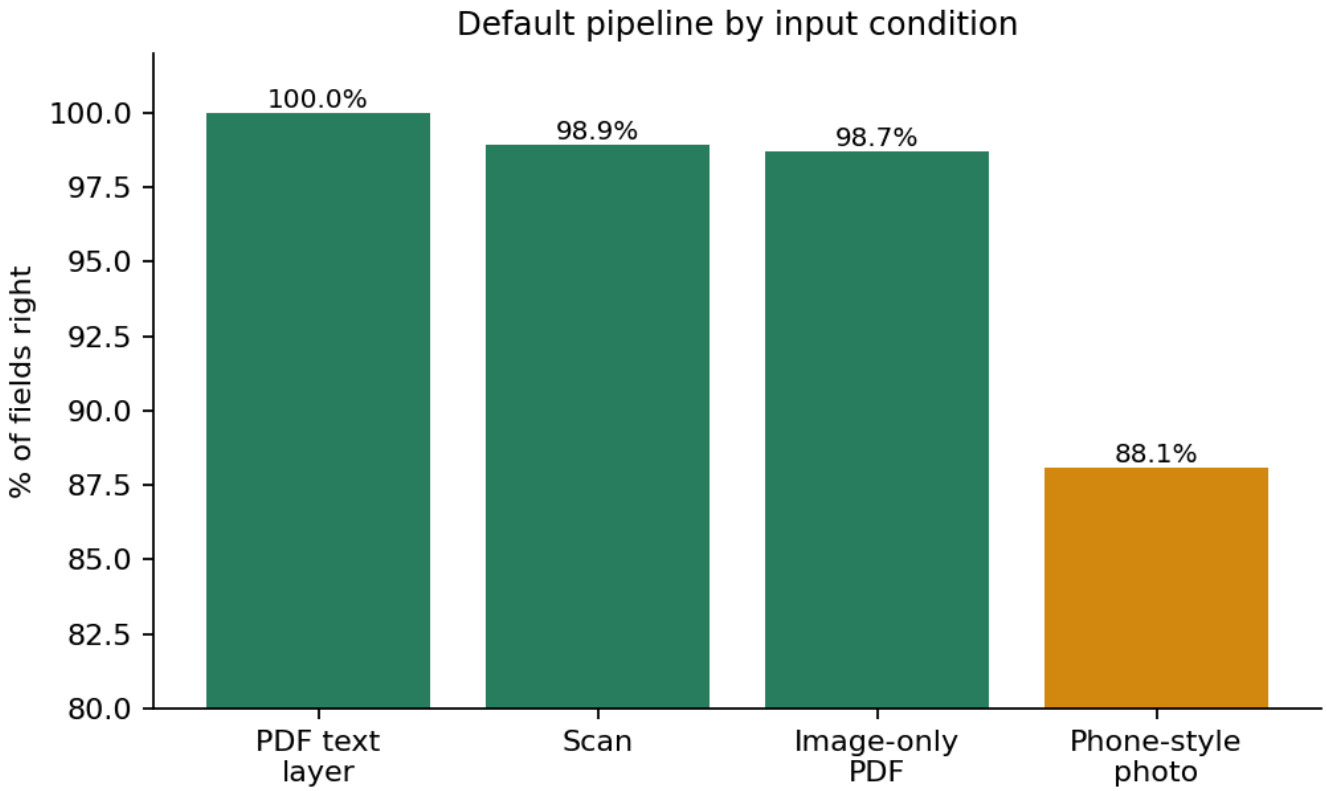
gemma3:4b is the strongest candidate on every automated axis and has an 84-case stress record with 0 garbled. qwen3:4b was disqualified on operational grounds (a hidden "thinking" mode neither documented switch could turn off through Ollama). MedGemma 27B was not tested (too large for the 8 GB laptop GPU). Automatic checks do not replace the manual 0/1 scorecard, which still needs a person.

11. Experiment: reading documents (OCR)

240 synthetic documents (two layouts, four conditions, about 40 fields each, answer key known), read by Tesseract alone, with upscaling, and with the two readers cross-checked.


OCR readers on 240 synthetic documents (about 40 fields each)
Fields read correctly
% of fields
100
98
96
94
92
90
94.6%
95.5%
96.8%
97.4%
Tesseract alone
Tesseract upscaled
Two readers (Tesseract + vision)
Two readers, upscaled (default)
Documents with every field right
% of documents
80
70
60
50
40
30
20
10
0
50.4%
57.9%
59.6%
66.7%
Tesseract alone
Tesseract upscaled
Two readers (Tesseract + vision)
Two readers, upscaled (default)


Wrong fields: the second reader turns silent errors into flagged ones
silent (wrong and not flagged)
wrong and flagged for a person
wrong fields (of about 9,600)
300
250
200
150
100
50
0
312
237
205
167
41
56
3
1
Tesseract alone
Tesseract upscaled
Two readers (Tesseract + vision)
Two readers, upscaled (default)



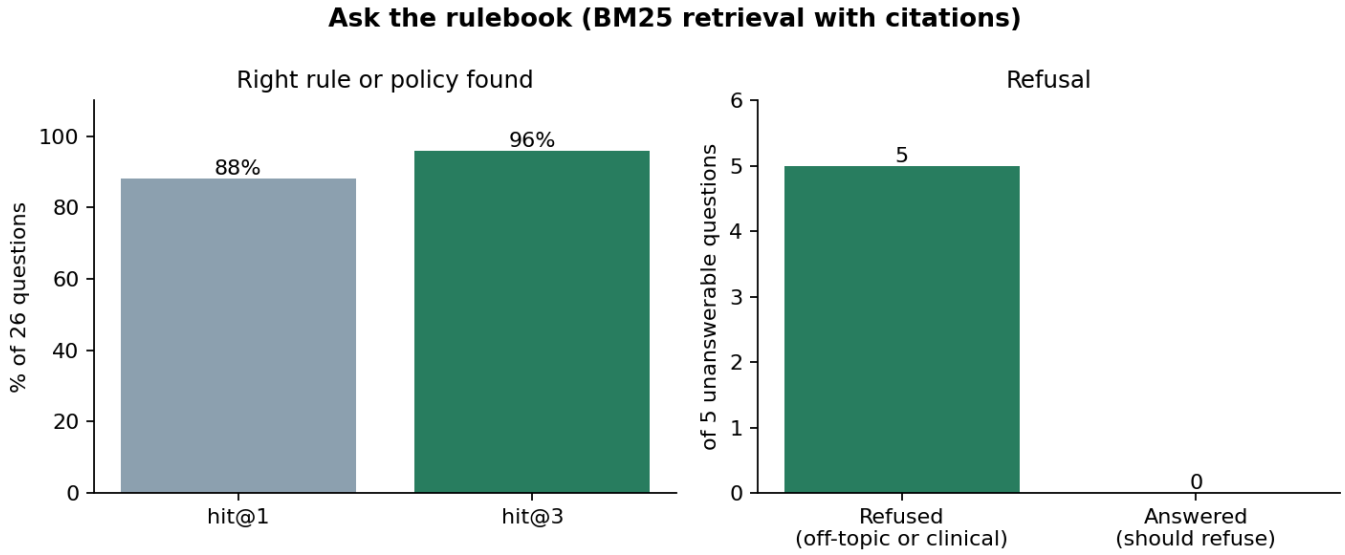
The default pipeline (two readers, upscaled) reads **97.4%** of fields correctly (digital PDFs 100%, scans 98.9%, image-only PDFs 98.7%, phone-style photos 88.1% with 12% of photos perfect). A *silent* error is a wrong field that nothing flagged: the second reader cuts them from 41 to **1** in 240 documents, and photos are the reason a person always compares the page before anything is submitted. The evaluation found and fixed two bugs (an empty date cell dropped a claim line; a two-number row must be flagged, not guessed). Other tests: 41 extractor/gate/reader unit tests and an end-to-end run in real Chrome. (The upload gate, the reader container and their attack tests are covered in the separate Privacy and Security Note, `docs/57`.) **Not tested:** real hospital paper, handwriting, stamps, and Arabic documents.

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12. Experiments: asking the rulebook (retrieval)

"Ask the rulebook" retrieves and quotes; by default nothing is generated, so nothing can be invented, and a question the documents cannot answer is refused. Sections 12.1 to 12.4 measure passage search; section 12.5 measures the knowledge graph and the answer cache added in front of it, and the optional model that picks a query or writes the cited facts up (every written sentence is checked against the facts and dropped if it fails).

12.1 Round 0: keyword search



BM25 over the text of the 15 rules and the payer policies: the right passage was in the top three for **96% of 26 questions** (top-1 88%, 5 of 5 off-topic refused).

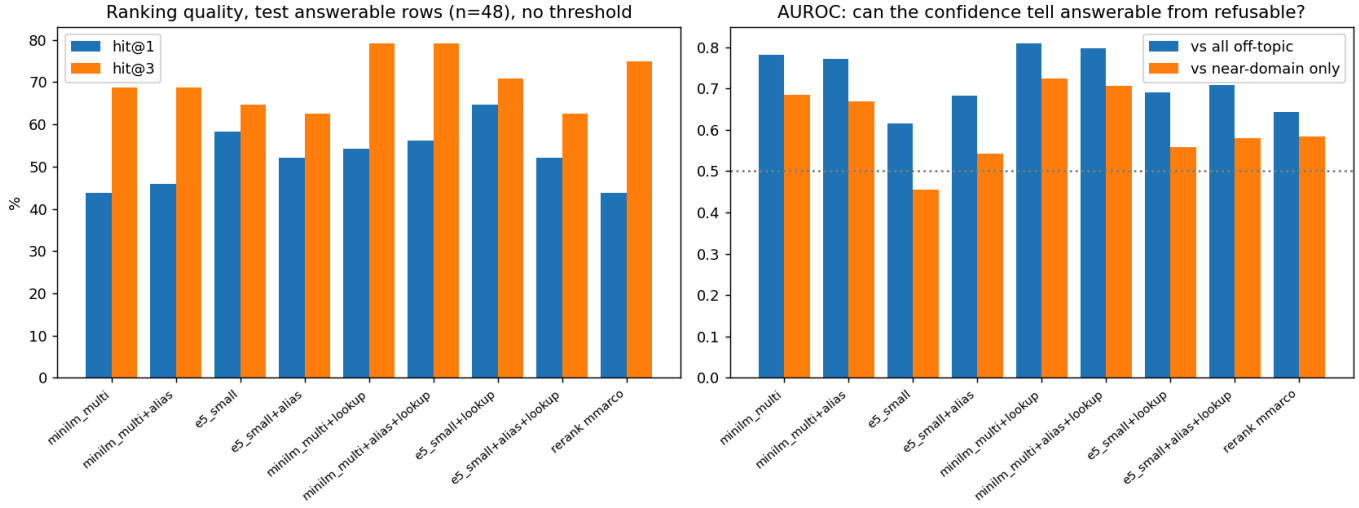
12.2 Experiment 52: keyword vs meaning vs a Laya gate (frozen question set, thresholds chosen on dev only)

Test half	BM25 (product)	Dense, English	Dense, multilingual	Hybrid	Laya gate on dense top 10
English paraphrases, hit@3 (n=21)	**5%**	76%	**81%**	62%	81%
...answerable but refused	76%	14%	5%	5%	0%
English, 0 shared words (n=8)	0%	50%	75%	38%	75%
French + Arabic (n=6)	0%	0%	**100%**	83%	83%
Original 26 questions	96%	96%	96%	96%	96%
Off-topic, pure (n=4) refused	100%	100%	100%	100%	75%
Off-topic, near the domain (n=4) refused	75%	50%	25%	25%	50%

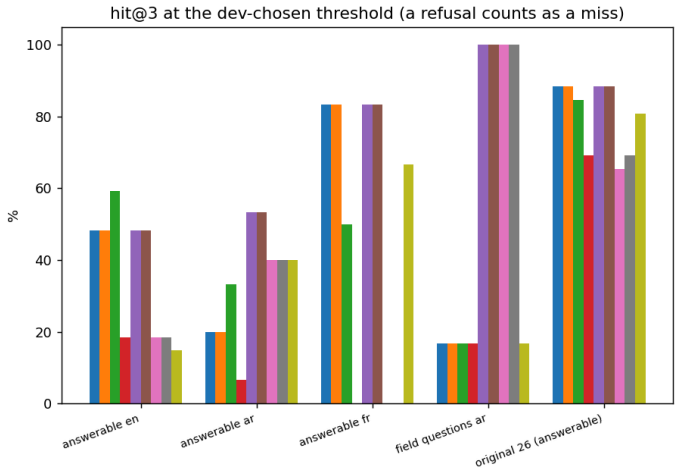
On operator-style wording the product's keyword search answered almost nothing. Dense search alone does the work (the BM25 hybrid is no better). Laya, used zero-shot as a gate, does not beat a plain similarity threshold (ECE 0.12; about 60 ms for no gain). The honest weakness of every dense variant: questions near the domain ("how do I appeal a rejected claim").

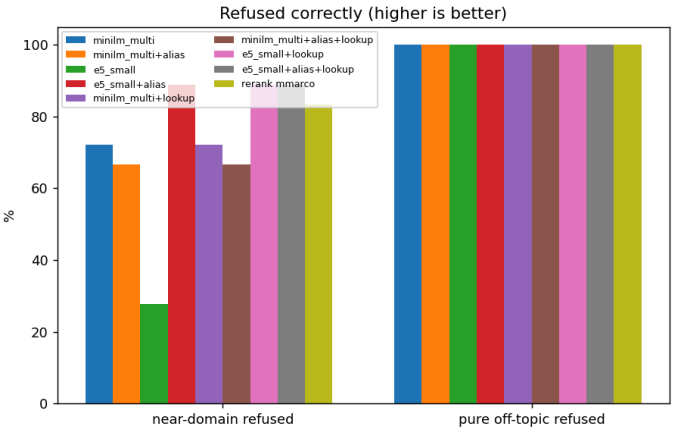
12.3 Round one of the upgrade (corpus of 156 passages, 180 questions; docs/55 section 3)

Test rows	MiniLM (shipped)	+ field lookup	e5-small	mmarco reranker gate
Ranking hit@3, threshold-free (n=48)	69%	**79%**	65%	75%
AUROC answerable vs off-topic	0.78	**0.81**	0.62	0.64
Near-domain refused (n=18)	72%	72%	28%	83%
Arabic field questions hit@3 (n=6)	17%	**100%**	17%	17%
Original 26: hit@3	88%	88%	85%	81%

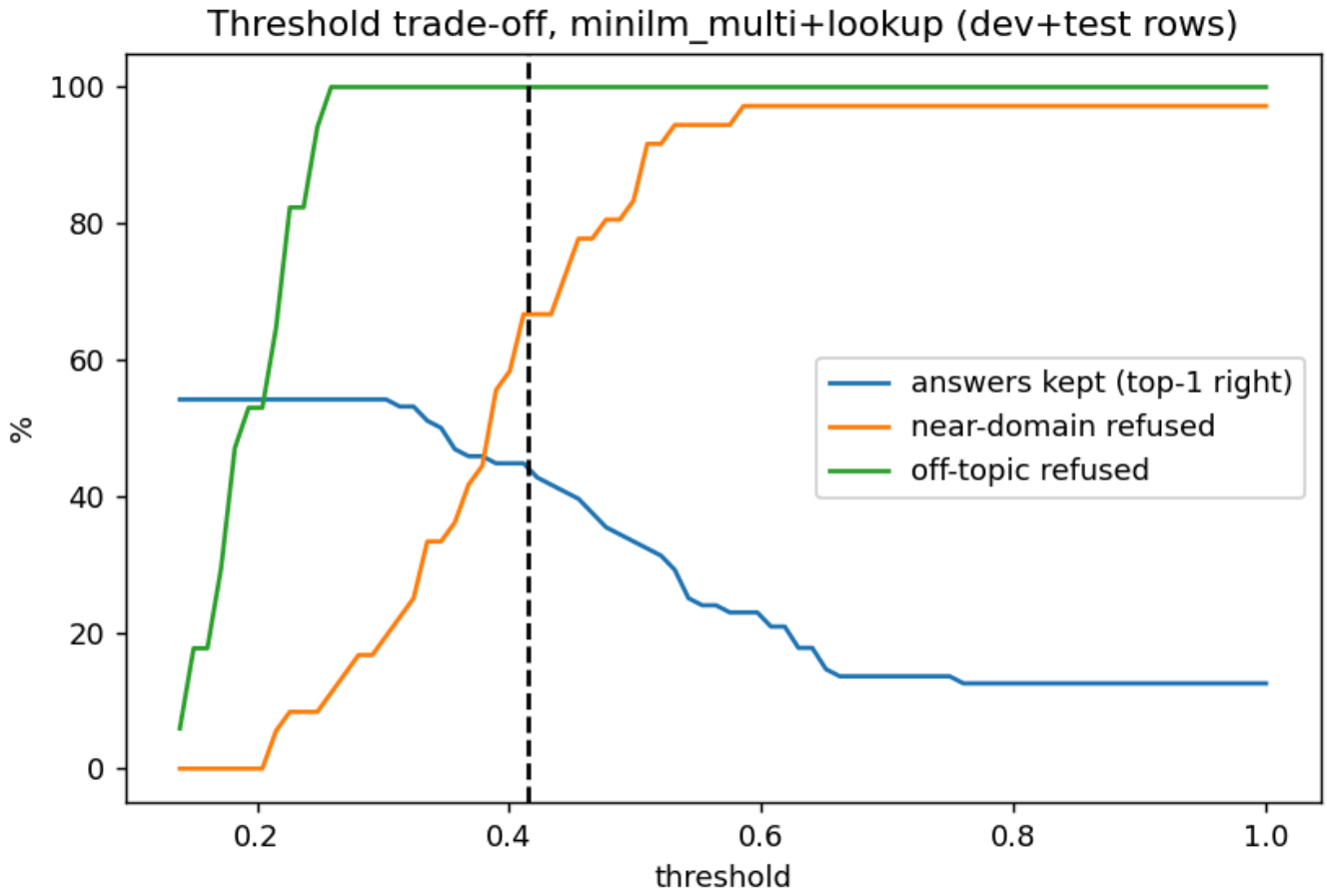

Ranking quality, test answerable rows (n=48), no threshold
hit@1
hit@3
%
minilm_multi
minilm_multi+alias
e5_small
e5_small+alias
minilm_multi+lookup
minilm_multi+alias+lookup
e5_small+lookup
e5_small+alias+lookup
rerank mmarco

AUROC: can the confidence tell answerable from refusable?
vs all off-topic
vs near-domain only
minilm_multi
minilm_multi+alias
e5_small
e5_small+alias
minilm_multi+lookup
minilm_multi+alias+lookup
e5_small+lookup
e5_small+alias+lookup
rerank mmarco


hit@3 at the dev-chosen threshold (a refusal counts as a miss)
%
answerable en
answerable ar
answerable fr
field questions ar
original 26 (answerable)


Refused correctly (higher is better)
minilm_multi
minilm_multi+alias
e5_small
e5_small+alias
minilm_multi+lookup
minilm_multi+alias+lookup
e5_small+lookup
e5_small+alias+lookup
rerank mmarco
%
near-domain refused
pure off-topic refused


Top-passage score on test rows; dashed = threshold chosen on dev
minilm_multi+lookup
minilm_multi+alias+lookup
minilm_multi
minilm_multi+alias
answerable
near-domain
pure

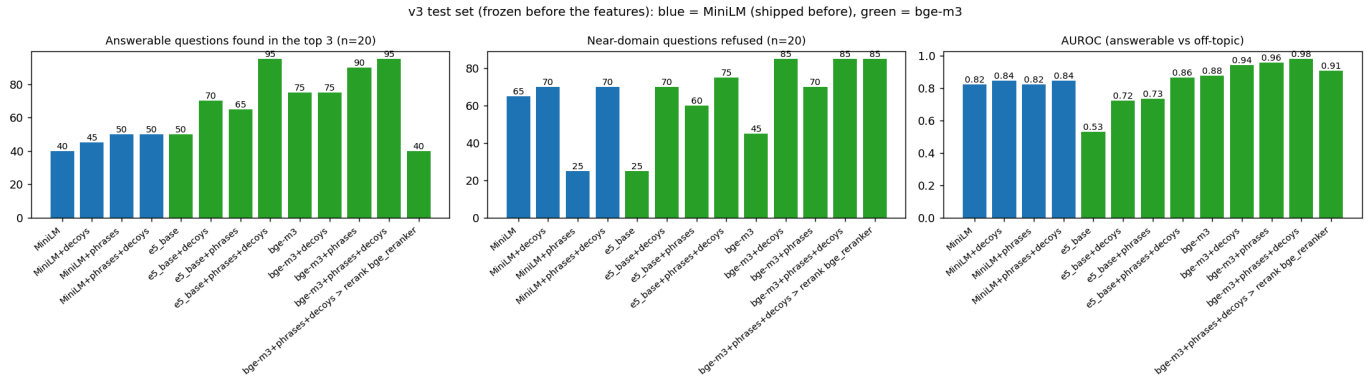


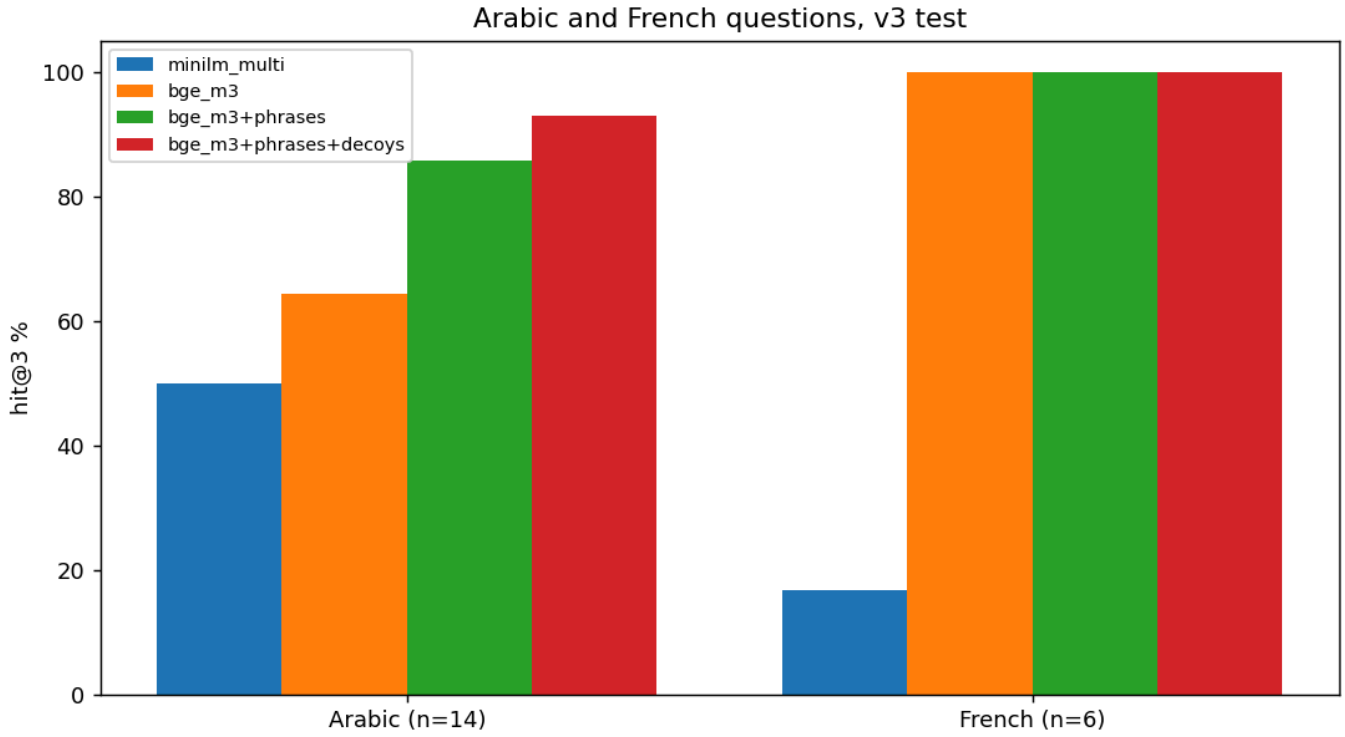
A dictionary lookup answers "what is the field ..." questions in Arabic (17% to 100%). Putting the Arabic field names into the embedded text gave no gain, so it is switched off.

12.4 Round two: bge-m3, Arabic and French phrases, out-of-scope topics (a test set frozen before the features were written)

Test: 20 answerable, 20 near-domain, 4 pure off-topic; thresholds chosen on dev rows only; GPU, fp16.

System	Threshold	Answerable hit@3	Arabic	French	Near-domain refused	Pure refused	AUROC
MiniLM (shipped before)	0.415	40%	50%	17%	65%	100%	0.82
MiniLM + phrases + decoys	0.295	50%	57%	33%	70%	100%	0.84
e5-base (bare)	0.80	50%	29%	100%	25%	100%	0.53
e5-base + phrases + decoys	0.79	95%	93%	100%	75%	75%	0.86
bge-m3 (bare)	0.48	75%	64%	100%	45%	100%	0.88
bge-m3 + phrases	0.515	90%	86%	100%	70%	100%	0.96
bge-m3 + decoys	0.45	75%	64%	100%	85%	100%	0.94
bge-m3 + phrases + decoys (shipped)	**0.485**	**95%**	**93%**	**100%**	**85%**	**100%**	**0.98**
bge-m3 + phrases + decoys, then bge-reranker-v2-m3 on the top five	0.505	40%	43%	33%	85%	100%	0.91





Phrases are Arabic and French ways of asking each rule, attached to the embedding only (never quoted). **Decoys** are out-of-scope topics (appeals, copayment, premiums, ...) as passages: a question whose closest passage is a decoy is refused and the topic is named. bge-m3 with both is the best. e5-base ranks as well but refuses worse. The reranker made ranking much worse (a probable cause, not tested: it scores the bare passage text without the aliases that carry the match), so it was not added.

The product reproduces the experiment (`rust/tools/rag_product_check.py`, real route, sidecar on CPU fp32, Rust cosine): answerable hit@3 0.95, near-domain refused 0.85, off-topic refused 1.00, refusals that name a topic at least 5.

Not settled: 20 questions per slice; the author wrote both the test and the features, so wording may overlap; no native Arabic speaker has reviewed the phrases; a CPU-only host needs about 2 GB more memory and a warm-up; the sidecar image was not rebuilt for bge-m3 (the script now uses the GPU when present).

12.5 Round three: a knowledge graph (KAG) and an answer cache (CAG) in front of passage search (docs/60)

Passage search quotes one passage at a time and cannot answer questions about how the rulebook's parts connect ("which rules can fire for SVC-IMAGE under EDU-PLUS?"). Two tiers were added in front of it. The **graph** is computed from the rule texts, the payer policies, the service catalogue and the field dictionary; it

answers a closed set of 11 query types with cited facts and needs no model. The **cache** keeps answers keyed by the question, the options, the policy-table version and a hash of the rulebook. An optional **model** (Gemma 3 4B through Ollama, `gemma3-ask:4b` with an 8,192-token context) can pick a query for questions the built-in planner does not understand and can write the facts up; its text is shown only if every sentence cites a fact and every id and number occurs in the facts.

Test. 60 questions (`experiments/kag_cag/questions.json`): 12 service-and-policy questions, 7 "which rule reads this policy attribute", 8 field-to-rule, 6 rule-to-field, 9 policy values, and providers, comparisons, severity, related rules, a list, and 8 French or Arabic questions. The gold for each is read from the data files or from a hand-made table of which rule reads which attribute (written from the rule texts, tier C); it is not taken from the graph code. A question counts as answered when every gold token (rule id, value, policy id) is in the returned text. RTX 4060 laptop, 8 GB.

Phase	Answered	Latency (median)
A. passage search only (before this work)	34 / 60	23 ms
B. graph + passage search	**57 / 60** (52 answered by the graph)	graph about 0 ms; passage search 31 ms
B, second ask of the same 60 (answer cache)	57 / 60	about 0 ms
C. B + the model picks a query when the planner cannot	57 / 60 (53 by the graph)	about 3.0 s when the model is asked
D. C + the model writes the answer	57 / 60; 44 write-ups shown, 15 refused by the check	3.7 s (graph facts), 7.0 s (passages)
E. D with the whole rulebook (about 3.6k tokens) at the front of every prompt	57 / 60; 43 shown, 16 refused	3.5 s, 7.1 s

By category, the graph moved service-and-policy questions from 0 of 12 to 12 of 12, rule-to-field from 0 of 6 to 6 of 6, comparisons from 0 of 2 to 2 of 2, and the list question from 0 of 1 to 1 of 1. The three misses are the yes/no wording "does EDU-BASIC require an authorization for SVC-LAB", the transitive "which rules depend on R007", and one Arabic question.

What the model added. Almost nothing on accuracy: one extra question handled by the graph. Its first version made things worse: on French and Arabic questions it could not understand it kept proposing the same query, "what rule R001 checks", copied from the prompt's example; every identifier existed, so the validator accepted it and the graph answered with confidence. Multilingual accuracy fell from 7 of 8 to 2 of 8 (phase C, 52 of 60 overall). The fix, an identifier in a proposed query must also appear in the question, restored 7 of 8. Written answers are slower than the facts they restate (3.5 to 7 s against 0 ms), and the check refused 15 of 59, so the cited facts remain the product and the write-up is a convenience. The rulebook at the front of the prompt did not change the outcome (43 against 44 shown). The prompt-prefix probe on Ollama (3,588 tokens): the second identical prefix took 275 ms to process against 518 ms the first time, but Ollama still reports the whole prompt as processed, so the large saving expected from a server with real prefix caching (vLLM) is not demonstrated.

Regression on the held-out sets of 12.4 (20 answerable and 24 off-topic questions in English, French and Arabic; `outputs/defense/kag_cag_regression.json`): the graph took **0 of 20** answerable and **0 of 24** off-topic questions, and the top three passages were unchanged for every answerable one. The first run took one off-topic question ("which documents does a patient need to bring to the clinic") because it shares a word with the rulebook; the planner now needs a policy, insurer or claim word beside a bare attribute word, and a test pins it.

Not settled. The 60 questions were written by the same author as the planner, so the 52 answered by the graph are not an out-of-sample figure; the French and Arabic questions and the regression sets are closer to one. The model is Gemma 3 4B on a laptop, not the server's model; the latencies are this laptop's; the cache lives in one process; the rule-to-attribute gold is hand-derived.

---

13. Experiment: Arabic and English claims

A table-driven step (`cg_core::localize`) runs before validation. It maps Arabic field names, enumerated values, digits, month names and service names to canonical English; records every change; keeps the original claim; and refuses what it cannot map (Hijri dates, ambiguous day/month, unknown field names or values, duplicate fields, Arabic letters, digits or invisible characters inside identifiers). Free text is untouched and English input is byte-identical.

Test	Result
Localize unit tests	16 pass
Arabic twin of every public claim vs the original	identical read-back and identical results for all 15 rules
End to end over HTTP (`rust/tools/arabic_e2e.py`)	8 of 8
Asynchronous 1,000-claim batch (`rust/tools/batch_e2e.py`)	all checks pass; request answered in 0.06 s; batch about 17.5 s

Not done: native-speaker review of every Arabic label and phrase (all written by the assistant); Arabic OCR data and the vision model's Arabic are unmeasured.

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14. Experiments: human review and workload

All evidence in this section is on **simulated reviewers** (a hidden true rate per rule, real findings recycled). No review time has been measured on a person.

14.1 Laya fine-tune (Kaggle, 2x T4, 6 epochs, about 41 minutes, 2,486 real labelled cases)

Test split, 150 unseen claims	Accuracy	Macro-F1	ECE
Base Laya, zero-shot	0.631	0.289	0.276
Fine-tuned	0.971	0.915	0.027
Always guess PASS	0.835	n/a	n/a

It learned the rulebook and is well calibrated, but recovers only 64% of real FAILs, and its labels are the engine's own answer key. Run after the engine it can only re-predict what the engine computed exactly, so it cannot save time. **Archived as an experiment, not wired.**

14.2 Reviewer workload report (read-only, wired)

Finding rules that waste reviews (20 histories, seeds 1 to 20): share of the two bad rules found after 300 / 600 / 1,200 / 2,500 / 5,000 decisions: 0% / 55% / 72.5% / 90% / 100%, with 0 good rules wrongly listed at every size. Findings that can get a lighter look (`skip_lower`): the stated 95% bound on wrong findings held in every history; a looser threshold covers far more findings but lets more wrong ones through (about 125 against a bound of 279 over 5,000 decisions at 0.85).

14.3 Claims processed with no person (20 random batches of 500)

Policy	No person needed	Findings a person still reviews	Automatic claims that were wrong
Today	0 of 500	645	0
Green claims only	167 (33%)	645	0 on this data
Green + light-review rules, `skip_lower` 0.95	174 (35%)	638	0.2

Policy	No person needed	Findings a person still reviews	Automatic claims that were wrong
Green + light-review rules, `skip_lower` 0.90	358 (72%)	420	6.5
Green + light-review rules, `skip_lower` 0.85	388 (78%)	370	8.3

With the options of docs/50 on the real pipeline and 600 labelled claims, per 500 claims: auto-clear alone decides 159 (32%) without a person; adding auto-request for all severities takes findings a person decides from 650 to 399 (-39%) and claims decided without a person to 196 (39%), 0 wrong against the answer key. The changes that alter who looks at what are **off by default** and wait for the mentor.

14.4 Routing by confidence (archived)

Learning from 2,000 decisions then reviewing 793 more: routing by reliability raised the share needing a senior from 76.2% to 87.5% (an added load, so it is **not wired**; archived on a tag). Poisoning: one adversarial senior who dismisses everything is stopped by the protections up to about 30% of decisions and when fully captured, but **not** at 40% to 80% (that is why activation needs two administrators and a replay report). Invariants that held: with the feature off, receipts are unchanged; applying a snapshot changes no lane, score, result hash or status over 200 claims.

14.5 Reviewer copilot, what-if and recheck (docs/51)

Deterministic summaries and correction suggestions (no model), a dry run that shows each rule before and after (cleared, still flagged, new problem), and "Correct and recheck" as a new claim version. Every sentence names a rule and the claim paths it rests on. All of it is traced; a what-if stores nothing.

14.6 Batches, requests and fills (docs/53)

Batches of up to 1,000 claims with a random 5% audit sample and quarantine of the whole batch when a sampled green claim is wrong; requests for missing information written by the system, with the officer confirming only the send; copilot fills the officer confirms; asynchronous intake with progress. **Not delivered anywhere:** the system has no outgoing channel; the mentor may have meant the patient, not the provider (open question).

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15. Security and privacy testing

The penetration test, the fuzzing, the mutation test, the deployment attack, the real-model red team, the browser attack, the flood and timing tests, the backup test and the dependency scans are reported in the separate **Privacy and Security Note** (`docs/57`), which also holds the threat model, the access-control and least-privilege design and the audit design. They are not repeated here.

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16. Deployment measurements

- **Proxy cost** (16 keep-alive connections, TLS in front): `/healthz` 6,200/s direct, 3,100/s nginx, 990/s Caddy; the signed-in inbox 421/s direct, 215/s nginx, 180/s Caddy. 421 signed-in requests a second from one process is about a hundred times the 200 requests a minute the target-server plan asks for. Docker Desktop adds a network hop that a Linux host does not, so these cost numbers overstate a proxy's cost on the target.

- **Concurrency of the explanation step:** section 9.4.
- **Desktop app:** the window starts its bundled server and the health check answers; unsigned, Windows only, not run on a clean PC. Current release: `desktop-v0.4.0`.

---

17. Limitations (read before quoting a number)

1. **A perfect public score cannot rank this system against a better one.** The answer key is deterministic and the public claims were available while the rules were built. What the sets can show is the absence of errors found, with the upper bounds above. Zero observed false positives is not zero true.
2. **No unseen test set was available to us.** The mentor-held set was not provided, so every accuracy figure is agreement on data we developed against.
3. **The oracle shares our reading of the rulebook.** Tiers B and C measure agreement and self-consistency, not accuracy against the real rulebook.
4. **Tier C is small and written by us** (140 labelled results in 123 claims).
5. **The rule categories are our own grouping**; the severity and all-rules macros are reported for that reason.
6. **Per-rule F1 can rest on very few failures:** 9 of 15 rules have none in the 50-claim split and the others as few as 3.
7. **The data is synthetic.** The mix of failures says nothing about the frequency of real claim errors. Real paper, handwriting, stamps and Arabic documents are untested.
8. **FHIR loses authorizations by design**, so R009 abstains on that path (F1 0.9745; 66.7% of clean claims get an unnecessary abstention).
9. **Latency is from one laptop**, one process, warm caches; it is not a load test, and the target server (RTX 3080, Linux) has not been used. The language benchmark's top group cannot be ordered because passes varied by a factor of two on that machine.
10. **The explanation step is not part of the detection metrics.** Its quality rests on mechanical proxies (useful, next step, value citation), small samples (36 tuning and 12 fresh cases per round), a hosted endpoint that drifts and one provider. Nobody has done the manual 0/1 scoring. Two of the decisions (round two, and the earlier E5) were judgements that overrode the pre-registered rule, and are recorded as such.
11. **The retrieval test sets are small and author-written.** 20 questions per slice; the same author wrote the test and the features; the Arabic and French are not reviewed by a native speaker. The 60 graph questions (section 12.5) were written by the planner's author too, so 52 answered by the graph is not an out-of-sample figure; the model is Gemma 3 4B on a laptop; the answer cache is per process; the prefix-cache saving was measured on Ollama only.
12. **The learning, routing and workload experiments use simulated reviewers.** No review time has been measured; the real effect of any workload saving is unknown. Confidence routing is archived because it raised senior load.
13. **The desktop app is unsigned**, Windows only, and was not run on a clean PC. The embedding sidecar image was not rebuilt for bge-m3.
14. **Not yet run on the target server** and nothing here has run on real hospital traffic.
15. **Nothing here measures review quality or decisions** by real people; the advisory extension rules (E001 to E103) are outside the official score.

---

18. Reproduce everything and where it lives

```
# detection (Python reference)
python scripts/evaluate_phase2.py --out-dir outputs/evaluation   # about 6 minutes
```

```
python scripts/render_eval_report.py --check                    # exits 1 if docs/29 is stale
# detection (Rust build)
python rust/tools/phase2_eval.py                                # writes
outputs/evaluation/rust_metrics.json
python rust/tools/compare_ingest.py                             # 35 ingestion checks
# AI explanation experiments (needs FEATHERLESS_API_KEY)
python scripts/run_experiments.py e1                            # also e2 to e10, resumable
python scripts/analyze_experiments.py                           # experiments/summary.json and
docs/figures/*.png
# language benchmark
python benchmarks/languages/make_report.py
# document reading
python rust/tools/doc_eval.py
# rulebook search
HF_HOME=D:/hf_cache python experiments/rag_v2/run.py
HF_HOME=D:/hf_cache python experiments/rag_v2/run_v3.py --models minilm_multi,e5_base,bge_m3 --
rerankers bge_reranker
python experiments/rag_v2/plot.py && python experiments/rag_v2/plot_v3.py
HF_HOME=D:/hf_cache python rust/tools/rag_product_check.py
# knowledge graph + answer cache (needs Ollama with: ollama create gemma3-ask:4b -f
experiments/kag_cag/Modelfile)
python experiments/kag_cag/build_questions.py
python experiments/kag_cag/run_eval.py && python experiments/kag_cag/run_regression.py
# Arabic, batches
python rust/tools/arabic_e2e.py && python rust/tools/batch_e2e.py
```

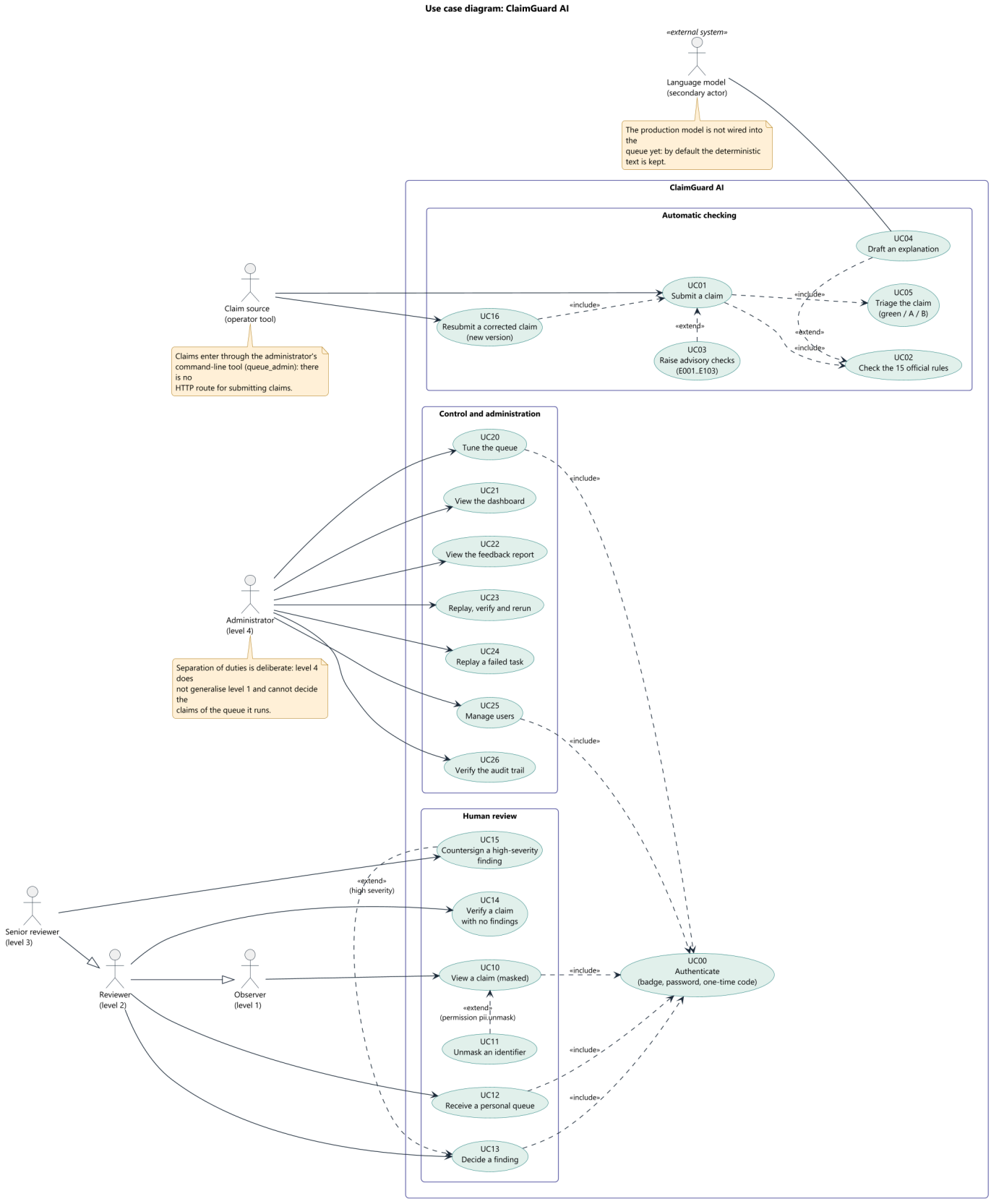
Accuracy numbers are deterministic (fixed seeds 20260927 and 20261004). Latency varies with the machine. The commit that produced each evidence file is recorded in its JSON.

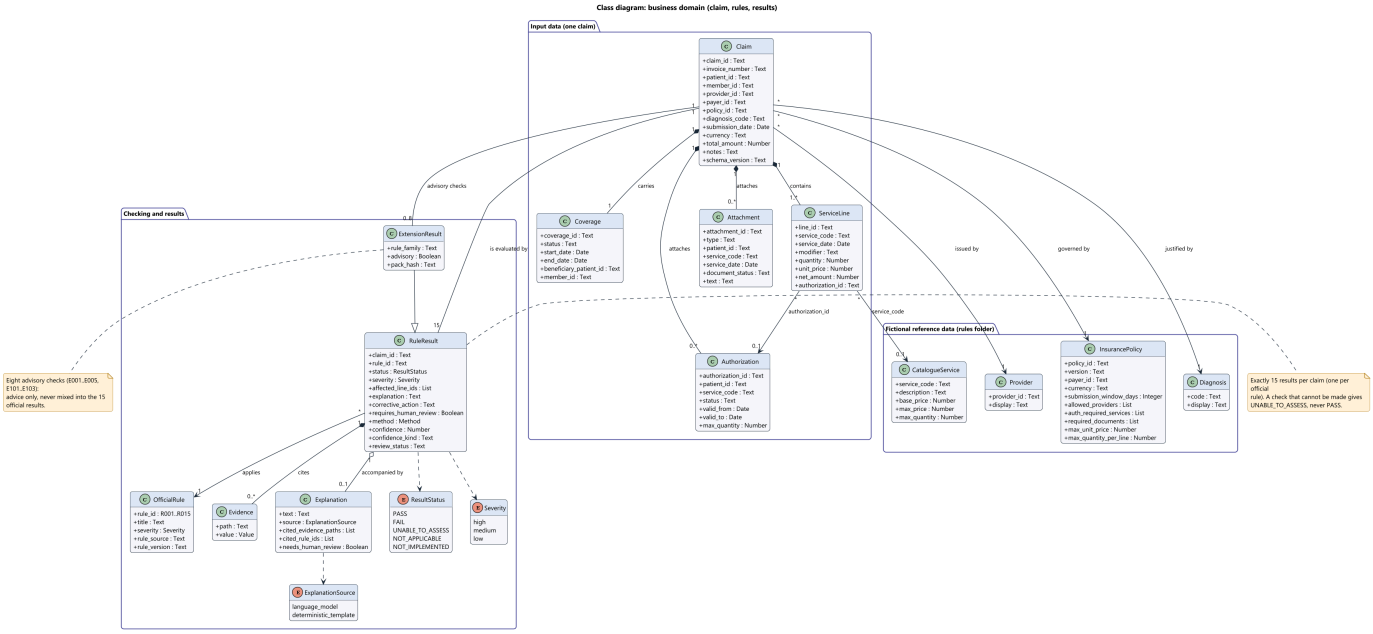
Topic	Document	Evidence
Detection, Python reference (nine sets)	docs/29	`outputs/evaluation/metrics.json`, `provenance.json`
Detection, Rust build	docs/46	`outputs/evaluation/rust_metrics.json`
Earlier evaluation report	docs/17	`outputs/yara_*_metrics.json`
Stress testing and judging coverage	docs/19	`scripts/status_coverage.py`
AI explanation experiments	docs/21	`experiments/raw/`, `experiments/summary.json`, `experiments/results_tables.md`, manual scoring sheets
Architecture comparison	docs/24, docs/25	`outputs/architecture_comparison/`
Language choice	docs/36	`benchmarks/languages/results/`
Documents (OCR), ledger, payer rules	docs/44, docs/45	`rust/tools/doc_eval.py`
Confidence routing (archived), workload	docs/48, docs/49, docs/50	`outputs/defense/automatic.json`, `experiments/laya_kaggle/`
Copilot, batches, nginx	docs/51, docs/53, docs/54	`outputs/defense/`
Rulebook search	docs/52, docs/55	`experiments/rag_semantic/`, `experiments/rag_v2/`, `outputs/defense/rag_product_check.json`
Knowledge graph and answer cache	docs/60	`outputs/defense/kag_cag_eval.json`, `outputs/defense/kag_cag_regression.json`, `experiments/kag_cag/`

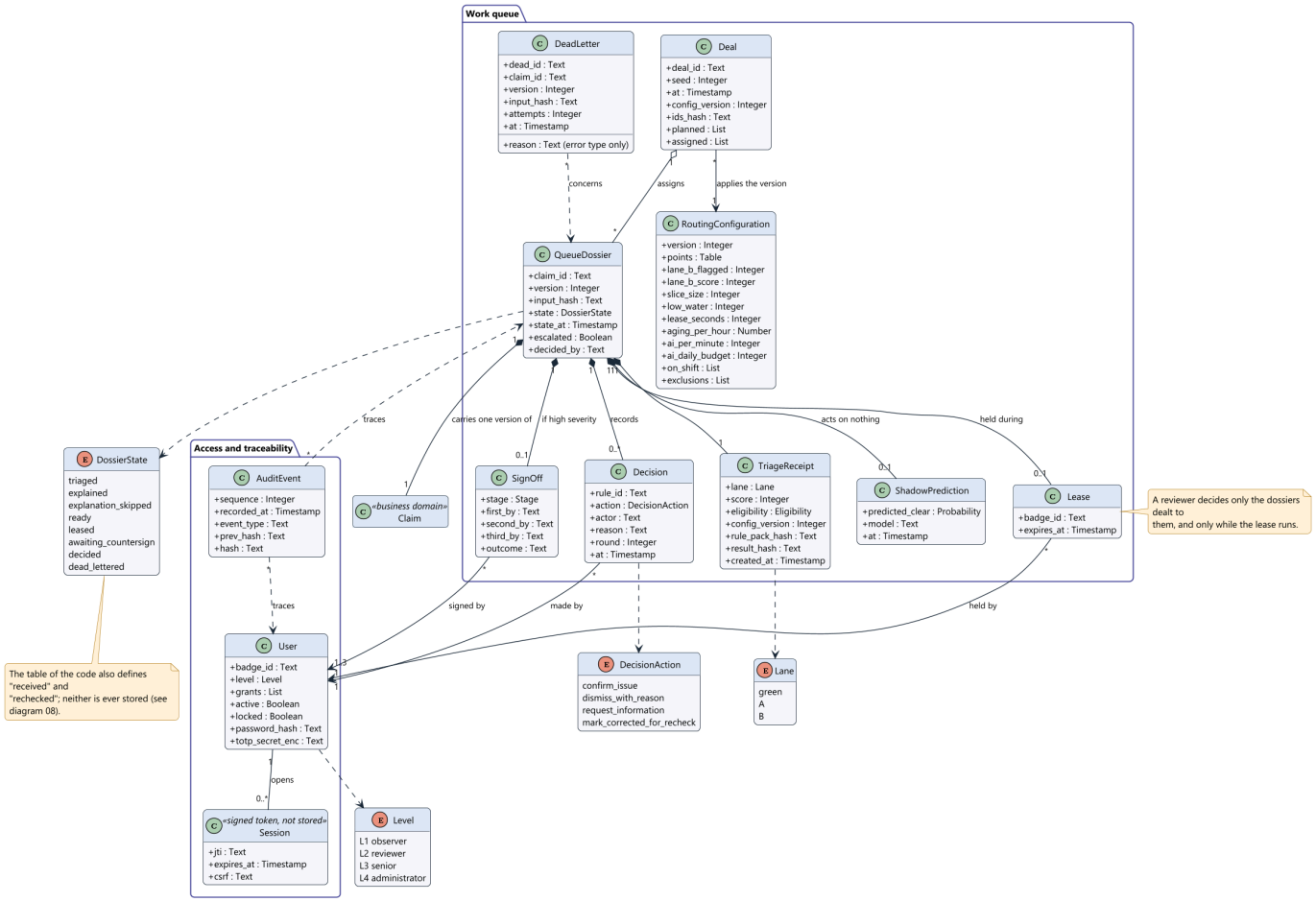
Topic	Document	Evidence
Arabic intake	docs/55	`outputs/defense/arabic_e2e.json`

19. Appendix: design diagrams (UML)

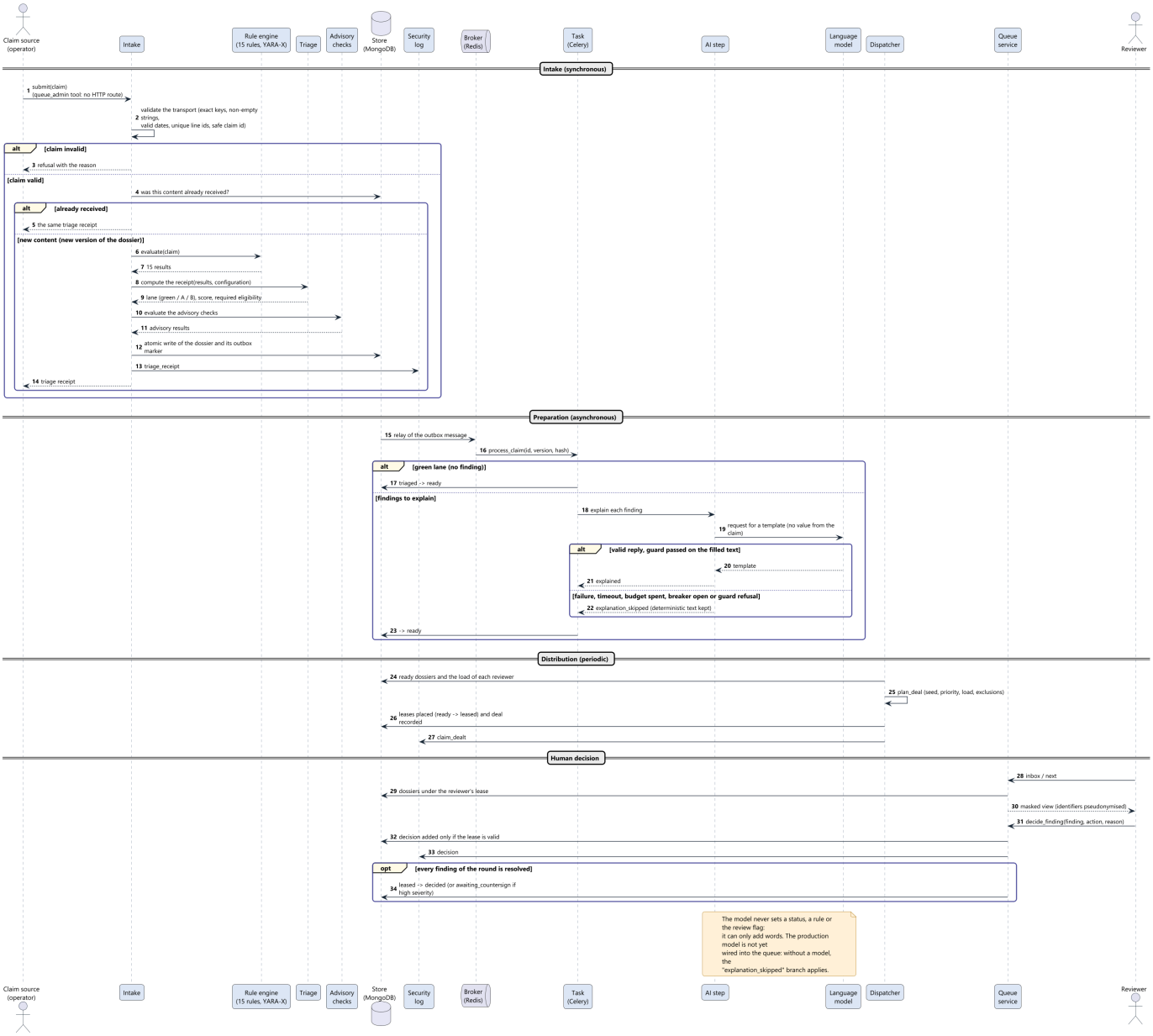
These are not experiment results; they document the system the experiments tested.



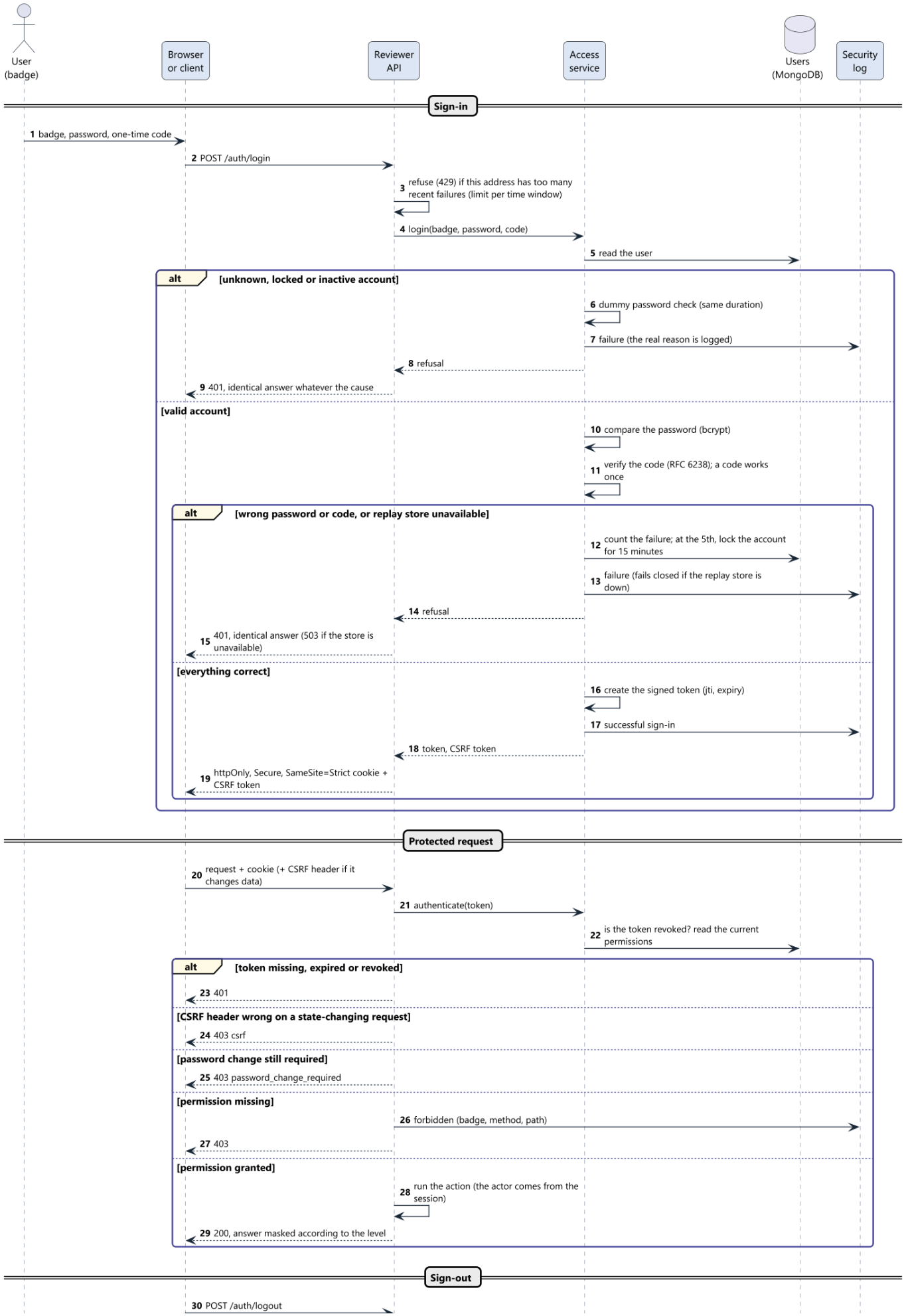

Class diagram: business domain (claim, rules, results)
Input data (one claim)
Claim
+claim_id : Text
+invoice_number : Text
+patient_id : Text
+member_id : Text
+provider_id : Text
+payer_id : Text
+policy_id : Text
+diagnosis_code : Text
+submission_date : Date
+currency : Text
+total_amount : Number
+notes : Text
+schema_version : Text
Coverage
+coverage_id : Text
+status : Text
+start_date : Date
+end_date : Date
+beneficiary_patient_id : Text
+member_id : Text
Attachment
+attachment_id : Text
+type : Text
+patient_id : Text
+service_code : Text
+service_date : Date
+document_status : Text
+text : Text
ServiceLine
+line_id : Text
+service_code : Text
+service_date : Date
+modifier : Text
+quantity : Number
+unit_price : Number
+net_amount : Number
+authorization_id : Text
Authorization
+authorization_id : Text
+patient_id : Text
+service_code : Text
+status : Text
+valid_from : Date
+valid_to : Date
+max_quantity : Number
carries
attaches
contains
advisory checks
is evaluated by
issued by
governed by
justified by
service_code
authorization_id
Checking and results
ExtensionResult
+rule_family : Text
+advisory : Boolean
+pack_hash : Text
RuleResult
+claim_id : Text
+rule_id : Text
+status : ResultStatus
+severity : Severity
+affected_line_ids : List
+explanation : Text
+corrective_action : Text
+requires_human_review : Boolean
+method : Method
+confidence : Number
+confidence_kind : Text
+review_status : Text
OfficialRule
+rule_id : R001..R015
+title : Text
+severity : Severity
+rule_source : Text
+rule_version : Text
Evidence
+path : Text
+value : Value
Explanation
+text : Text
+source : ExplanationSource
+cited_evidence_paths : List
+cited_rule_ids : List
+needs_human_review : Boolean
ResultStatus
PASS
FAIL
UNABLE_TO_ASSESS
NOT_APPLICABLE
NOT_IMPLEMENTED
Severity
high
medium
low
ExplanationSource
language_model
deterministic_template
applies
cites
accompanied by
Fictional reference data (rules folder)
CatalogueService
+service_code : Text
+description : Text
+base_price : Number
+max_price : Number
+max_quantity : Number
Provider
+provider_id : Text
+display : Text
InsurancePolicy
+policy_id : Text
+version : Text
+payer_id : Text
+currency : Text
+submission_window_days : Integer
+allowed_providers : List
+auth_required_services : List
+required_documents : List
+max_unit_price : Number
+max_quantity_per_line : Number
Diagnosis
+code : Text
+display : Text
Eight advisory checks (E001..E005, E101..E103); advice only, never mixed into the 15 official results.
Exactly 15 results per claim (one per official rule). A check that cannot be made gives UNABLE_TO_ASSESS, never PASS.

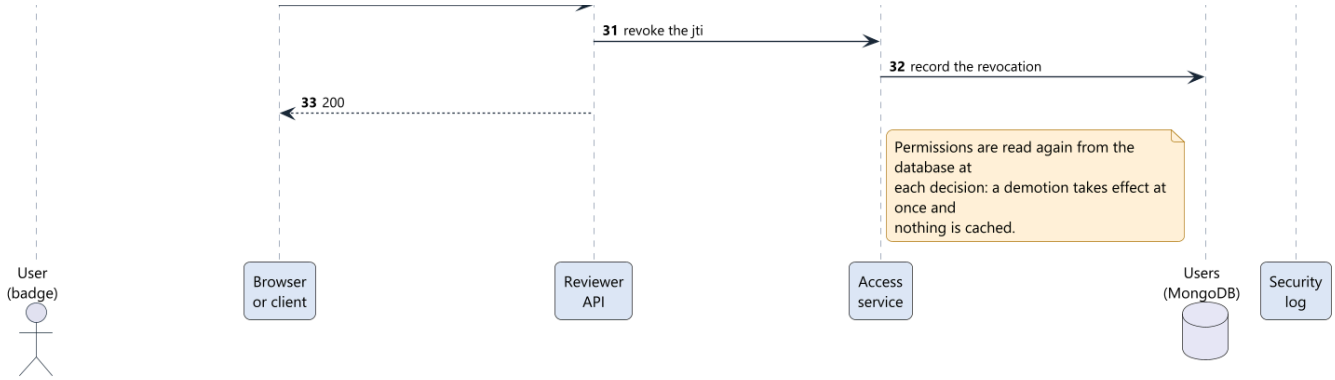

Class diagram: work queue, decisions, access and audit
Work queue
DeadLetter
+dead_id : Text
+claim_id : Text
+version : Integer
+input_hash : Text
+attempts : Integer
+at : Timestamp
+reason : Text (error type only)
Deal
+deal_id : Text
+seed : Integer
+at : Timestamp
+config_version : Integer
+ids_hash : Text
+planned : List
+assigned : List
concerns
assigns
applies the version
QueueDossier
+claim_id : Text
+version : Integer
+input_hash : Text
+state : DossierState
+state_at : Timestamp
+escalated : Boolean
+decided_by : Text
RoutingConfiguration
+version : Integer
+points : Table
+lane_b_flagged : Integer
+lane_b_score : Integer
+slice_size : Integer
+low_water : Integer
+lease_seconds : Integer
+aging_per_hour : Number
+ai_per_minute : Integer
+ai_daily_budget : Integer
+on_shift : List
+exclusions : List
traces
carries one version of
if high severity
records
acts on nothing
held during
SignOff
+stage : Stage
+first_by : Text
+second_by : Text
+third_by : Text
+outcome : Text
Decision
+rule_id : Text
+action : DecisionAction
+actor : Text
+reason : Text
+round : Integer
+at : Timestamp
TriageReceipt
+lane : Lane
+score : Integer
+eligibility : Eligibility
+config_version : Integer
+rule_pack_hash : Text
+result_hash : Text
+created_at : Timestamp
ShadowPrediction
+predicted_clear : Probability
+model : Text
+at : Timestamp
Lease
+badge_id : Text
+expires_at : Timestamp
A reviewer decides only the dossiers dealt to them, and only while the lease runs.
DossierState
triaged
explained
explanation_skipped
ready
leased
awaiting_countersign
decided
dead_lettered
Access and traceability
AuditEvent
+sequence : Integer
+recorded_at : Timestamp
+event_type : Text
+prev_hash : Text
+hash : Text
«business domain» Claim
traces
User
+badge_id : Text
+level : Level
+grants : List
+active : Boolean
+locked : Boolean
+password_hash : Text
+totp_secret_enc : Text
opens
«signed token, not stored» Session
+jti : Text
+expires_at : Timestamp
+csrf : Text
Level
L1 observer
L2 reviewer
L3 senior
L4 administrator
signed by
made by
held by
DecisionAction
confirm_issue
dismiss_with_reason
request_information
mark_corrected_for_recheck
Lane
green
A
B
The table of the code also defines "received" and "rechecked"; neither is ever stored (see diagram 08).

Sequence diagram: from submission to the human decision

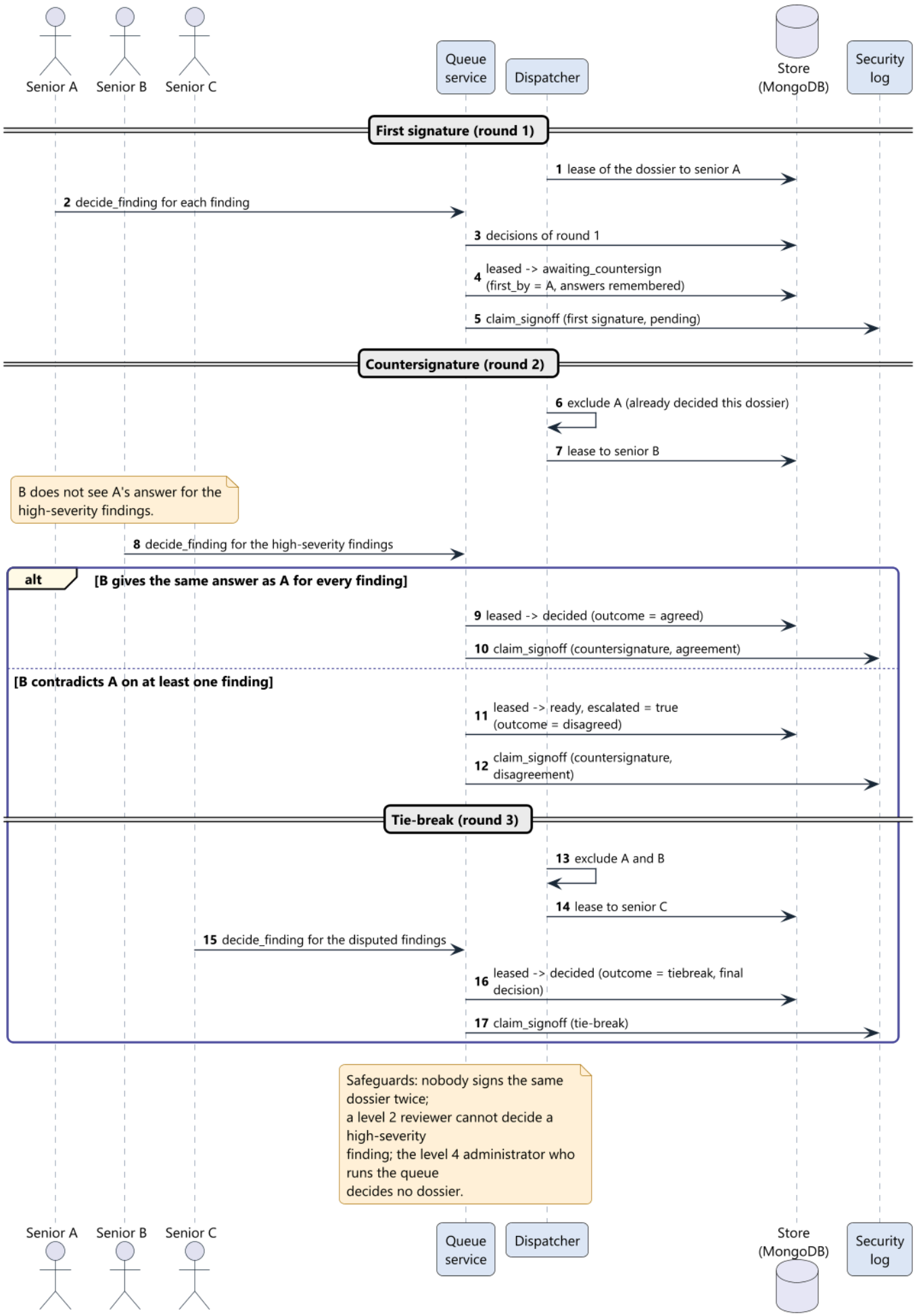


Sequence diagram: sign-in and access control of a request

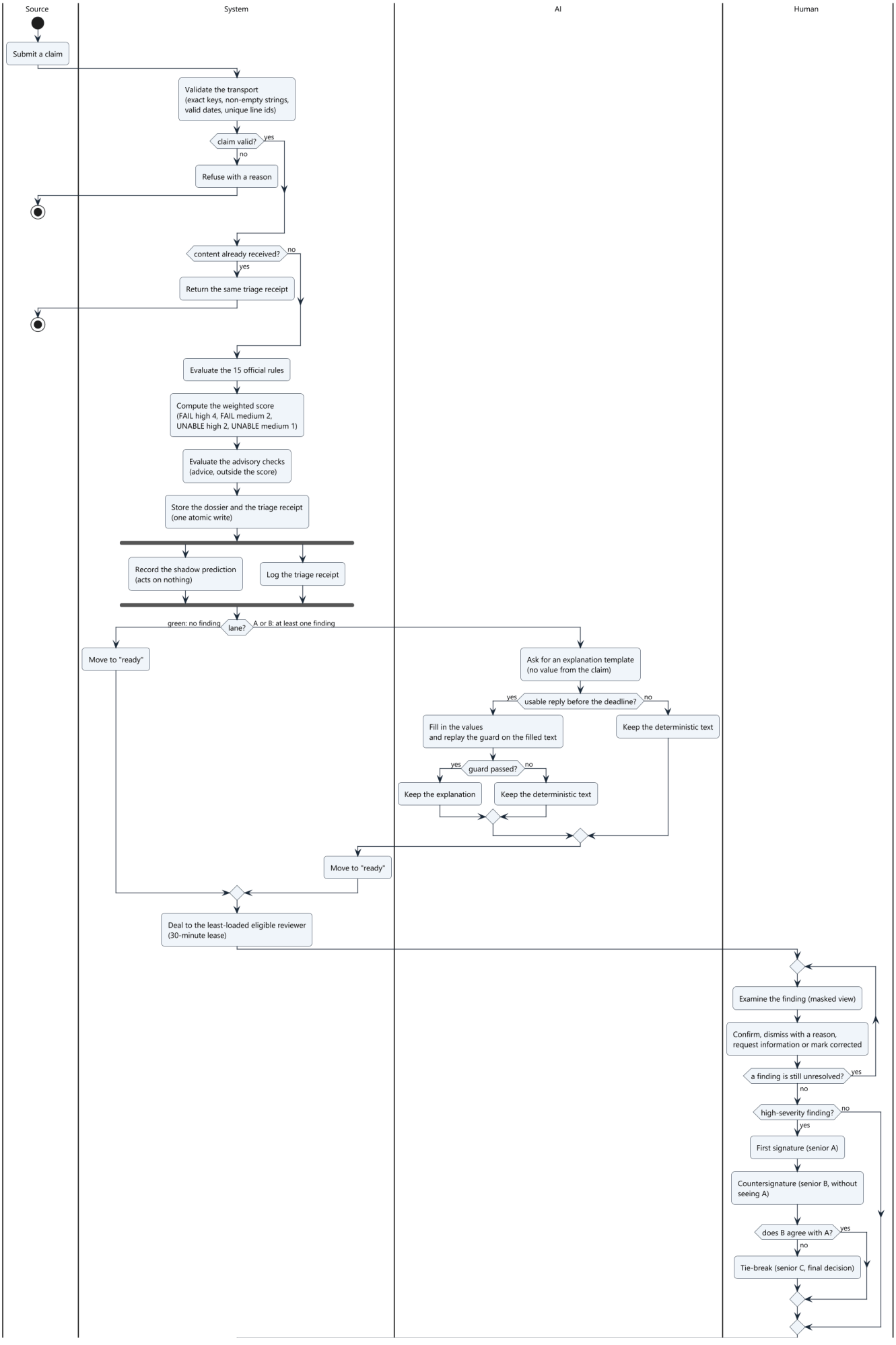



31 revoke the jti
32 record the revocation
33 200
Permissions are read again from the database at
each decision: a demotion takes effect at once and
nothing is cached.
User (badge)
Browser or client
Reviewer API
Access service
Users (MongoDB)
Security log

Sequence diagram: two-person sign-off of a high-severity finding



Activity diagram: processing a claim, from intake to the decision





State machine diagram: life cycle of a dossier in the queue

Defined in the code's transition table but never used:
"received" (only the origin of the first logged event) and
"rechecked" (decided -> rechecked -> triaged). No code path
moves a dossier into "rechecked" today.

dead_lettered

intake writes the dossier directly in this state
(the event log records received -> triaged)

replayed by the administrator (level 4)
(once only)

failure after the retries
(back-off between attempts)

Active dossier

triaged

explanation kept by the guard

AI unavailable, guard refusal,
budget spent or deadline passed

green lane (no finding)

explained

explanation_skipped

ready

deal (lease placed)

lease expired or handed back,
or sign-off disagreement (escalation)

Every transition is one event added to the log; a
transition from the wrong state is refused.

leased

first signature
(high severity)

lease to another senior

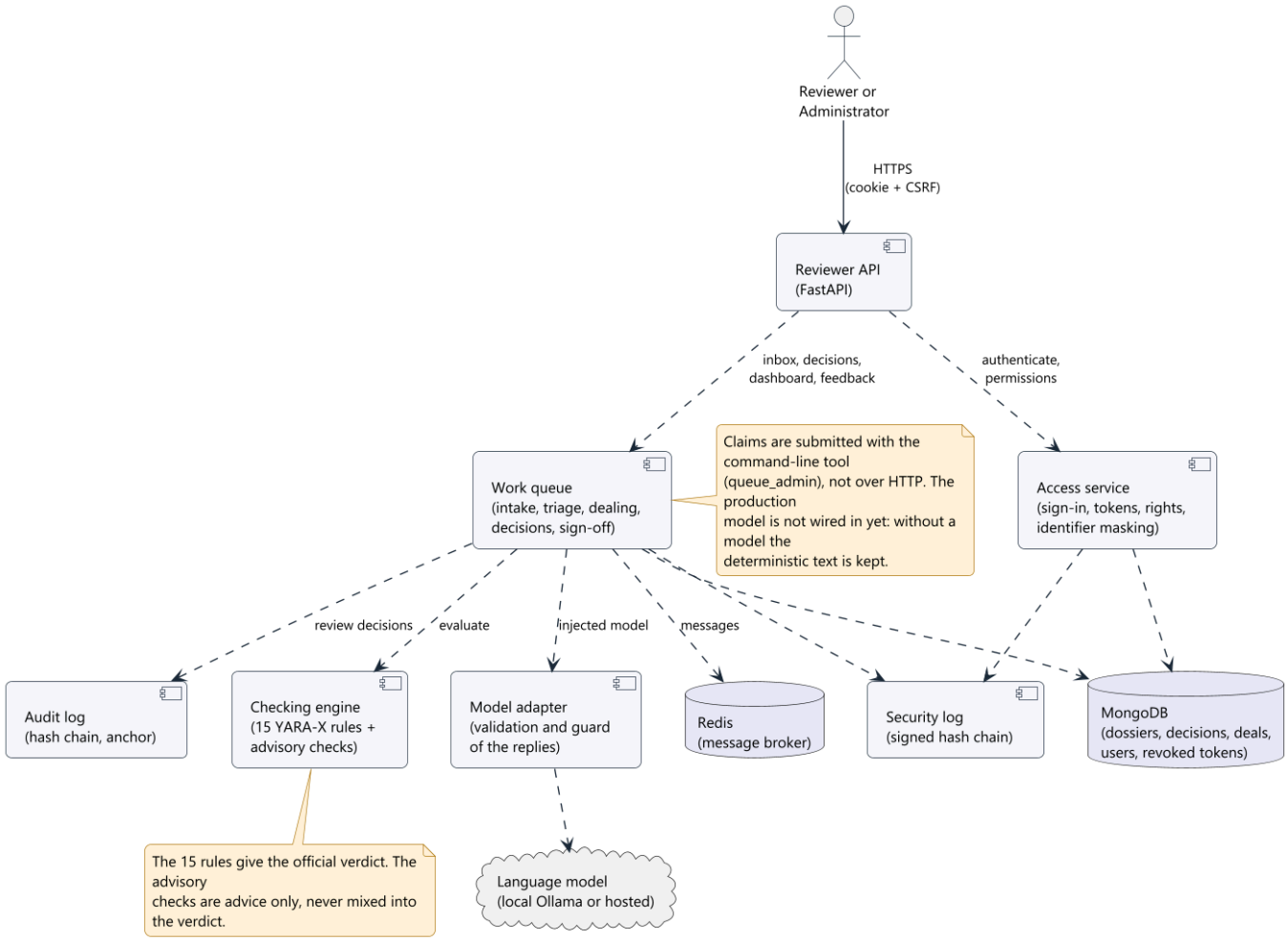
every finding resolved

awaiting_countersign

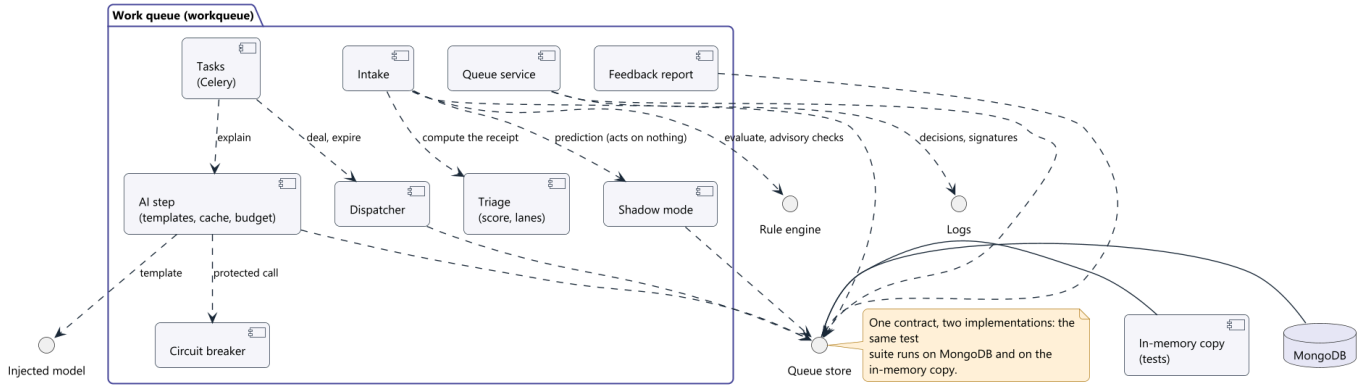
decided

A decided dossier is never changed. A corrected claim
submitted again is a NEW document (version n+1) that
starts at "triaged"; the old version stays "decided".

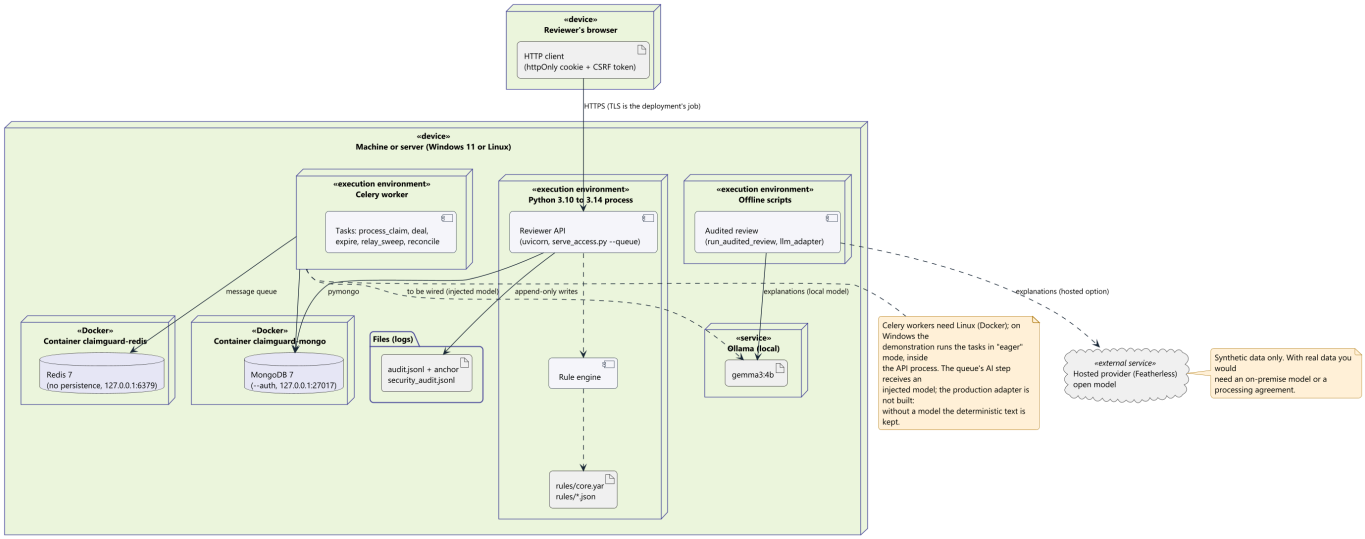
Component diagram (overview): ClaimGuard AI



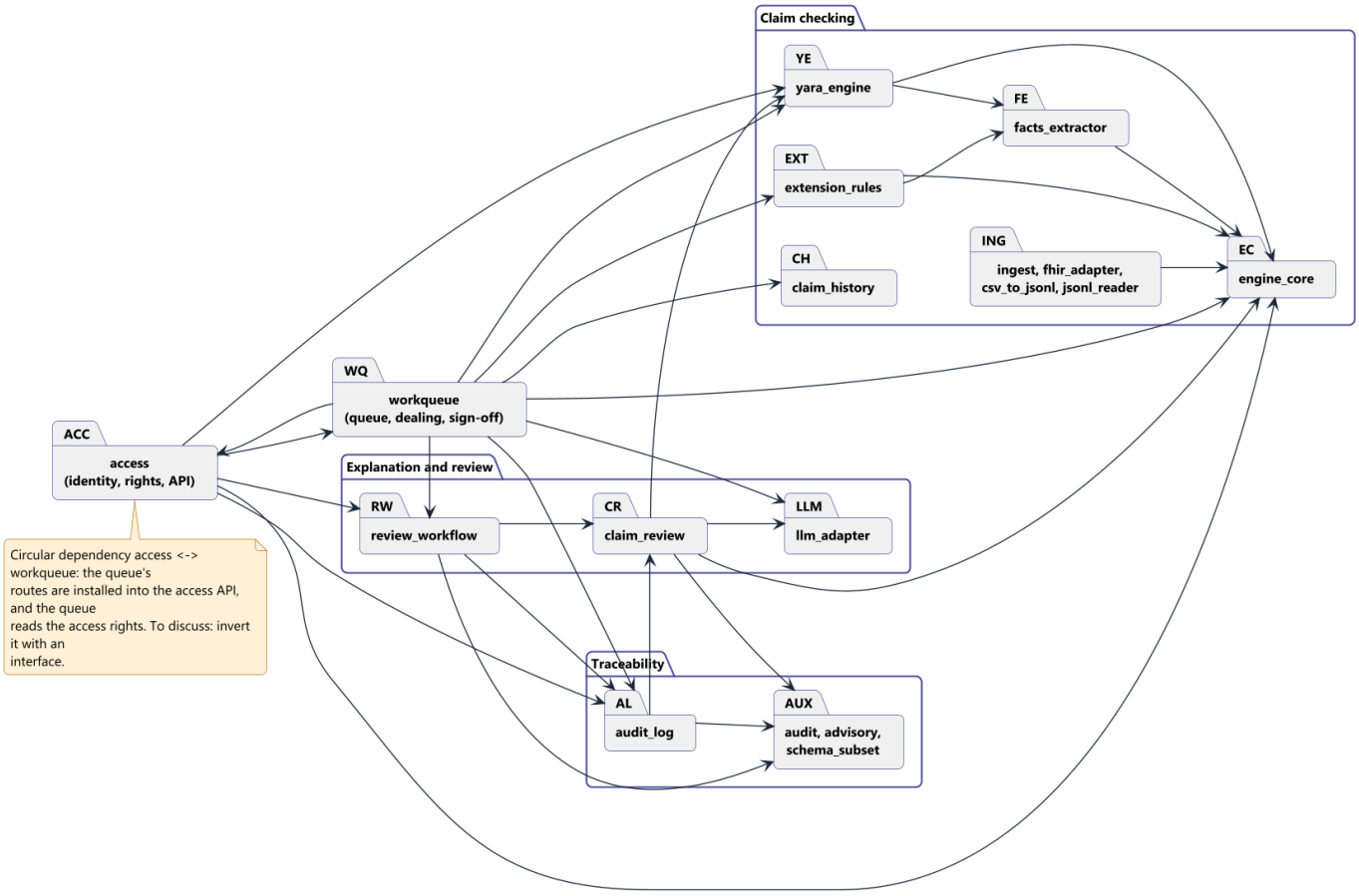
Component diagram (detail): inside the work queue



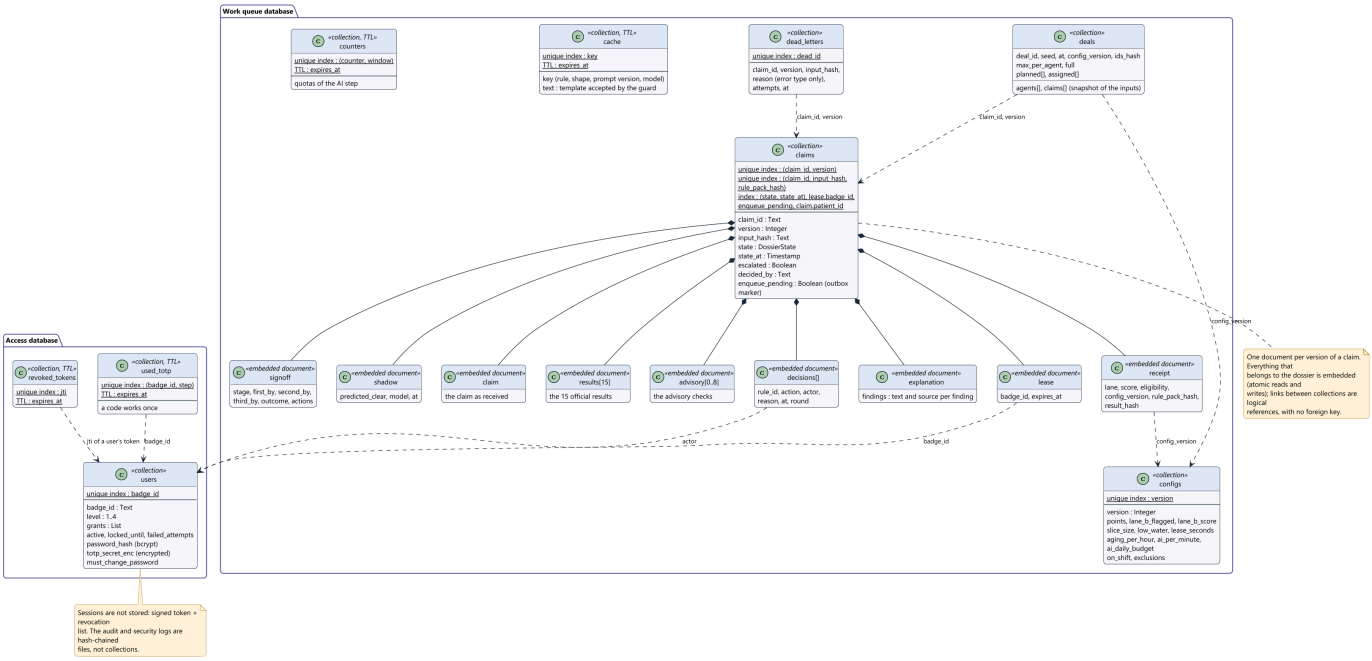
Deployment diagram: demonstration machine and target



Package diagram: real dependencies of the code (src folder)



NoSQL data model (MongoDB): collections, embedded documents and references



Sequence diagram: failure and recovery of a dossier in the pipeline

